Multi-Text Representation Learning for Patent Validity Prediction

Nghia Tran Pham

October 2026

Abstract

Predicting patent validity litigation outcomes requires combining information from text fields that serve distinct roles and may depend on one another for interpretation. Using a newly constructed dataset of UK patent litigation, I investigate four questions: which text fields provide useful predictive information, whether they should be encoded separately or jointly, how to represent texts whose meaning depends on other texts, and how to aggregate information across multiple challenges into an overall validity prediction. To characterize textual dependencies, I define maximum dependency depth as the length of the longest chain of dependencies. I then construct expanded text representations that explicitly incorporate relevant content from the texts on which they depend. To aggregate information across challenges, I develop a multiple-instance learning architecture tailored to patent validity litigation. The architecture uses cross-attention, constructing queries from representations of the challenged patent claim and keys and values from representations of the texts cited against it. Empirical results show that the proposed architecture predicts better than the architectures used in the literature in this setting.

1 Introduction

Patents can be economically important intangible assets. They allow their owners to protect their market position by excluding competitors from using the patented invention, or to generate revenue by selling patent rights. Empirical evidence links patents to firms' market value (Hall et al., 2005) and shows that patent litigation outcomes can have substantial financial consequences (Panattoni, 2011). Predicting these outcomes can therefore help assess the legal uncertainty surrounding patent rights. Since patent validity depends in part on the content of the patent itself, natural language processing may help predict litigation outcomes. The main goal of this paper is to predict patent litigation outcomes using natural language processing.

My dataset contains all U.K. patent litigation judgments from 1995 through the end of August 2026, covering 405 cases and 561 patent invalidity events. Since judgments are usually not written in a format suitable for analysis, my first contribution is to develop LLM prompts for extracting structured information from published judgments. I provide predictions at two levels, corresponding to ex-ante and ex-post information sets defined relative to the hearing. Here, the ex-post information set refers to information available after the hearing but before the judgment is issued.

A patent consists of two parts: the description, which explains the invention, and the claims, which define the scope of protection. In litigation, validity is assessed at the level of individual claims. However, which claims will be challenged may not be known before the hearing. Hence, at the ex-ante-hearing level, I predict the outcomes while restricting the information set to the text of the patent under attack, without assuming knowledge of the disputed claims.

At the ex-post-hearing level, I predict outcomes for individual grounds of attack against a claim, conditional on the claims challenged, the grounds pursued, and the documents relied upon at the hearing. A ground of attack is a legal basis for challenging the validity of a claim. I focus on insufficiency, novelty, and obviousness, the three most common grounds of attack in the dataset.

Both of these predictions can be important for different reasons. The ex-ante level is important for a patent owner because it indicates broadly which parts of their patent portfolio are weak and susceptible to attack. Since the owner also has to pay renewal costs, this information also enters into their calculation of which patents are worth retaining. For a trader, ex-ante prediction helps them trade on information available at the time of the first hearing. At the post-hearing level, the owner would know which attacks may be dangerous and how to prepare for them. For the challenger, it indicates which attacks are likely to be successful. For the trader, it allows them to trade on post-hearing information by taking a position shortly before the judgment is issued.

Here, my ex-post information set differs most sharply from those used in the existing literature, including PANORAMA (Lim et al., 2025) and FiNE-Patents (Knappich et al., 2026). My approach assumes a worst-case scenario for prediction, in which no additional information about the judge's analysis is used. This distinction is important for my goals: observers would not have access to the judge's analysis until the judgment is issued.

In finding the best predictive models, a few empirical questions naturally arise. First, how should a model represent text whose meaning depends on other text? A dependent patent claim incorporates limitations from the claims to which it refers. For example, claim 2 may add a restriction to claim 1, while claim 3 may refer to either claim 1 or claim 2 and add a further restriction. The literal text of a dependent claim therefore does not state all the limitations that define its scope.

I address this problem by introducing the concept of an *expanded claim*. With the exception of Gao et al. (2024), the existing literature has largely handled claim dependency either by restricting attention to independent claims or by treating dependent claims in the same way as independent claims. I instead represent the dependency structure as a tree, in which each path from a dependent claim eventually terminates at an independent claim. A dependent claim is then expanded by tracing each of these paths and incorporating the text of the claims on which it depends. This produces a self-contained representation that explicitly includes the limitations inherited through the dependency structure. Intuitively, one might expect prediction accuracy to deteriorate as dependency depth increases, since deeper claims require information from a longer chain of claims. In my test set, however, I do not observe such a decline.

Second, which texts provide useful predictive information? Adding more text does not necessarily improve predictive performance. For example, in arithmetic reasoning tasks, Shi et al. (2023) show

that adding irrelevant information can reduce the accuracy of large language models. These findings motivate an empirical assessment of which additional texts improve prediction in the patent-litigation setting. To my knowledge, however, no existing study makes the systematic comparison of text-input combinations its central research question in patent-validity prediction. FiNE-Patents (Knappich et al., 2026) compares some input configurations through ablation experiments, but identifying which combinations of texts should be used is not its primary objective; further, these ablation experiments do not exhaust all comparisons. I compare models using different combinations of the challenged claim, the complete claim set, the patent description, and the corresponding texts of the cited prior-art patents.

Third, how should these texts be combined? One approach concatenates the texts and encodes them jointly; another encodes each text separately and then combines the resulting representations. When multiple texts are supplied as a single input to a large language model, they interact through self-attention. However, Chen et al. (2019) show that separately encoding a document's title and using it to guide the document representation can outperform simple concatenation.

The results favour encoding text fields separately and then combining their representations across all prediction tasks. One possible explanation is the substantial imbalance in text length. For example, the claim is much shorter than other text fields, yet they are equally important. Concatenating them into a single input may therefore make it harder for the model to preserve information specific to the claim within the resulting representation. Separate encoding preserves a distinct representation of each source before combining their information for prediction.

Fourth, given the attacks brought against a patent claim, how can we predict its overall validity? I formulate this task as a multiple-instance learning (MIL) problem: each bag contains the individual attacks against a claim, and the bag label indicates whether the claim is invalid. To address this task, I develop a new architecture, which constitutes the main contribution of this paper. The architecture encodes the challenged claim separately from the texts associated with each attack. These representations then interact through cross-attention, with queries derived from the claim representation and keys and values derived from the representations of the associated texts. The intuition is that the relevance of the available evidence depends on the claim being challenged. By using the claim as the query, the model learns which information to emphasize and how to combine evidence across attacks into an overall validity prediction.

The remainder of the paper is organized as follows. The remainder of Section 1 reviews the related literature. Section 2 describes the U.K. patent system and the institutional and legal setting. Section 3 explains the construction of the dataset, the outcome definitions, and the training and evaluation samples. Section 4 describes the prediction models and evaluation metrics. Section 5 presents the empirical results at the patent, claim, and ground-of-attack levels. Section 6 concludes.

1.1 Related Literature

1.1.1 Machine-learning literature

Prediction settings. Relatively few studies directly predict outcomes in patent litigation. Much of the related literature instead examines patent-office decisions on whether to grant a patent or reject an application on a particular ground (Lo and Chu, 2021; Ashtor, 2022; Knappich et al., 2025; Lim et al., 2025; Knappich et al., 2026). Studies of litigation outcomes include Janicke and Ren (2006), Cowart et al. (2014), and Hicks (2021). These earlier studies employ logistic regression, trees, or TF–IDF, without evaluating transformer-based language models. To my knowledge, my paper is the first study to evaluate transformer-based language models for predicting patent-validity outcomes in a litigation setting.

Information available for prediction. A second distinction concerns the information available when predictions are made. Raghupathi et al. (2018) and Raghupathi (2026) apply text-mining and classification methods to published judicial opinions in pharmaceutical patent cases. These texts contain the court's reasoning, which is unavailable before judgment. A related distinction arises in patent examination: PANORAMA's patentability-classification task supplies passages from prior-art documents that the examiner cited in the office action (Lim et al., 2025). Although valuable in their own right, these exercises do not directly address the problem faced by traders and lawyers seeking to assess patent vulnerability before the relevant decision is issued.

Hicks (2021) provides the closest comparison to my study in terms of the prediction task, predicting judicial validity outcomes from claim text without relying on the court's written reasoning. He studies U.S. district-court cases filed in 2008 and 2009, considering specifications based on either the complete claim set or the asserted claims. He examines validity outcomes at both the patent and ground-of-attack levels, including novelty, obviousness, and indefiniteness. However, his textual inputs do not include the patent description or cited prior-art documents. My study extends the range of texts examined and evaluates their predictive value using alternative representations and models. Published judgments supply outcome labels and identify the claims, grounds pursued, and documents relied upon at the hearing, while judicial reasoning and statements of the outcome are excluded from the predictive inputs.

Representing dependent claims. Claim dependencies create an additional challenge because the literal text of a dependent claim does not state all the limitations defining its scope. Gao et al. (2024) explicitly model these dependencies when predicting novelty rejections from USPTO prosecution data. They decompose claims into textual nodes and connect them through a graph representing relationships within claims and across parent and ancestor claims. Fixed text embeddings initialize the nodes, and graph neural networks combine information across connected nodes to produce a prediction. PEDANTIC (Knappich et al., 2025) addresses the problem using a different approach. They use a retrieval-augmented agent that can obtain relevant passages from parent claims when interpreting the target claim.

Multiple-instance learning. MIL provides a framework for predicting outcomes for bags of instances (Dietterich et al., 1997; Maron and Lozano-Pérez, 1998). A prominent application is whole-slide image classification, where information from image patches is aggregated into a slide-level prediction. Ilse et al. (2018) develop attention-based pooling of instance representations, while TransMIL incorporates morphological and spatial correlations among patches (Shao et al., 2021).

In my setting, each bag represents a claim and contains the attacks brought against it. Every instance within the bag contains exactly the same challenged claim and full claim set, while the grounds of challenge and associated texts may vary. This sharing of entire input components is distinct from correlations among image patches and motivates an explicit separation between information common to the bag and information specific to each attack. Similar settings appear in Sathe and Park (2021) and Kumar et al. (2023). The closest architectural connection is Xu et al. (2025), who use cross-attention with the query as Q and the bag instances as K and V. My architecture follows a similar principle, using the challenged claim as the query and the attack-specific representations as keys and values.

1.1.2 Economics literature

The economics literature provides two motivations for predicting patent-validity outcomes: patents can generate substantial value for their owners, and the loss of protection can affect competition and subsequent innovation. These studies establish the economic consequences of patent protection; my paper examines whether patent texts and related information can predict whether challenged claims survive.

The first branch examines the value of patents to their owners. Pakes (1986) estimates annual returns from patent protection at 11–16% of R&D expenditure, with half accruing to 5–10% of patents. Hall et al. (2005) find that, conditional on R&D and patent stocks, one additional citation per patent is associated with approximately 3% higher firm value at the sample mean. Kogan et al. (2017) find evidence of creative destruction: high-value patents benefit their owners, while capital and labor move from non-innovating firms toward innovating firms. Farre-Mensa et al. (2020) show that patent grants accelerate startup growth and improve access to external financing.

The second branch studies the economic consequences of losing patent protection. Marco (2005) estimates that invalidity decisions reduce owners' stock-market value by approximately 1.4%. Panattoni (2011) finds cumulative abnormal market-value losses of slightly more than $1 billion for branded pharmaceutical firms losing patent litigation in the sample. Following patent expiration and generic entry, Suh et al. (2000) find that originators' dollar sales fall by approximately 12% by the fourth year and their revenue market share falls to 78% of its initial level. Branstetter et al. (2016) show that generic entry following successful patent challenges reduces prices and incumbents' market shares while increasing competition and consumer surplus. Galasso and Schankerman (2015) find that invalidation increases follow-on innovation, measured by subsequent citations, in computers, electronics, and medical instruments, with no significant effect in drugs, chemicals, or mechanical technologies.

2 Overview of the UK Patent System

2.1 Institutions and procedural settings

In England and Wales, first-instance patent litigation is heard principally by the Patents Court and the Intellectual Property Enterprise Court (IPEC). The Patents Court generally handles larger and more complex disputes, while IPEC provides a streamlined and cost-controlled procedure for smaller, less complex disputes. The UKIPO also determines certain patent disputes through the Comptroller, acting as an administrative tribunal rather than a court. Appeals from most decisions of the Comptroller lie to the Patents Court. Judgments of the Patents Court and IPEC may be appealed, with permission, to the Court of Appeal, with a further appeal potentially reaching the UK Supreme Court.[1]

Questions concerning patentability or validity principally arise in three settings. First, before grant, the UKIPO assesses whether an application satisfies the statutory requirements for patentability. Second, after grant, a person may apply to the court or the Comptroller to revoke a patent. Third, a defendant in infringement proceedings may challenge the patent's validity, usually by counterclaiming for revocation. The first setting concerns the patentability of an application, while the latter two concern the validity of an existing patent; the substantive requirements largely overlap.

2.2 The specification and the claim-level nature of validity

A patent specification contains a description of the invention, claims, and any relevant drawings. The description explains the invention and how it may be performed, while the claims define the scope of protection. An independent claim stands on its own. A dependent claim incorporates the limitations of an earlier claim and adds further limitations. Validity is assessed claim by claim, so some claims may survive while others are invalidated.

Throughout this paper, I use *Leo Pharma v Sandoz* as a running example.[2] The dispute concerned European Patent (UK) EP0679154B1, covering calcipotriol monohydrate and pharmaceutical compositions containing it. Example 2.1 presents its five claims in abbreviated form.

Example 2.1 (Independent and dependent claims)**.**

1. Calcipotriol monohydrate.

2. A pharmaceutical composition containing the compound of claim 1.

3. A pharmaceutical composition according to claim 2 that is a cream.

4. A pharmaceutical composition according to claim 2 that is a gel.

[1] The institutional descriptions draw on UK Intellectual Property Office, nd, secs. 18, 72, 74, and 97, Courts and Tribunals Judiciary, ndb, Courts and Tribunals Judiciary, nda, and UK Supreme Court, nd.

[2] *Leo Pharma A/S and another v Sandoz Ltd* [2009] EWHC 996 (Pat), 15 May 2009. `https://www.bailii.org/ew/cases/EWHC/Patents/2009/996.html`.

5. A pharmaceutical composition according to any one of claims 2–4, with an active-component content of 1–100 μg/g of the composition.

Claim 1 independently defines the compound. Claim 2 incorporates claim 1 and adds the requirement of a pharmaceutical composition. Claims 3 and 4 restrict claim 2 to particular dosage forms, while claim 5 adds a concentration restriction to any one of claims 2–4.

Invalidating claim 1 does not automatically invalidate claims 2–5: their additional limitations may support novelty and inventiveness. Each challenged claim must therefore be assessed on its own limitations and against the grounds raised.

2.3 Grounds of invalidity

Section 72(1) of the Patents Act 1977 states the exhaustive statutory grounds on which a granted patent may be revoked. These grounds are divided into the following seven substantive categories plus entitlement, which I will not use in my analysis. The reason is that entitlement is an ownership dispute rather than an assessment of the claimed invention's technical validity.

Before introducing the seven grounds of invalidity, it is useful to define three terms: the priority date, prior art, and the person skilled in the art (UK Intellectual Property Office, nd, secs. 2, 3, 5, and 14).

The *priority date* is the cut-off date used to decide what earlier information can be used against an invention. It is usually the filing date of the patent application.

Prior art is information that was already available to the public before the priority date. It does not have to be an earlier patent or even something described as an invention. It may be a journal article, a product sold to the public, a conference presentation, a public demonstration, a website, or any other public disclosure anywhere in the world.

The *person skilled in the art* is a hypothetical person who is competent in the relevant technical field but is not unusually inventive.

The seven categories are as follows:

1. **Lack of novelty (anticipation).** A claim lacks novelty if a single prior-art disclosure reveals every feature of the claimed invention and enables a person skilled in the art to perform it. Separate disclosures cannot be combined to establish anticipation.

2. **Lack of inventive step (obviousness).** A claim is invalid if the claimed invention would have been obvious to a person skilled in the art, given the relevant prior art and common general knowledge at the priority date.

3. **Insufficiency.** The patent must disclose the invention clearly and completely enough for a person skilled in the art, using common general knowledge, to perform it across the scope of the claim without undue burden.

4. **Lack of industrial applicability.** The invention must be capable of being made or used in industry, understood broadly to include practical activities such as agriculture.

5. **Excluded subject matter.** Some types of subject matter, such as scientific theories and discoveries as such, cannot be patented even if they are novel and non-obvious.

6. **Added matter.** A patent may be revoked if the matter disclosed in the granted specification extends beyond the disclosure of the application as filed.

7. **Extension of protection by post-grant amendment.** This ground arises when a patent has been amended after grant. An amendment is impermissible if it extends the protection conferred by the patent as granted.

A claim may face several attacks and survive some while failing others. One successful attack is sufficient to invalidate the claim.

3 Data collection and description

3.1 Sample scope and judgment collection

I construct a dataset of cases concerning patentability or validity decided by the Patents Court of the High Court of Justice of England and Wales and the Intellectual Property Enterprise Court (IPEC) between 1995 and August 2026.

Scope of the sample. The sample covers three types of proceedings: applications for revocation; infringement proceedings involving a counterclaim challenging patent validity; and appeals to the Patents Court from UKIPO decisions. The third category includes appeals against the refusal of a patent application and appeals against decisions concerning the revocation of a granted patent.

I exclude judgments of the Court of Appeal and the UK Supreme Court to avoid counting the same underlying dispute again at a subsequent stage. Including both the lower-court judgment and the appellate judgment would generate repeated observations with substantially overlapping patent and party characteristics, as well as the grounds of attack.

Sources and inclusion criteria. I identify cases using vLex, a legal research database, and cross-check its listings against BAILII, the British and Irish Legal Information Institute's publicly accessible repository of legal decisions. Some vLex entries contain the full judgment, while others provide only the case title or bibliographic information. When neither source provides the full judgment, I search CaseMine, another legal research database, and other available sources.

Where a full judgment is missing, I reconstruct information using a subsequent Court of Appeal judgment. The appellate judgment is used only as a source of information about the earlier proceedings; the observation and recorded outcome remain those of the Patents Court or IPEC.

A case is retained only if the available material provides the information required to complete the case-level record. This includes the names of the parties (the claimant and defendant, or the appellant and respondent), the identity of the patent or application in dispute, and the outcome.

For 1995–2003, at most four otherwise eligible cases are missing from the sample in any given year. Coverage of eligible cases is complete for 2004–August 2026.

A case may satisfy these inclusion criteria even when the available material does not identify the specific claims challenged. Such cases are retained for patent-level analysis but do not contribute observations to the claim-level sample. Subsection 3.2 describes the observation units.

Sample size. The full court sample contains 405 cases: 364 in the Patents Court and 41 in the IPEC. These cases yield 561 patent-level observations, each corresponding to a patent or application considered in a particular case. The observations cover 549 distinct patents or applications, with 12 patents appearing in two proceedings each. For these repeated patents, the proceedings differ materially in the claims challenged or the prior art relied upon to support an obviousness challenge.

The sample includes both cases in which invalidity was conceded and cases in which patentability or validity was contested and determined by the court. Detailed claim-level sample sizes and the frequency of attacks on each ground are reported in Subsection 3.2.

Once a judgment has been obtained, I use four structured LLM prompts to extract outcome labels and prediction inputs. The full prompts are provided in Appendix D. Table 1 summarizes their inputs and functions.

Table 1: Overview of the extraction prompts

No.	Input	Main purpose	Summary
1	Judgment	Extract case metadata and information needed to construct numerical features.	Records court, parties, dates, patent details, and patent-level outcomes, and retrieves citation information.
2	Judgment, claim texts, and initial extraction records	Identify challenged claims, grounds of attack, and outcomes.	Produces attack-level records, links novelty and obviousness attacks to prior-art units, and records claim dependencies and expanded claim text.
3	Patent publication	Clean the patent text.	Verifies the publication and version, then extracts the English description and claims separately without translation or paraphrasing. Removes bibliographic material, repeated headers and footers, non-English duplicates, tables, drawings, and unusable formula or chemical-structure text.
4	Judgment and prior-art references	Clean the prior-art text.	Verifies the cited document and version and applies the same cleaning rules. Retains full documents or relevant chapters rather than only judge-selected passages. Uses English sources where available and literal translation otherwise.

Below I provide an example of the output of the second prompt. In *Leo Pharma v San-*

doz (Example 2.1), each of claims 1–4 faced one novelty attack based on the patent application WO1991012807A1 and two obviousness attacks based separately on applications WO1991012807A1 and WO1987000834A1. The novelty attack alleged that following Example 4 would inevitably produce a cream containing calcipotriol monohydrate; the obviousness attacks relied on routine formulation and crystallisation in light of the prior art and common general knowledge. The second extraction prompt therefore outputs four novelty and eight obviousness attack-level observations, with none for the other validity grounds.

3.2 Units of observation and outcome coding

At the patent level, an observation corresponds to a patent or application considered in a particular case. I classify the owner or applicant as losing if all challenged claims are invalidated or refused, and winning if at least one challenged claim survives. The reason is that a follow-up check using ChatGPT indicated that every patent with at least one surviving challenged claim remained in force following the dispute.

At the claim level, I include claims when their validity or patentability is challenged on at least one ground and the judgment determines the outcome. Claims mentioned only as technical background, through dependency references, or in relation to infringement are not included on that basis alone. Individual claim outcomes are not inferred solely from patent revocation.

Original and amended claim versions are recorded separately when both are assessed; otherwise, only the assessed version is included.

An attack-level observation corresponds to a claim version, a ground of challenge, and, for novelty and obviousness, a particular prior-art unit. Prior-art documents assessed jointly form one unit, even if extracted in separate rows; distinct units generate separate attacks. Outcomes are coded separately for each attack, with 1 indicating a loss for the owner or applicant and 0 indicating a win. A claim may therefore survive one attack while failing another.

Table 2 reports the frequency of each ground across 354 cases, 479 patent observations, and 1,776 claims.

Table 2: Frequency of invalidity attacks

	Novelty	Obviousness	Added matter	Industrial applicability	Excluded subject matter	Extension by amendment	Insufficiency
Cases with attack	207	317	119	7	42	13	136
Share of cases (%)	58.5	89.5	33.6	2.0	11.9	3.7	38.4
Patents with attack	241	429	145	7	56	15	172
Share of patents (%)	50.3	89.6	30.3	1.5	11.7	3.1	35.9
Claims with attack	753	1,519	384	35	175	27	622
Share of claims (%)	42.4	85.5	21.6	2.0	9.9	1.5	35.0

Denominators: 354 cases, 479 patent observations, and 1,776 claims.

The same patent is counted separately when considered in different judgments.

For the ground-specific prediction exercises, I focus on novelty, obviousness, and insufficiency, the three grounds with the largest numbers of challenged claims. At least one of the three focal

grounds is raised in 332 of the 354 cases involving a substantive invalidity attack, corresponding to 93.8% of such cases.

These grounds also capture most overall claim-level outcomes. Among the 1,698 claims challenged on at least one of the three focal grounds, considering the outcomes on these grounds alone yields the same overall claim-level classification for 1,666 claims, or 98.1%. Only 32 claims have an overall outcome different from that implied by the focal grounds alone. Each of these claims survives the challenges brought against it on the focal grounds but is invalidated for added matter.

3.3 Claim dependencies and expanded claim representations

A dependent claim incorporates the limitations of one or more earlier claims and therefore generally cannot be interpreted from its own text alone. For a given target claim, its dependency structure can be visualized by unfolding the dependency paths into a tree. The target claim forms the root, each branch follows a reference to an earlier claim, and each path terminates at an independent claim. An earlier claim may appear on more than one branch when different dependency paths lead to it.

Definition 3.1 (Maximum dependency depth)**.** A claim's *maximum dependency depth* is the number of dependency links in the longest path from that claim to an independent claim. An independent claim therefore has maximum dependency depth zero. For a dependent claim, the depth equals one plus the greatest maximum dependency depth among its direct parent claims.

Formally, let $\mathcal{P}(c)$ denote the set of direct parent claims referenced by claim c, including all alternatives specified by references such as "any preceding claim." The maximum dependency depth of claim c, denoted by $d(c)$, is

$$d(c) = \begin{cases} 0, & \text{if } \mathcal{P}(c) = \varnothing, \\ 1 + \max\limits_{p \in \mathcal{P}(c)} d(p), & \text{if } \mathcal{P}(c) \neq \varnothing. \end{cases}$$

An *expanded claim* is a self-contained representation that explicitly incorporates the limitations inherited through this dependency structure. The claim-level extraction prompt instructs the LLM to trace every dependency path from the target claim to an independent claim and incorporate the relevant wording along that path. Where the dependency references specify alternative paths, the expanded text preserves those alternatives. Each alternative includes both the limitations inherited along its path and the limitations added by the target claim. For example, the expanded version of claim 5 in Example 2.1 is:

> A pharmaceutical composition containing calcipotriol monohydrate, under any one of the following alternatives: (i) no additional limitation as to dosage form; (ii) the pharmaceutical composition is a cream; or (iii) the pharmaceutical composition is a gel; with an active-component content of 1–100 μg/g of the composition.

Greater dependency depth requires information to be combined across longer chains of claims, potentially increasing the difficulty of interpretation and prediction. I therefore conduct the main specification search and architecture selection using claims with a maximum dependency depth of at most two. After selecting the architectures, I evaluate the same model specifications on the full claim sample without imposing a dependency-depth restriction. This exercise assesses whether architectures selected using claims with simpler dependency structures also perform well when deeper claims are included, without conducting a separate architecture search for the unrestricted sample.

3.4 Training, validation, and evaluation protocol

I select models and architectures using an approximately 80/20 training–validation split of the 1995–2022 court sample. For the ex-ante and individual attack-level analyses, I assess the stability of shortlisted specifications using Monte Carlo cross-validation (MCCV) before evaluating them on the two holdout test sets (Arlot and Celisse, 2010). MCCV uses 25 or 50 random case-level 80/20 splits, depending on the task, with all observations from the same case kept together. Models are refitted on each split, while architectures, specifications, and hyperparameters remain fixed. Repeated splitting assesses sensitivity to sample partitioning, which can substantially affect estimates from a single run of k-fold cross-validation (Moss et al., 2018).

For each selected ex-ante or individual attack-level architecture, I also evaluate a version without numerical features to assess prediction using text alone. The claim-level aggregation comparison uses the dedicated model-selection split reported in Section 5.3.

I then evaluate the shortlisted specifications on two holdout sets. The temporal holdout contains 59 patent observations from 44 cases decided in 2023–August 2026: 38 in the Patents Court and 6 in the IPEC. It tests whether models trained on past decisions predict outcomes in future disputes.

The institutional holdout contains 85 patent observations from 82 UKIPO cases. These proceedings are decided by hearing officers rather than judges, providing a test of generalization across institutions and decision-makers. I exclude eight UKIPO cases (eight patent observations) whose Patents Court appeals are already included in the training and validation sample.

4 Models and evaluation metrics

For patent-level and attack-level prediction, I compare three groups of methods: classical models, models using frozen transformer embeddings, and transformer models fine-tuned with low-rank adaptation (LoRA). I then introduce multiple-instance learning architectures for predicting overall claim validity across multiple attacks. The main text explains the intuition behind these methods, with additional mathematical details provided in Appendix B.

4.1 Classification heads

A model first maps each text field into a numerical representation vector. The resulting representation may also be combined with additional numerical features before being passed to a task-specific prediction head. The linear classification heads considered here comprise logistic regression, support vector machines (SVMs), and L_2-regularized least-squares classifiers.

Their generic training objective is

$$(\widehat{w}, \widehat{b}) = \arg\min_{w,b} \left\{ \frac{\sum_{i=1}^{n} a_{y_i} s_i \, \ell_i(w, b)}{\sum_{i=1}^{n} s_i} + \lambda R(w) \right\}, \qquad s_i = n_{p(i)}^{-\gamma}, \tag{1}$$

where w is the coefficient vector, b is the intercept, ℓ_i is the classifier-specific loss, and a_{y_i} and s_i are the class and sample weights. The penalty $R(w)$, with strength λ, discourages overfitting. Classifier-specific losses and penalties are provided in Appendix B.

Class weighting. I compare uniform and balanced class weights. Balanced weights give greater weight to observations from less frequent outcome classes.

Sample weighting. A patent group is defined by the case and patent together, and $n_{p(i)}$ is the number of training observations in the group containing observation i. The parameter $\gamma \in [0, 1]$ controls the adjustment for group size. With $\gamma = 0$, observations receive equal weight; with $\gamma = 1$, sample weights sum to one within each patent group. Intermediate values partially reduce the influence of patents contributing more observations.

4.2 Classical (non-neural) models

Numerical-only models and Extra Trees. I also investigate how accurately outcomes can be predicted without textual features. For this exercise, I use only the numerical features and apply the classification heads described above. In addition, I consider the Extremely Randomized Trees classifier, commonly referred to as Extra Trees (**?**).

TF–IDF representation. TF–IDF (**?**) represents each text field as a numerical vector, with one component for each term in the vocabulary. Each component reflects the frequency of that term in the text, multiplied by an inverse document frequency weight. This weight discounts terms that appear in many texts and gives greater weight to terms that appear in fewer texts.

The resulting vector is concatenated with the numerical features and passed to one of the classification heads described in the previous subsection for training and prediction.

Complement Naive Bayes. For each outcome class, Complement Naive Bayes (**?**) constructs a profile of term use from texts outside that class, referred to as its complement. In my binary setting, the complement of class 1 consists of texts from class 0, and vice versa.

To construct each profile, the classifier sums the TF–IDF weights of each term across texts in the complement. At prediction time, the classifier selects the class whose complement provides the poorest explanation for the text's pattern of term use.

Word2Vec. Word2Vec (**?**) learns vector representations of individual words from their surrounding contexts. I train these embeddings from scratch using two variants: continuous bag-of-words (CBOW), which predicts a word from its surrounding words, and skip-gram, which predicts surrounding words from a central word. After training, I represent each text field by averaging its word vectors, concatenate the resulting vector with the numerical features, and pass it to a classification head.

4.3 Frozen embeddings

I represent each text field using frozen embeddings from `Qwen3-Embedding-8B` (Zhang et al., 2025). Only the downstream layers and classification head are trained using outcome labels.

My applications frequently involve several related text fields. For example, predicting whether a patent claim is obvious may require information from both the patent and the relevant prior art. In addition to representing each text field separately, it may therefore be useful to explicitly represent the relationship between them. I consider three methods for combining the resulting embeddings. In each method, the combined representation is concatenated with numerical features before classification.

Simple concatenation. I concatenate the text embeddings, optionally adding a hidden layer with 32–128 units before the classification head.

Interaction features. For selected embedding pairs e_{ia} and e_{ib}, I augment concatenation with the element-wise product $e_{ia} \odot e_{ib}$ and absolute difference $|e_{ia} - e_{ib}|$, capturing alignment and differences, respectively (Conneau et al., 2017; Reimers and Gurevych, 2019). I also add either cosine similarity alone or five scalar measures: cosine similarity, mean absolute difference, Euclidean distance, maximum absolute difference, and mean element-wise product. The augmented representation is passed directly to the classification head or through a hidden layer.

Projected fusion. I apply a separate trainable nonlinear projection to each text field's embedding to obtain a lower-dimensional, task-specific representation (Ngiam et al., 2011). The projected embeddings are normalized and concatenated, optionally augmented with the interaction features above.

4.4 LoRA tuning

The final family of models uses Low-Rank Adaptation (LoRA). A pretrained language model has already learned general language representations from a large corpus. Fine-tuning adapts

these representations to the prediction task through further training on labeled data. Full fine-tuning updates all pretrained parameters together with the classification head, which might be computationally expensive and may be prone to overfitting in this setting, given the small number of labeled observations relative to the number of model parameters. I therefore use LoRA to adapt the model with substantially fewer trainable parameters (Hu et al., 2022).

Two key parameters for LoRA tuning are r and α. The rank r controls both the number of trainable parameters and the flexibility of the adaptation. A larger rank r introduces more trainable parameters and allows greater flexibility. The parameter α controls the magnitude of the LoRA adjustment relative to the pretrained weights and does not change the number of trainable parameters.

I use the same LoRA adapter for the text fields. The resulting representations, together with any frozen embeddings, are combined using the combination schemes described in Section 4.3.

4.5 Claim-level validity across multiple attacks

Given the attacks brought against a claim, how should their information be aggregated to predict overall validity? Among the insufficiency and obviousness attacks retained in this study, a claim survives only if every attack fails. This gives a multiple-instance learning formulation.

Let $x = (c, s)$ contain the expanded claim text c and the full claim set s. A claim facing K attacks is represented by the bag

$$X = \{(x, a_1), \ldots, (x, a_K)\},$$

where a_i contains the text associated with attack i: the patent description for insufficiency, or the cited prior-art text for obviousness. Thus, every instance shares the same claim-side texts x, while the attack-specific texts may vary.

Let $y_i = 1$ if attack i succeeds and $y_i = 0$ otherwise. The claim-level invalidity label is

$$y = \begin{cases} 0, & \sum_{i=1}^{K} y_i = 0, \\ 1, & \sum_{i=1}^{K} y_i > 0. \end{cases} \tag{2}$$

The attack outcomes define the bag label; they are not inputs to the prediction model.

Attention blocks. Let U, V be any matrix. Following Lee et al. (2019), define the Multihead Attention Block (MAB) as

$$\mathrm{MAB}(U, V) = \mathrm{LayerNorm}\big(H + \mathrm{rFF}(H)\big), \tag{3}$$

$$H = \mathrm{LayerNorm}\big(U + \mathrm{Multihead}(U, V, V)\big), \tag{4}$$

where rFF is a feedforward transformation applied independently to each row, with shared parameters. The first argument supplies the queries, while the second supplies the keys and values. The Set Attention Block (SAB) is

$$\mathrm{SAB}(U) = \mathrm{MAB}(U, U). \tag{5}$$

It allows the rows of U to interact through self-attention.

Claim-query cross-attention. I encode the expanded claim, full claim set, and attack-specific texts separately. Let U denote the representations derived from the expanded claim and full claim set, and let $V = [A_1; \ldots; A_K]$ collect the attack-specific representations, where each row A_i encodes the text associated with attack i. These are combined through

$$Z = \mathrm{MAB}(U, V), \tag{6}$$

where U supplies the queries and V supplies the keys and values. This allows the claim-side information to guide the aggregation of information across attacks.

The intuition is that the relevance of an attack depends on the claim being challenged. Using claim-side representations as queries allows the model to learn how much weight to assign to each attack when constructing the representation used for the overall prediction. Separate encoding retains a distinct representation of each text before this interaction. This design choice is informed by the separate-versus-joint encoding comparisons reported in Subsections 5.2.1 and 5.2.2.

Ground-type encoding and classification. To distinguish insufficiency from obviousness, I also consider specifications with learned ground-type embeddings. Let $g_i \in \{\mathrm{INS}, \mathrm{OBV}\}$ denote the ground of attack i, and let $t_{\mathrm{INS}}, t_{\mathrm{OBV}} \in \mathbb{R}^{1\times p}$ be learned vectors, where p is the dimension of each attack representation A_i. The attack representations become

$$\widetilde{A}_i = A_i + t_{g_i}, \qquad i = 1, \ldots, K. \tag{7}$$

Writing $\widetilde{V} = [\widetilde{A}_1; \ldots; \widetilde{A}_K]$, Equation (6) becomes

$$Z = \mathrm{MAB}(U, \widetilde{V}).$$

The output Z is flattened, concatenated with numerical features, and passed to a classification head to predict claim invalidity.

Self-attention. In the self-attention specifications, the expanded-claim and full-claim-set representations interact through an SAB before they are fused to construct the claim-side query U.

I compare seven supervised specifications:

- Individual-attack predictions aggregated using mean or max probability pooling (IND).

- Attention-based MIL pooling without ground-type encoding (ILSE).
- Attention-based MIL pooling with ground-type encoding (ILSE+T).
- Claim-query cross-attention without ground-type encoding or self-attention (CUS).
- Claim-query cross-attention with ground-type encoding and without self-attention (CUS+T).
- Claim-query cross-attention with self-attention and without ground-type encoding (CUS+SAB).
- Claim-query cross-attention with ground-type encoding and self-attention (CUS+T+SAB).

Appendix B.3 defines mean and max probability pooling and the attention and gated-attention operators of Ilse et al. (2018).

4.6 Evaluation metrics

Because outcome classes are often imbalanced, accuracy alone can be misleading. I select models using validation-set Matthews correlation coefficient (MCC), with macro-F1 and then accuracy as secondary criteria when MCC values are very similar. I also report balanced accuracy, ROC–AUC, claim-weighted accuracy, patent-weighted accuracy, and class-1 precision and recall as supplementary diagnostics. All definitions are in Appendix B.4.

5 Results

Table 3 summarizes the prediction samples. I follow the evaluation protocol in Section 3.4. For holdout evaluation, I refit each shortlisted specification on all training and validation observations, with training and evaluation using the same dependency-depth restriction where applicable.

Table 3: Prediction sample sizes and owner-win shares

		Training–validation		Court holdout		UKIPO holdout	
Task	Depth	N	Win (%)	N	Win (%)	N	Win (%)
Ex-ante	—	502	39.6	59	47.5	85	52.9
Insufficiency	≤ 2	442	61.5	46	54.3	37	62.2
	Full	622	61.1	64	54.7	60	56.7
Obviousness	≤ 2	1,342	55.7	228	67.5	232	81.5
	Full	1,789	53.3	306	58.8	379	85.5
Novelty	≤ 2	637	70.8	—	—	—	—
	Full	863	71.0	—	—	—	—
Claim-level validity	≤ 2	987	41.3	133	54.1	172	72.7
	Full	1,374	40.2	183	45.4	293	75.8

Observations are patents for ex-ante prediction, claims for insufficiency and claim-level validity, and claim–prior-art attacks for obviousness and novelty. Ex-ante wins include partial survival. For claim-level validity, a win means survival against every retained insufficiency and obviousness attack. Full depth imposes no dependency-depth restriction. Novelty models are not evaluated on the holdouts.

5.1 Ex-ante level

Owner loss is the majority class in the training–validation sample, giving an always-owner-loss classifier an accuracy of 60.4%.

In this setting, the best-performing models are TF–IDF and Word2Vec. Both the LoRA and frozen-embedding models collapsed to guessing the majority class (owner loss). One explanation is that the ex-ante task excludes information that is central to the legal determination. As shown in Section 3, obviousness is by far the most frequently raised ground of invalidity, followed by novelty. Both grounds depend critically on the relationship between the claimed invention and the relevant prior art, which is not included in the ex-ante information set.

Both models (TF–IDF and Word2Vec) use representations of both the description and the full claim set. For ex-ante prediction, at least in this setting, I conclude that both of these are equally important. Their performance on the original validation split is reported in Table C.1 in Appendix C.

Monte Carlo cross-validation. Table 4 summarizes performance across 50 case-grouped validation splits. In general, it seems that the addition of numerical features does not always lead to improvement in performance.

Table 4: Median performance of the shortlisted classical models across 50 case-grouped validation splits

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Precision (owner loss)	Recall (owner loss)
TF–IDF, without numerical features	**0.631**	**0.640**	**0.597**	**0.212**	**0.598**	**0.676**	**0.767**
	(0.062)	(0.055)	(0.056)	(0.115)	(0.059)	(0.042)	(0.072)
TF–IDF, with numerical features	0.619	0.600	0.582	0.164	0.581	0.667	0.683
	(0.056)	(0.049)	(0.050)	(0.102)	(0.051)	(0.039)	(0.074)
Word2Vec + linear SVM (L_1), without numerical features	0.590	0.580	0.572	0.150	0.565	0.667	0.583
	(0.068)	(0.061)	(0.061)	(0.122)	(0.059)	(0.060)	(0.100)
Word2Vec + linear SVM (L_1), with numerical features	0.600	0.580	0.574	0.157	0.568	0.670	0.615
	(0.064)	(0.061)	(0.060)	(0.120)	(0.059)	(0.053)	(0.091)

Entries report medians; standard deviations across the same 50 splits are shown in parentheses below.

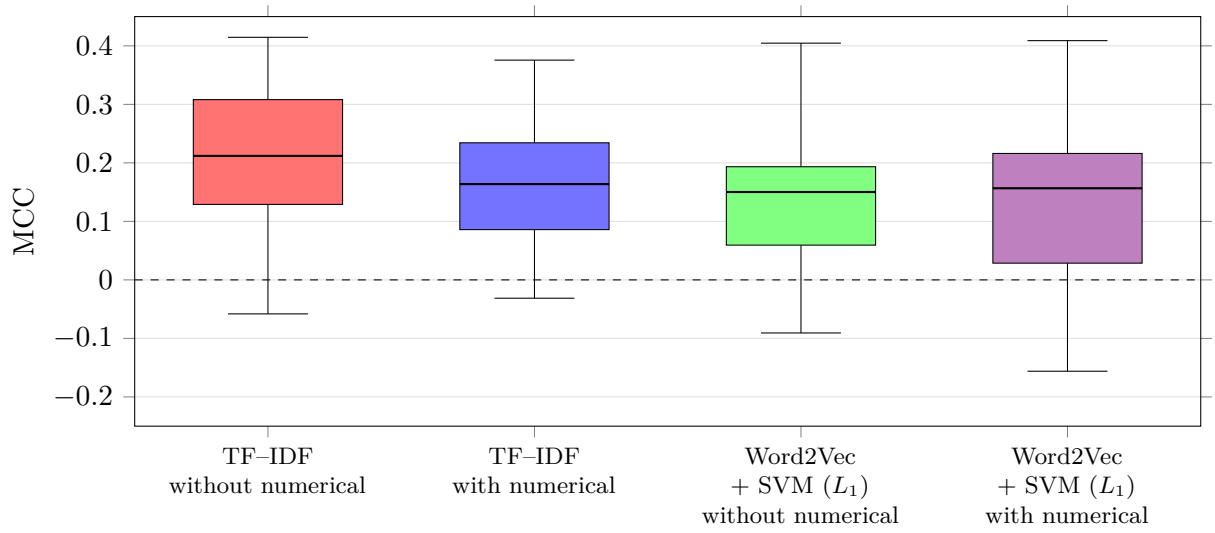


Figure 1: Distribution of MCC across 50 case-grouped validation splits. Boxes show the interquartile range, horizontal lines show medians, and whiskers show the observed minimum and maximum.

External test performance. Table 5 reports the court results; Tables C.7 and C.8 in Appendix C report the UKIPO and combined results. I also evaluate GPT-5.6 Sol as a zero-shot benchmark using the prompt in Section E.1.

Table 5: External test performance: Patents Court and IPEC

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Precision (owner loss)	Recall (owner loss)
TF–IDF, without numerical features	0.584	0.627	0.624	0.250	0.624	0.636	0.677
TF–IDF, with numerical features	0.540	0.610	0.608	0.217	0.608	0.625	0.645
Word2Vec + linear SVM (L_1), without numerical features	0.616	0.610	0.608	0.217	0.608	0.625	0.645
Word2Vec + linear SVM (L_1), with numerical features	**0.705**	**0.678**	**0.678**	**0.356**	**0.678**	**0.700**	0.677
ChatGPT (GPT-5.6 Sol), zero-shot	0.684	0.542	0.523	0.071	0.452	0.538	**0.903**

Depending on the evaluation (MCCV versus the test set), numerical features might improve or worsen the models' performance. Nevertheless, all supervised specifications outperform zero-shot ChatGPT in MCC, macro-F_1, and accuracy. Generalization to UKIPO is weaker.

5.2 Ex-post level

5.2.1 Insufficiency attack

I predict insufficiency outcomes using claims from 133 cases at depth ≤ 2 and 136 cases at full depth (Table 3).

I consider four textual inputs: the literal claim, the expanded claim, the complete claim set, and the patent description. Every specification includes either the literal or expanded version of the challenged claim, while the complete claim set and description are included or excluded depending on the architecture.

The legal structure of insufficiency suggests that the description should be particularly important.

The relevant question is not whether the claim appears strong in isolation, but whether the description provides sufficient information to perform the invention defined by that claim. I therefore expect the strongest architectures to compare the description with the challenged claim.

For an independent claim, the literal text may be enough on its own. For a dependent claim, however, the literal text omits the limitations inherited from its parent claims. Those limitations can be supplied either directly through the expanded claim (Subsection 3.3) or indirectly by providing both the literal claim and the complete claim set. This reasoning motivates three principal input combinations: the expanded claim and description; the literal claim, complete claim set, and description; and the expanded claim, complete claim set, and description.

The initial architecture search identified two leading input combinations: the expanded claim and description, and the expanded claim, complete claim set, and description. I retain these shortlisted architectures and re-estimate them on the deduplicated insufficiency sample, using dropout 0.1 for both models throughout original validation, repeated validation, and external-test refitting. All supervised original-validation, repeated-validation, and external-test results reported below use these corrected fits. The majority-class accuracy benchmark in the corrected development sample is 61.5% at depth ≤ 2 and 61.1% at full depth.

Selected architectures. I retain two frozen-embedding architectures. INS-A1 uses the expanded claim and patent description, projecting each embedding into 32 dimensions and combining them with pairwise interaction features and numerical inputs before an MLP classifier, following the projected-fusion specification in Subsection 4.3. INS-A2 directly concatenates the expanded-claim, complete-claim-set, and description embeddings with numerical inputs, then uses an MLP with a 128-dimensional hidden layer.

Both models use equal patent weights ($\gamma = 1$), selected on the original validation split. For original and repeated validation, I train for 20 epochs and select the checkpoint with the highest validation MCC, using the earliest epoch in a tie. Initial learning rates are 0.0003 for INS-A1 and 0.001 for INS-A2. Both models use dropout 0.1 and the first 16,384 description tokens. INS-A1 uses up to 1,024 expanded-claim tokens; INS-A2 uses up to 2,048 expanded-claim tokens and 4,096 complete-claim-set tokens. I also evaluate versions without numerical features.

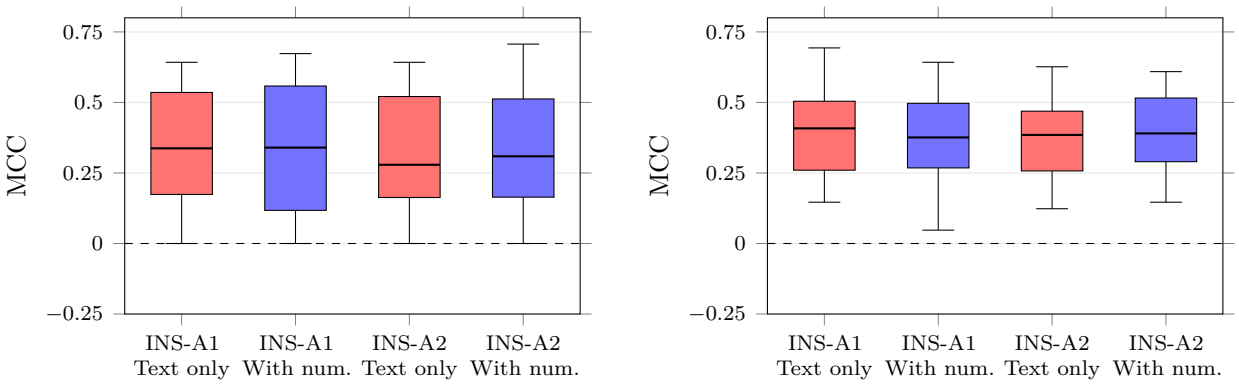
Figure C.1 and Table C.2 in Appendix C present the architectures and original validation results, respectively.

Monte Carlo cross-validation. Table 6 summarizes 25 case-grouped validation splits. Overall, incorporating numerical features does not consistently improve predictive performance. Increasing dependency depth does not appear to reduce performance.

Table 6: Insufficiency: median performance across validation splits

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 (25 splits)								
INS-A1, text only	**0.715** (0.105)	**0.704** (0.077)	**0.648** (0.100)	0.337 (0.193)	**0.653** (0.124)	0.734 (0.063)	0.679 (0.208)	**0.424** (0.235)
INS-A1, text + numerical	0.705 (0.089)	0.695 (0.090)	0.606 (0.111)	**0.340** (0.238)	0.590 (0.141)	0.729 (0.067)	**0.714** (0.266)	0.320 (0.237)
INS-A2, text only	0.703 (0.120)	0.663 (0.087)	0.606 (0.094)	0.279 (0.207)	0.594 (0.114)	**0.747** (0.069)	0.659 (0.229)	0.400 (0.183)
INS-A2, text + numerical	0.710 (0.115)	0.675 (0.084)	0.621 (0.102)	0.309 (0.214)	0.623 (0.132)	0.727 (0.072)	**0.714** (0.267)	0.324 (0.224)
Panel B: Full depth (25 splits)								
INS-A1, text only	0.692 (0.074)	0.711 (0.067)	0.679 (0.073)	**0.408** (0.160)	0.681 (0.086)	**0.743** (0.065)	0.667 (0.158)	0.462 (0.157)
INS-A1, text + numerical	**0.717** (0.074)	0.711 (0.070)	0.644 (0.083)	0.376 (0.173)	0.645 (0.104)	0.732 (0.051)	**0.719** (0.187)	0.455 (0.187)
INS-A2, text only	0.701 (0.076)	**0.719** (0.062)	0.684 (0.070)	0.385 (0.145)	0.684 (0.084)	0.718 (0.051)	0.682 (0.133)	**0.500** (0.160)
INS-A2, text + numerical	0.702 (0.072)	**0.719** (0.060)	**0.689** (0.067)	0.390 (0.136)	**0.685** (0.079)	0.714 (0.050)	0.684 (0.146)	0.490 (0.163)

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Entries report medians; sample standard deviations across the 25 splits are shown in parentheses. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100). Bold indicates column maxima within each panel at the reported precision.



(a) Depth ≤ 2 (b) Full depth

Figure 2: Insufficiency: distribution of MCC across 25 case-grouped validation splits for each depth regime. Boxes show the interquartile range, horizontal lines show medians, and whiskers show the observed minimum and maximum.

External test performance. Table 7 reports the court results; Tables C.9 and C.10 in Appendix C report the UKIPO and combined results. The zero-shot ChatGPT benchmark uses the prompt in Section E.2.

Table 7: Insufficiency: external-test performance on court observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 46$)								
INS-A1, text only	0.680	0.543	0.538	0.077	0.538	0.625	0.500	0.476
INS-A1, text + numerical	0.707	0.609	0.606	0.211	0.606	0.653	0.571	0.571
INS-A2, text only	0.665	0.565	0.562	0.124	0.562	0.639	0.524	0.524
INS-A2, text + numerical	**0.709**	**0.630**	**0.630**	**0.258**	**0.629**	**0.688**	0.591	**0.619**
ChatGPT (GPT-5.6 Sol), zero-shot	0.704	0.587	0.559	0.155	0.522	0.642	**0.625**	0.238
Panel B: Full depth ($n = 64$)								
INS-A1, text only	0.767	0.656	0.635	0.305	0.627	0.700	0.706	0.414
INS-A1, text + numerical	**0.793**	**0.750**	**0.730**	**0.519**	**0.729**	**0.780**	**0.882**	0.517
INS-A2, text only	0.745	0.672	0.691	0.411	0.665	0.640	0.591	**0.897**
INS-A2, text + numerical	0.765	0.656	0.662	0.324	0.656	0.640	0.600	0.724
ChatGPT (GPT-5.6 Sol), zero-shot	0.611	0.516	0.489	-0.026	0.457	0.631	0.429	0.207

All supervised models in Panel A are trained and evaluated at depth ≤ 2; all supervised models in Panel B are trained and evaluated at full depth (≤ 100). ChatGPT is evaluated zero-shot on the same observations. Bold indicates column maxima within each panel at the reported precision.

On the court holdout, the benefit of numerical features varies across models and depth regimes, while increasing dependency depth does not appear to reduce performance. All models with numerical features outperform ChatGPT in MCC, accuracy, and macro-F_1.

Generalization to UKIPO is weaker. At both depths, all four supervised specifications predict survival for every claim, yielding an MCC of zero and zero recall for owner losses. ChatGPT also fails to identify any owner losses.

Separate versus combined encoding. I next examine whether the texts should be encoded separately or concatenated before encoding. I consider three configurations: the literal claim, complete claim set, and description (INS-X1); the expanded claim, complete claim set, and description (INS-X2); and the expanded claim and description (INS-X3). For each configuration, I concatenate the selected texts and encode them jointly using a frozen Qwen-8B encoder. The resulting single embedding is combined with numerical features and passed to an MLP classification head.

Table 8 compares all three combined-encoding configurations with both separate-encoding baselines, INS-A1 and INS-A2, all with numerical features. INS-A1 outperforms all combined-encoding models in MCC, macro-F_1, and accuracy at both depths. INS-A2 also outperforms these models on all three metrics at depth ≤ 2, while its full-depth results are mixed.

Table 8: Insufficiency: separate versus combined encoding on court observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 46$)								
INS-A1, separate	0.707	0.609	0.606	0.211	0.606	0.653	0.571	0.571
INS-A2, separate	**0.709**	**0.630**	**0.630**	**0.258**	**0.629**	**0.688**	**0.591**	**0.619**
INS-X1, combined	0.638	0.565	0.562	0.124	0.562	0.632	0.524	0.524
INS-X2, combined	0.621	0.543	0.542	0.084	0.542	0.611	0.500	0.524
INS-X3, combined	0.632	0.522	0.514	0.029	0.513	0.549	0.474	0.429
Panel B: Full depth ($n = 64$)								
INS-A1, separate	**0.793**	**0.750**	**0.730**	**0.519**	**0.729**	**0.780**	**0.882**	0.517
INS-A2, separate	0.765	0.656	0.662	0.324	0.656	0.640	0.600	**0.724**
INS-X1, combined	0.694	0.672	0.656	0.334	0.653	0.653	0.700	0.483
INS-X2, combined	0.676	0.672	0.659	0.332	0.658	0.660	0.682	0.517
INS-X3, combined	0.699	0.656	0.638	0.302	0.633	0.647	0.684	0.448

INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100).

Tables C.15 and C.16 in Appendix C report the UKIPO and pooled-sample results. On UKIPO, performance is weaker: MCC is zero for the separate-encoding models and negative for the combined-encoding models at both depths.

Overall, the separate-encoding models outperform combined-encoding models. These findings support retaining distinct representations of the claim and description and combining them through a learned head. Separate representations may help preserve claim-specific information when the description is much longer than the claim.

5.2.2 Obviousness

Observations are claim–prior-art attacks, as defined in Subsection 3.2. I exclude attacks relying on more than three prior-art documents to stay within the Qwen encoder's approximately 32k-token budget, removing two observations from the training–validation sample. The retained sample covers 853 distinct claims from 259 cases at depth ≤ 2 and 1,157 claims from 262 cases at full depth (Table 3). Some cases in Table 2 are excluded because the full text of their prior-art references could not be located.

For the obviousness analysis, I consider five potential textual inputs: the literal claim, the expanded claim, the complete claim set, the patent description, and the prior art. Every specification includes either the literal or expanded version of the challenged claim, while the inclusion of the other textual inputs varies across architectures.

The legal structure of obviousness suggests that the prior art should be present in any well-specified architecture. A claim is obvious if the claimed invention would have been obvious to a skilled person in light of the prior art. I therefore expect the strongest architectures to compare the challenged claim with the relevant prior art.

The patent description may also be informative because some claim terms can be understood

only by reference to the description. Its inclusion creates a trade-off, however. For claims that can be understood without such reference, the lengthy description may add noise and increase the risk of overfitting.

Finally, as in the previous section, I expect the optimal architecture to include one of the three representations of the challenged claim: the expanded claim; the literal claim together with the complete claim set; and the expanded claim together with the complete claim set.

The results are consistent with this intuition. The two best-performing architectures use, respectively, the literal claim and the complete claim set together with the prior art, and the expanded claim and the complete claim set together with the prior art. Models that use the literal claim and prior art without the complete claim set, as well as architectures that exclude the prior art, perform only slightly better than the majority-class baseline.

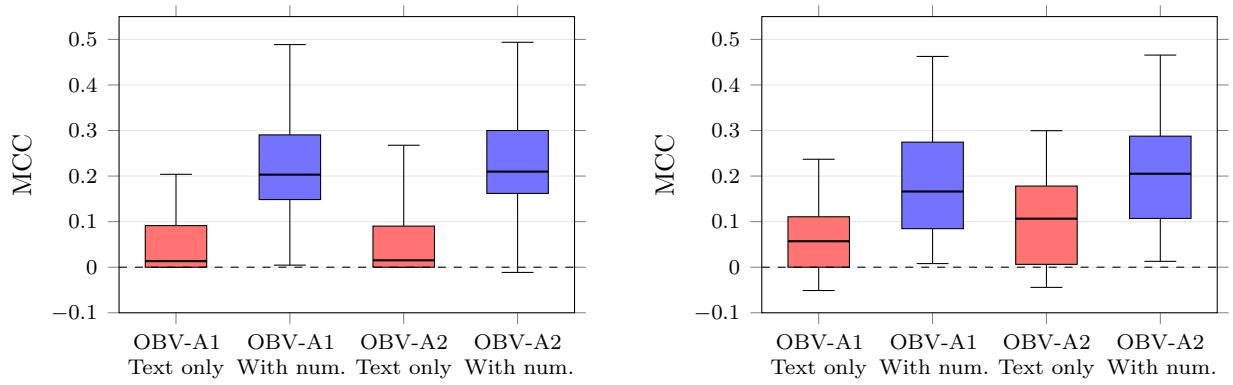
Selected architectures. OBV-A1 uses the literal claim, while OBV-A2 uses the expanded claim. Both LoRA-tune the encoders for the challenged claim and complete claim set ($r = 2$, $\alpha = 4$), keeping the prior-art encoder frozen. The embeddings are concatenated with numerical features and passed directly to a linear classification head. Both models use sample weighting with $\gamma = 1/16$, which maximizes MCC on the original validation split. Table C.3 in Appendix C reports performance on that split.

Monte Carlo cross-validation. Table 9 summarizes 50 case-grouped validation splits for each depth regime. Figure 3 shows the corresponding MCC distributions for both depth regimes. Numerical features are important for performance.

Table 9: Obviousness: median performance across validation splits

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 (50 splits)									
OBV-A1, text only	0.544 (0.088)	0.556 (0.030)	0.502 (0.012)	0.013 (0.060)	0.370 (0.038)	0.504 (0.032)	0.558 (0.034)	0.500 (0.399)	0.017 (0.038)
OBV-A1, text + numerical	0.591 (0.071)	0.601 (0.057)	0.585 (0.059)	0.203 (0.109)	0.579 (0.079)	0.588 (0.059)	0.588 (0.049)	**0.586** (0.077)	0.574 (0.244)
OBV-A2, text only	0.564 (0.089)	0.554 (0.034)	0.503 (0.016)	0.015 (0.073)	0.370 (0.045)	0.511 (0.036)	0.557 (0.035)	0.477 (0.414)	0.009 (0.044)
OBV-A2, text + numerical	**0.599** (0.072)	**0.603** (0.057)	**0.594** (0.061)	**0.210** (0.117)	**0.581** (0.075)	**0.594** (0.060)	**0.591** (0.050)	0.569 (0.075)	**0.617** (0.234)
Panel B: Full depth (50 splits)									
OBV-A1, text only	0.533 (0.090)	0.546 (0.034)	0.506 (0.023)	0.057 (0.072)	0.367 (0.066)	0.489 (0.043)	0.555 (0.030)	0.555 (0.346)	0.025 (0.085)
OBV-A1, text + numerical	0.588 (0.078)	0.580 (0.058)	0.579 (0.062)	0.166 (0.122)	0.563 (0.075)	0.593 (0.059)	0.564 (0.050)	0.549 (0.044)	0.670 (0.212)
OBV-A2, text only	0.542 (0.093)	0.553 (0.039)	0.520 (0.028)	0.107 (0.092)	0.410 (0.069)	0.506 (0.046)	0.552 (0.029)	**0.667** (0.332)	0.068 (0.082)
OBV-A2, text + numerical	**0.595** (0.081)	**0.596** (0.060)	**0.589** (0.061)	**0.205** (0.120)	**0.580** (0.076)	**0.602** (0.054)	**0.568** (0.054)	0.557 (0.055)	**0.700** (0.194)

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100). Entries report medians across 50 case-grouped validation splits; sample standard deviations are shown in parentheses. Bold indicates column maxima within each panel at the reported precision.



(a) Depth ≤ 2 (b) Full depth

Figure 3: Obviousness: distribution of MCC across 50 case-grouped validation splits for each depth regime. Boxes show the interquartile range, horizontal lines show medians, and whiskers show the observed minimum and maximum.

External test performance. Table 10 reports the court results; Tables C.11 and C.12 in Appendix C report the UKIPO and combined results. The three-document limit excludes 24 UKIPO attacks at full depth, 23 of which also belong to the depth-≤ 2 sample. For holdout prediction, training stops at epoch 2 of the six-epoch learning-rate schedule. The zero-shot ChatGPT benchmark uses the prompt in Section E.3.

Table 10: Obviousness: external-test performance on court observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 228$)									
OBV-A1, text only	0.415	**0.675**	0.500	0.000	0.403	**0.677**	**0.711**	0.000	0.000
OBV-A1, text + numerical	0.647	0.614	0.535	0.075	0.535	0.610	0.698	0.383	0.311
OBV-A2, text only	0.417	**0.675**	0.500	0.000	0.403	**0.677**	**0.711**	0.000	0.000
OBV-A2, text + numerical	0.654	0.636	0.559	0.125	**0.559**	0.642	0.709	**0.424**	0.338
ChatGPT (GPT-5.6 Sol), zero-shot	**0.676**	0.500	**0.598**	**0.211**	0.498	0.499	0.448	0.382	**0.878**
Panel B: Full depth ($n = 306$)									
OBV-A1, text only	0.442	0.582	0.494	-0.068	0.368	0.577	0.668	0.000	0.000
OBV-A1, text + numerical	**0.738**	**0.618**	**0.654**	**0.323**	**0.615**	**0.611**	0.615	**0.522**	0.857
OBV-A2, text only	0.452	0.585	0.497	-0.048	0.369	0.580	**0.678**	0.000	0.000
OBV-A2, text + numerical	**0.738**	0.588	0.624	0.261	0.585	0.570	0.586	0.500	0.825
ChatGPT (GPT-5.6 Sol), zero-shot	0.690	0.578	0.627	0.291	0.566	0.580	0.471	0.494	**0.905**

All supervised models in Panel A are trained and evaluated at depth ≤ 2; all supervised models in Panel B are trained and evaluated at full depth (≤ 100). ChatGPT is evaluated zero-shot on the same observations. Bold indicates column maxima within each panel at the reported precision.

Overall, without numerical features, the models are not better than predicting the majority class. Unlike in the previous sections, ChatGPT's performance is largely in line with that of my models.

Generalization to UKIPO is weaker: every supervised specification has negative MCC at both depths, whereas ChatGPT has positive MCC.

Separate versus combined encoding. I also examine combined encoding for obviousness. OBV-X1 combines the literal claim, complete claim set, and prior-art material, while OBV-X2 replaces the literal claim with the expanded claim. Both specifications include numerical features.

Table 11 compares these specifications with OBV-A1 and OBV-A2 on the court holdout. Tables C.17 and C.18 in Appendix C report the UKIPO and combined-sample results.

Table 11: Obviousness: separate versus combined encoding on court observations

Model	Training depth	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 228$)										
OBV-A1, separate	2	0.647	0.614	0.535	0.075	0.535	0.610	0.698	0.383	0.311
OBV-A2, separate	2	**0.654**	**0.636**	**0.559**	**0.125**	**0.559**	**0.642**	**0.709**	**0.424**	**0.338**
OBV-X1, combined	2	0.229	0.412	0.347	-0.296	0.351	0.397	0.444	0.143	0.162
OBV-X2, combined	2	0.226	0.412	0.347	-0.296	0.351	0.397	0.444	0.143	0.162
Panel B: Full depth ($n = 306$)										
OBV-A1, separate	100	**0.738**	**0.618**	**0.654**	**0.323**	**0.615**	**0.611**	**0.615**	**0.522**	**0.857**
OBV-A2, separate	100	**0.738**	0.588	0.624	0.261	0.585	0.570	0.586	0.500	0.825
OBV-X1, combined	100	0.198	0.343	0.315	-0.375	0.312	0.303	0.369	0.174	0.159
OBV-X2, combined	100	0.197	0.340	0.313	-0.380	0.310	0.300	0.366	0.172	0.159

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100). Bold indicates column maxima within each panel at the reported precision.

The court results favour the separate-encoding specifications. The intuition is the same as in the insufficiency section: when the texts are encoded together, the much longer prior-art material may dominate the resulting representation, making it harder for the model to retain information specific to the claim.

5.2.3 Novelty

I predict novelty outcomes at the claim–prior-art attack level. Table 3 reports the samples; attacks whose prior-art text could not be obtained are excluded. I compare combinations of the literal or expanded claim, complete claim set, patent description, and prior art. The three shortlisted models use frozen embeddings and all include prior art; their specifications and detailed results appear in Appendix C.3.

Promising performance on the initial validation split does not translate into strong discrimination across 50 case-grouped splits. The highest median MCC is 0.148 at depth ≤ 2 and 0.123 at full depth (Table C.6). Given this weak repeated-validation performance, I do not evaluate the novelty models on the holdouts.

5.3 Claim-level validity across multiple attacks

I next predict whether a claim is invalid when all retained insufficiency and obviousness attacks against it are considered together. A claim is coded as invalid if at least one of these attacks succeeds. The sample contains claims with at least one eligible attack; obviousness attacks citing more than three prior-art documents are excluded before claims and their labels are assembled. Table 3 reports the samples.

Table 12 reports performance on the court holdout. Definitions of the model labels are in 4.5.

Table 12: Claim-level validity: external-test performance on court observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 133$ claims)								
IND	0.542	0.571	0.559	0.124	0.554	0.613	0.543	0.410
ILSE	0.750	0.639	0.653	0.320	0.634	**0.644**	0.575	0.820
ILSE+T	**0.794**	0.624	0.642	0.308	0.614	0.600	0.559	0.852
CUS	0.737	0.692	0.690	0.380	0.690	0.582	0.661	0.672
CUS+T	0.770	0.699	0.696	0.393	0.696	0.567	**0.678**	0.656
CUS+SAB	0.768	0.549	0.521	0.057	0.471	0.559	0.524	0.180
CUS+T+SAB	0.719	**0.722**	**0.728**	**0.457**	**0.722**	0.576	0.662	0.803
ChatGPT (GPT-5.6 Sol), zero-shot	0.687	0.534	0.559	0.149	0.499	0.533	0.495	**0.869**
Panel B: Full depth (≤ 100; $n = 183$ claims)								
IND (mean)	0.624	0.579	0.577	0.154	0.577	0.599	0.619	0.600
ILSE	0.672	0.607	0.583	0.194	0.564	0.559	0.600	0.840
ILSE+T	0.771	0.650	0.622	0.315	0.596	0.601	0.620	0.930
CUS	0.687	0.585	0.548	0.158	0.484	0.513	0.573	**0.940**
CUS+T	0.653	0.590	0.554	0.173	0.493	0.531	0.577	**0.940**
CUS+SAB	0.730	0.617	0.584	0.245	0.540	0.567	0.595	**0.940**
CUS+T+SAB	**0.826**	**0.661**	**0.635**	**0.334**	**0.616**	**0.624**	**0.630**	0.920
ChatGPT (GPT-5.6 Sol), zero-shot	0.699	0.607	0.582	0.195	0.561	0.552	0.599	0.850

All supervised models include numerical features. IND (mean) averages the probabilities from INS-A1 and OBV-A2 over all retained attacks on a claim. ILSE and ILSE+T use attention-based MIL; the four custom specifications use claim-query cross-attention. SAB denotes self-attention over the two claim representations before query fusion. Each supervised specification uses an independent refit on all development observations at the matching depth. Bold indicates column maxima within each panel at the reported precision.

Overall, we see that our custom models outperform both the ILSE models from the literature and ChatGPT. Furthermore, adding type encoding always improves performance. The effect of self-attention is somewhat less straightforward, but the best-performing models always include it.

6 Conclusion

In this paper, I investigate whether patent texts and numerical features can predict outcomes in UK patent litigation. I construct a dataset linking judicial decisions to patents, individual claims, grounds of challenge, and prior-art references, and examine prediction under different information sets. The analysis addresses four questions: which texts are informative for each prediction task, how dependent claims should be represented, whether different texts should be encoded separately or combined before encoding, and how evidence from multiple attacks should be aggregated into a claim-level validity prediction.

The results show that the models with best predictive performance depends on both the legal ground and the representation of the available information. Classical text models remain competitive in the ex-ante patent-level analysis, while the insufficiency and obviousness analyses provide evidence of predictive information in representations tailored to the relevant ground. Numerical features generally improve performance on the test set. The encoding comparisons favour preserving separate representations and combining them through task-specific heads. My efforts to improve the representation of dependent claims yield mixed but promising results.

My main contribution is a novel architecture for aggregating information across attacks to predict claim-level validity. The architecture follows an intuitive approach, using cross-attention

between the target claim and the texts associated with the attacks against it: prior art or the patent description. I show that this architecture outperforms both the baseline architecture from the literature and ChatGPT.

These findings also identify limits to the approach. Prediction remains difficult in the novelty analyses, and performance on the court holdout does not consistently carry over to the UKIPO sample. The latter result limits the extent to which models estimated on court disputes can be treated as general measures of patent strength across institutional settings.

The first extension to this paper is to test whether pre-judgment predictions contain trading signals not already reflected in stock prices. The present U.K. sample contains too few relevant litigation events to reliably assess whether a strategy based on these predictions would be profitable after transaction costs.

The second extension is to compare predicted vulnerability to insufficiency between challenged and otherwise similar unchallenged claims, examining whether challengers target claims that the model also identifies as weaker. Interpreting these comparisons requires accounting for selection into challenges, since outcomes for unchallenged claims are unobserved.

A Definition of Numerical Features

Table A.1: Definition and use of numerical features

Feature	Definition	Used at
`claimant_entity`	Legal status of the claimant. It equals 1 if the claimant is an individual, a private company, or a non-profit organization such as a university; 2 if the claimant is a publicly listed company; and 3 if the claimant is a public authority, specifically the Comptroller.	Patent and claim
`defendant_entity`	Legal status of the defendant. It equals 1 if the defendant is an individual, a private company, or a non-profit organization such as a university; 2 if the defendant is a publicly listed company; and 3 if the defendant is a public authority, specifically the Comptroller.	Patent and claim
`claimant_headquarter`	Location of the claimant's headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.	Patent and claim
`defendant_headquarter`	Location of the defendant's headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.	Patent and claim
`patent_age`	Age of the patent, measured in days from the patent's priority date to the date of the first judgment in the case. Formally, it is calculated as the first judgment date minus the priority date.	Patent and claim
`forward_citation`	Number of times the patent had been cited by other patents as of the date of the first judgment. Citations occurring after that date are not counted.	Patent and claim
`backward_citation`	Number of earlier patents cited by the patent.	Patent and claim
`claimant_is_owner`	Indicator for the relationship between the claimant and the patent owner. It equals 0 if the claimant is the patent owner and 1 otherwise.	Patent and claim
`amended`	Indicator for whether the claim is an amended version. It equals 0 if the claim was not amended and 1 if it was amended.	Claim only
`dependency_depth`	Maximum number of dependency links in any path connecting the claim to an independent claim. An independent claim has depth zero. The depth of a dependent claim is one plus the greatest depth among its parent claims, as defined in Definition 3.1.	Claim only

B Derivation of models in Section 4

B.1 Regularization and loss functions for classification heads

Regularization For a coefficient vector $w = (w_1, \ldots, w_d)^\top$, the L_1 (lasso) regularization function takes the form

$$R_{L_1}(w) = \|w\|_1 = \sum_{j=1}^{d} |w_j|.$$

The L_2 (ridge) regularization function takes the form

$$R_{L_2}(w) = \|w\|_2^2 = \sum_{j=1}^{d} w_j^2.$$

Elastic-net regularization is a convex combination of the L_1 and L_2 penalties:

$$R_{\mathrm{EN}}(w) = \rho\|w\|_1 + (1-\rho)\|w\|_2^2, \qquad \rho \in [0,1].$$

Here, $\rho = 1$ corresponds to lasso regularization, while $\rho = 0$ corresponds to ridge regularization.

Loss function for logistic regression. Logistic regression models the probability that observation i belongs to class 1 as

$$p_i = P(y_i = 1 \mid x_i) = \sigma(w^\top x_i + b) = \frac{1}{1 + \exp[-(w^\top x_i + b)]}.$$

The probability assigned to the observed outcome $y_i \in \{0, 1\}$ is

$$P(y_i \mid x_i) = p_i^{y_i}(1-p_i)^{1-y_i}.$$

Taking the negative logarithm gives the loss for a single observation i:

$$\ell_i(w, b) = -\log P(y_i \mid x_i) = -\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right].$$

The total weighted and regularized loss is

$$\mathcal{L}_{\mathrm{logit}}(w, b) = -\sum_{i=1}^{n} a_{y_i} s_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] + \lambda R(w).$$

The first term is the weighted binary cross-entropy loss. Minimizing this term is equivalent to maximizing the weighted conditional log-likelihood of the observed outcomes: the model is rewarded for assigning high probability to the observed class and penalized for confident but incorrect predictions. The second term regularizes the coefficient vector to reduce overfitting. I consider L_2, L_1, and elastic-net regularization.

At prediction time, an observation is assigned to class 1 if $p_i > 0.5$ and to class 0 otherwise.

Loss function for support vector machines. For a linear SVM, let

$$f_i = w^\top x_i + b$$

denote the decision score for observation i. I transform the binary outcome $y_i \in \{0, 1\}$ into

$$\widetilde{y}_i = 2y_i - 1 \in \{-1, +1\}.$$

The product $\widetilde{y}_i f_i$ is the functional margin: it is positive when the observation is correctly classified and negative when it is misclassified.

The SVM seeks to classify each observation with a functional margin of at least one:

$$\widetilde{y}_i f_i \geq 1.$$

To allow violations of this requirement, introduce a nonnegative slack variable ξ_i such that

$$\widetilde{y}_i f_i \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

For given w and b, the smallest feasible slack is $\max\{0, 1 - \widetilde{y}_i f_i\}$. This gives the hinge loss for a single observation:

$$\ell_i(w, b) = \max\{0, 1 - \widetilde{y}_i f_i\} = \max\left\{0,\ 1 - (2y_i - 1)(w^\top x_i + b)\right\}.$$

The loss is zero when the observation is correctly classified with a functional margin of at least one. It is positive when the observation lies inside the margin or is misclassified, with larger violations receiving larger penalties.

The total weighted and regularized loss is

$$\mathcal{L}_{\text{SVM}}(w, b) = \sum_{i=1}^{n} a_{y_i} s_i \max\left\{0,\ 1 - (2y_i - 1)(w^\top x_i + b)\right\} + \lambda R(w).$$

The first term is the weighted hinge loss. The second term regularizes the coefficient vector and favors a simpler decision boundary. I consider both L_2 and L_1 regularization for the linear SVM.

At prediction time, an observation is assigned to class 1 if $w^\top x_i + b > 0$ and to class 0 otherwise.

I also consider an SVM with a radial basis function kernel, referred to as the RBF-SVC. It applies the same hinge-loss principle in an implicitly transformed representation space using the kernel

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|_2^2\right).$$

The kernel measures the similarity between observations according to the distance between their representations and allows the SVM to construct nonlinear decision boundaries. A smaller γ makes each observation influence a larger region of the representation space and generally produces a smoother, more global decision boundary. A larger γ makes each observation's influence more local

and permits a more complex decision boundary.

Loss function for the ridge-regularized least-squares classifier. Using the transformed labels $\widetilde{y}_i = 2y_i - 1 \in \{-1, +1\}$, the loss for a single observation i is the squared difference between its transformed label and the linear decision score:

$$\ell_i(w, b) = \left(\widetilde{y}_i - w^\top x_i - b\right)^2.$$

The total weighted and regularized objective is

$$\min_{w,b} \left\{ \sum_{i=1}^{n} a_{y_i} s_i \left(\widetilde{y}_i - w^\top x_i - b\right)^2 + \lambda R(w) \right\}.$$

Thus, unlike logistic regression and SVMs, the L_2-regularized least-squares classifier uses squared-error loss. I use L_2 regularization only, with $R(w) = \|w\|_2^2$ and λ controlling its strength.

The fitted classifier produces the linear decision score

$$g(x) = w^\top x + b.$$

An observation is assigned to class 1 if $g(x) > 0$ and to class 0 otherwise.

B.2 Classical models

Extra Trees. Extra Trees is an ensemble of randomized decision trees. For each tree, I first specify a maximum depth. A greater depth allows the tree to represent more complex relationships and generally fit the training data more closely, but it can also increase the risk of overfitting.

Consider a node containing a set of training observations S, and let d be the number of numerical features. The algorithm randomly selects a subset of the d features. For each selected feature j, it then randomly generates a candidate threshold τ. Each candidate split (j, τ) divides the observations into

$$S_L(j, \tau) = \{i \in S : x_{ij} \leq \tau\}$$

and

$$S_R(j, \tau) = \{i \in S : x_{ij} > \tau\}.$$

Thus, observations with values at or below the threshold move to the left child, while those with values above the threshold move to the right child.

For a node A, its Gini impurity is

$$G(A) = 1 - \sum_{c \in \{0,1\}} p_c(A)^2,$$

where $p_c(A)$ is the share of observations in A belonging to class c. A node has zero impurity when all its observations belong to the same class. Among the randomly generated candidate splits, the

algorithm chooses the pair (j, τ) that maximizes the reduction in impurity,

$$\Delta G(j, \tau) = G(S) - \frac{|S_L|}{|S|} G(S_L) - \frac{|S_R|}{|S|} G(S_R).$$

This procedure is repeated recursively until a stopping condition is reached, such as the maximum tree depth, a minimum number of observations in a node, or a node containing observations from only one class. A terminal node, or leaf, predicts according to the class distribution of the training observations it contains. Extra Trees constructs many such randomized trees and obtains its final prediction by aggregating their predictions. Randomizing both the candidate features and their thresholds reduces the dependence of the final prediction on any single fitted tree.

Derivation of TF–IDF. Let the corpus contain n texts and let

$$\mathcal{V} = \{v_1, \ldots, v_d\}$$

denote the selected vocabulary of d terms. TF–IDF maps text i into a d-dimensional vector

$$x_i^{\text{TFIDF}} = (\text{tfidf}_{i1}, \ldots, \text{tfidf}_{id})^\top,$$

where each dimension corresponds to one term in the vocabulary. For term v_j, let

$$\text{tf}_{ij}$$

denote the number of times that the term appears in text i, and let

$$\text{df}_j = \sum_{i=1}^{n} \mathbf{1}\{\text{tf}_{ij} > 0\}$$

denote the number of texts containing that term. Its TF–IDF weight is

$$\text{tfidf}_{ij} = \text{tf}_{ij} \log\left(\frac{n}{\text{df}_j}\right).$$

The term-frequency component assigns greater weight to terms that occur frequently within a text, while the inverse-document-frequency component discounts terms that appear in many texts and therefore provide relatively little information for distinguishing among them.

After vectorization, the TF–IDF representation is concatenated with other features z_i,

$$\widetilde{x}_i = \begin{bmatrix} x_i^{\text{TFIDF}} \\ z_i \end{bmatrix},$$

and the resulting vector is passed to one of the classification heads described in the previous subsection for training and prediction.

Complement Naive Bayes. Each text i, regardless of its outcome class, is represented using the common vector

$$x_i = (x_{i1}, \ldots, x_{id})^\top,$$

where x_{ij} is the TF–IDF weight of term v_j in text i.

For each class c, the classifier aggregates term weights over texts that do *not* belong to that class:

$$N_{cj}^{\text{comp}} = \sum_{i : y_i \neq c} x_{ij}.$$

It then estimates the relative frequency of term v_j in the complement of class c as

$$q_{cj} = \frac{\alpha + N_{cj}^{\text{comp}}}{\alpha d + \sum_{k=1}^{d} N_{ck}^{\text{comp}}},$$

where $\alpha > 0$ is a smoothing parameter. In the binary setting, q_{1j} is therefore estimated using texts from class 0, while q_{0j} is estimated using texts from class 1.

A new text is transformed into the same TF–IDF vector

$$x = (x_1, \ldots, x_d)^\top$$

and assigned to the class

$$\widehat{y} = \arg\min_{c} \sum_{j=1}^{d} x_j \log q_{cj}.$$

Derivation of Word2Vec. Let $\mathcal{V}$ denote the learned vocabulary and let r denote the embedding dimension. Each word $w \in \mathcal{V}$ is associated with two vectors,

$$v_w \in \mathbb{R}^r \qquad \text{and} \qquad u_w \in \mathbb{R}^r,$$

corresponding to its input and output representations, respectively. The distinction arises because a word can be used either to predict another word or as the word being predicted.

After tokenization and removal of words not retained in $\mathcal{V}$, text i is represented by the sequence

$$(w_{i1}, \ldots, w_{iT_i}),$$

where T_i is the number of retained token occurrences. For the word in position t, let

$$C_{it} = \{s : 1 \leq s \leq T_i,\ 0 < |s - t| \leq m\}$$

denote the positions within a context window of at most m words on either side.

I consider both the skip-gram and continuous bag-of-words (CBOW) variants of Word2Vec. Under skip-gram, the central word w_{it} is used to predict each surrounding word w_{is}, $s \in C_{it}$. The

probability of observing context word o given central word c is

$$P(o \mid c) = \frac{\exp\left(u_o^\top v_c\right)}{\sum_{w \in \mathcal{V}} \exp\left(u_w^\top v_c\right)}.$$

Thus, a context word receives a higher probability when its output representation has a larger dot product with the input representation of the central word. The skip-gram loss is

$$\mathcal{L}^{\mathrm{SG}} = -\sum_{i=1}^{n} \sum_{t=1}^{T_i} \sum_{s \in C_{it}} \log P(w_{is} \mid w_{it}).$$

CBOW reverses the prediction task. It first averages the input representations of the surrounding words,

$$h_{it} = \frac{1}{|C_{it}|} \sum_{s \in C_{it}} v_{w_{is}},$$

and uses the resulting context representation to predict the central word. The corresponding probability is

$$P(w_{it} \mid C_{it}) = \frac{\exp\left(u_{w_{it}}^\top h_{it}\right)}{\sum_{w \in \mathcal{V}} \exp\left(u_w^\top h_{it}\right)},$$

and the CBOW loss is

$$\mathcal{L}^{\mathrm{CBOW}} = -\sum_{i=1}^{n} \sum_{t=1}^{T_i} \log P(w_{it} \mid C_{it}).$$

In either variant, minimizing the loss increases the similarity between words that frequently occur in the same local contexts. For computational efficiency, my implementation replaces the full softmax calculation with negative sampling (**?**).

After training, I retain the input embedding v_w of each word. To obtain a fixed-dimensional representation of text i, I consider two pooling methods. Unweighted mean pooling averages the embeddings of all token occurrences in the text:

$$x_i^{\mathrm{W2V}} = \frac{1}{T_i} \sum_{t=1}^{T_i} v_{w_{it}}.$$

I also consider TF–IDF-weighted mean pooling. Let a_{it} denote the inverse-document-frequency weight of token w_{it}. The text representation is

$$x_i^{\mathrm{W2V}} = \frac{\sum_{t=1}^{T_i} a_{it} v_{w_{it}}}{\sum_{t=1}^{T_i} a_{it}}.$$

After vectorization, the representation is concatenated with other features z_i,

$$\widetilde{x}_i = \begin{bmatrix} x_i^{\mathrm{W2V}} \\ z_i \end{bmatrix},$$

and the resulting vector is passed to one of the classification heads described in the previous subsection for training and prediction.

B.3 Multiple-instance pooling

Mean and max probability pooling (IND). Let $p_i \in [0,1]$ be the predicted probability that attack i succeeds, obtained from the corresponding attack-level model. For a claim facing $K \geq 1$ attacks, mean and max pooling produce the scores

$$\widehat{p}_{\text{mean}} = \frac{1}{K}\sum_{i=1}^{K} p_i, \qquad \widehat{p}_{\max} = \max_{1\leq i\leq K} p_i. \tag{8}$$

The claim is predicted to be invalid when the chosen score is at least 0.5. Mean pooling averages the predicted attack probabilities, whereas max pooling uses the largest probability. These operators pool probabilities rather than text embeddings. The reported IND (mean) specification averages the probabilities from INS-A1 and OBV-A2 across the retained attacks.

Attention-based MIL pooling. Following Ilse et al. (2018), let $h_i \in \mathbb{R}^m$ be the representation of instance i. In this application, each instance combines the shared claim-side representations with its attack-specific representation. The pooled bag representation is

$$b = \sum_{i=1}^{K} \alpha_i h_i, \qquad \alpha_i = \frac{\exp(s_i)}{\sum_{j=1}^{K}\exp(s_j)}. \tag{9}$$

The attention weights are nonnegative and sum to one. For the attention operator without a gate, the score is

$$s_i = w^\top \tanh(W h_i), \tag{10}$$

where $W \in \mathbb{R}^{\ell\times m}$ and $w \in \mathbb{R}^{\ell}$ are learned parameters. The pooled representation, together with numerical features, is passed to a classification head.

Gated-attention MIL pooling. The gated variant of Ilse et al. (2018) replaces the score in Equation (10) with

$$s_i = w^\top \left[\tanh(W h_i) \odot \sigma(R h_i)\right], \tag{11}$$

where $R \in \mathbb{R}^{\ell\times m}$ is an additional learned matrix, σ is the element-wise sigmoid function, and $\odot$ denotes element-wise multiplication. The sigmoid gate modulates each component of the transformed instance representation before its attention score is computed. The normalization and pooling in Equation (9) remain unchanged. Ground-type encoding is a separate design choice: the suffix +T denotes ground-type information, not gating.

B.4 Evaluation metrics

Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. These are the four elements of the confusion matrix.

Matthews correlation coefficient. MCC measures the correlation between the predicted and observed binary outcomes while incorporating all four cells of the confusion matrix:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

MCC equals 1 for perfect prediction, 0 when the predictions have no overall association with the observed outcomes, and -1 for perfect disagreement. Because it jointly accounts for performance on both classes, MCC is my principal evaluation metric.

Macro-F1. For class c, the F1 score is the harmonic mean of precision and recall and can be written as

$$\text{F1}_c = \frac{2TP_c}{2TP_c + FP_c + FN_c},$$

where TP_c, FP_c, and FN_c are computed by treating class c as the positive class. I report macro-F1,

$$\text{F1}_{\text{macro}} = \frac{1}{2}\left(\text{F1}_0 + \text{F1}_1\right),$$

which gives equal weight to predictive performance on the two outcome classes.

Accuracy. Accuracy is the fraction of observations assigned to the correct class:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Accuracy provides an intuitive measure of the overall proportion of correct predictions. However, it can be misleading when one class is substantially more frequent because a classifier can obtain relatively high accuracy by predominantly predicting the majority class.

Precision for class 1. Precision measures how often the model is correct when it predicts that the owner loses:

$$\text{Precision}_1 = \frac{TP}{TP + FP}.$$

Recall for class 1. Recall measures how often the model successfully identifies cases in which the owner actually loses:

$$\text{Recall}_1 = \frac{TP}{TP + FN}.$$

Balanced accuracy. Balanced accuracy gives equal importance to accuracy within each of the two classes:

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right).$$

The two terms are the true-positive and true-negative rates. Balanced accuracy therefore indicates whether the classifier performs comparably across the two classes, rather than obtaining its accuracy primarily from the majority class.

ROC–AUC. ROC–AUC measures how well a classifier ranks positive observations above negative observations across all possible classification thresholds. Let s_i denote the prediction score for observation i. ROC–AUC can be written as

$$\text{ROC-AUC} = P(s_i > s_j \mid y_i = 1,\ y_j = 0) + \frac{1}{2}P(s_i = s_j \mid y_i = 1,\ y_j = 0).$$

It is therefore the probability that a randomly selected positive observation receives a higher score than a randomly selected negative observation, with half credit assigned to ties. A value of 0.5 corresponds to random ranking, while a value of 1 corresponds to perfect ranking.

Group-weighted accuracy. The preceding metrics do not account for unequal numbers of observations across patents. Ordinary claim-level accuracy gives greater aggregate influence to patents for which more claims are observed. I therefore report patent-weighted accuracy, which assigns equal aggregate weight to each patent regardless of the number of challenged claims it contains. This adjustment is particularly important because claim outcomes within the same patent are highly correlated: correctly predicting one claim often coincides with correctly predicting other claims from the same patent. Without patent weighting, patents containing many observed claims may therefore exert disproportionate influence on the reported performance.

For novelty and obviousness, an additional issue arises. A single claim may face several distinct attacks, each based on a different prior-art unit. A prior-art unit may consist of one document or several documents assessed jointly, and each claim–prior-art-unit pair constitutes one attack-level observation. Ordinary attack-level accuracy therefore gives greater aggregate weight to claims facing more attacks. I also report claim-weighted accuracy, which assigns equal aggregate weight to each claim while continuing to evaluate predictions at the attack level.

Let

$$c_i = \mathbf{1}\{\widehat{y}_i = y_i\}$$

indicate whether observation i is classified correctly. Let $n_{g(i)}$ denote the number of observations associated with the same claim as observation i, and let $n_{p(i)}$ denote the total number of observations associated with the same patent and case. The two measures are

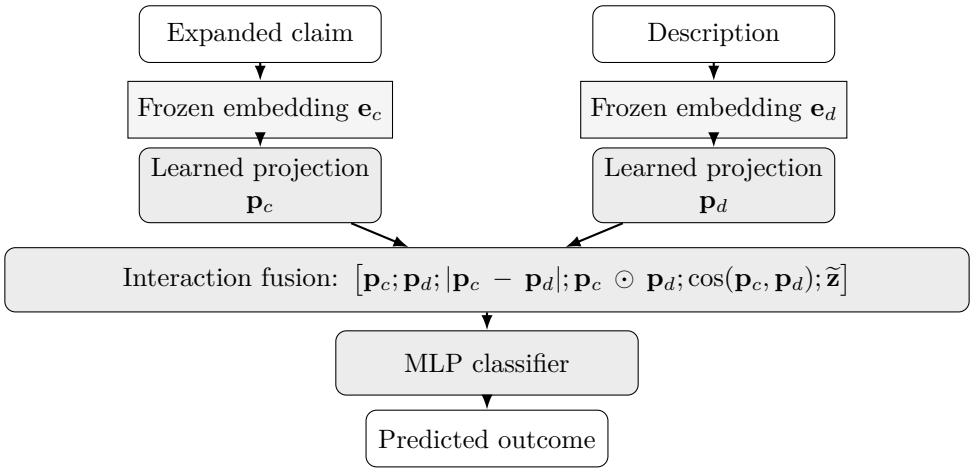
$$\text{Acc}_{\text{claim}} = \frac{\sum_i c_i/n_{g(i)}}{\sum_i 1/n_{g(i)}}, \qquad \text{Acc}_{\text{patent}} = \frac{\sum_i c_i/n_{p(i)}}{\sum_i 1/n_{p(i)}}.$$

C Supplementary Results and Model Architectures

This appendix presents the selected insufficiency architectures, performance on the original model-selection validation splits, novelty model specifications and repeated-validation results, and additional external-test results for the UKIPO and combined court and UKIPO samples.

C.1 Insufficiency model architectures

(a) Projected interaction fusion, illustrated using INS-A1



(b) Simple concatenation, illustrated using INS-A2

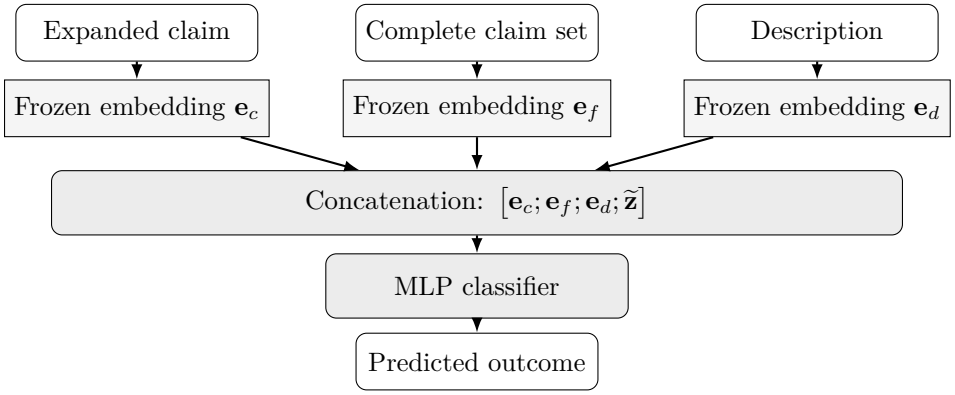


Figure C.1: Frozen-embedding architecture families. Panel (a) illustrates projected interaction fusion using the two text fields in INS-A1. Each frozen embedding is mapped into a learned, normalized, task-specific representation before the projected embeddings and their interactions are passed to an MLP classifier. The same construction extends to specifications containing one or more than two text fields by including all available projected embeddings and all unordered pairs. Panel (b) illustrates the simple-concatenation architecture using INS-A2. Here the frozen embeddings are concatenated directly, without learned text projections or explicit interaction features. In both panels, $\widetilde{\mathbf{z}}$ denotes either the numerical-feature vector or a learned projection of that vector. Projection dimensions, included text fields, and classifier dimensions vary across model specifications.

C.2 Original validation performance

C.2.1 Ex-ante patent-level prediction

Table C.1: Performance of the shortlisted classical models on the original model-selection validation split

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Precision (owner loss)	Recall (owner loss)
TF–IDF, without numerical features	0.655	0.612	0.601	0.213	0.595	0.609	**0.764**
TF–IDF, with numerical features	0.608	0.534	0.529	0.059	0.529	0.559	0.600
Word2Vec + linear SVM (L_1), without numerical features	0.673	0.660	0.658	0.316	0.658	**0.679**	0.691
Word2Vec + linear SVM (L_1), with numerical features	**0.692**	**0.670**	**0.664**	**0.334**	**0.665**	0.672	0.745

Precision and recall treat owner loss (class 1) as the positive class.

C.2.2 Insufficiency

The table below reports the corrected results on the original validation splits at both depths, retaining their case assignments. Precision and recall refer to owner loss (class 1). At depth ≤ 2, MCC ranges from 0.583 to 0.667 and accuracy from 0.800 to 0.835. At full depth, MCC ranges from 0.211 to 0.446. INS-A1 with numerical features achieves the highest MCC in both panels. These single-split results differ from the repeated-validation medians and illustrate sensitivity to the particular cases held out.

Table C.2: Insufficiency: performance on the original validation splits

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 85$)								
INS-A1, text only	**0.851**	0.800	0.796	0.583	0.791	0.813	0.714	**0.781**
INS-A1, text + numerical	0.836	**0.835**	0.781	**0.667**	**0.802**	**0.854**	**1.000**	0.562
INS-A2, text only	0.830	0.812	**0.806**	0.605	**0.802**	0.816	0.735	**0.781**
INS-A2, text + numerical	0.844	0.812	0.762	0.599	0.778	0.823	0.900	0.562
Panel B: Full depth ($n = 118$)								
INS-A1, text only	0.652	0.686	0.664	0.338	0.666	0.723	0.634	**0.542**
INS-A1, text + numerical	0.668	**0.737**	**0.700**	**0.446**	**0.705**	**0.744**	**0.774**	0.500
INS-A2, text only	**0.728**	0.678	0.627	0.310	0.620	0.672	0.708	0.354
INS-A2, text + numerical	0.653	0.636	0.595	0.211	0.591	0.662	0.581	0.375

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. The original case assignments are retained after deduplication: 357 training and 85 validation observations at depth ≤ 2, and 504 training and 118 validation observations at full depth. All metrics use the checkpoint with the highest validation MCC within each run; ties select the earliest epoch. Precision and recall refer to owner loss on the insufficiency ground. Bold indicates column maxima within each panel at the reported precision.

C.2.3 Obviousness

Table C.3: Obviousness: performance on the original depth-≤ 2 validation split

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
OBV-A1, text only	0.615	0.593	0.522	0.128	0.421	0.494	0.548	0.778	0.057
OBV-A1, text + numerical	0.661	0.690	0.689	0.374	0.686	0.659	0.576	0.622	0.683
OBV-A2, text only	0.609	0.597	0.525	0.154	0.423	0.500	0.551	**0.875**	0.057
OBV-A2, text + numerical	**0.662**	**0.703**	**0.705**	**0.406**	**0.701**	**0.690**	**0.590**	0.633	**0.715**

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. The original validation sample contains 290 attack-level observations. All metrics in a row use the same checkpoint, selected by the highest validation MCC within that run; ties select the earliest epoch. Six epochs are available for each specification. Precision and recall treat owner loss (class 1) as the positive outcome. Bold indicates column maxima at the reported precision.

C.2.4 Claim-level validity

The table below reports the same seven specifications evaluated on the external claim-level test sets. The mean-pooling baseline uses the rerun INS-A1 component. These are single-split model-selection results; ChatGPT is evaluated only on the external test sets and is therefore not included here.

Table C.4: Claim-level validity: performance on the original depth-≤ 2 validation split

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (claim invalid)	Recall (claim invalid)
IND (mean)	**0.715**	0.670	**0.654**	0.313	**0.655**	0.585	**0.701**	0.754
ILSE	0.552	0.645	0.588	0.256	0.556	0.628	0.631	0.941
ILSE+T	0.562	**0.685**	0.638	0.346	0.628	0.666	0.665	0.924
CUS	0.655	0.660	0.602	0.302	0.572	0.657	0.638	0.958
CUS+T	0.655	0.680	0.621	**0.370**	0.592	0.621	0.648	**0.983**
CUS+SAB	0.504	0.631	0.598	0.215	0.595	0.669	0.648	0.797
CUS+T+SAB	0.590	0.665	0.623	0.292	0.615	**0.685**	0.658	0.881

All seven specifications use numerical features and the same 203 validation claims from 57 cases and 76 patent–proceeding groups; the training set contains 784 claims. For the six MIL specifications, the checkpoint is selected by maximum claim-level validation MCC over six epochs, with the earliest epoch selected in an exact tie. The selected epochs are 1 for ILSE, CUS, CUS+SAB, and CUS+T+SAB; 2 for CUS+T; and 6 for ILSE+T. IND (mean) uses the rerun INS-A1 component and the retained OBV-A2 component, selected separately by attack-level validation MCC at epochs 13 of 20 and 1 of 6, respectively. Their attack probabilities are averaged within each claim. Precision and recall treat claim invalidity on at least one retained attack as the positive outcome. Patent-weighted accuracy averages within-patent claim accuracies, with patents grouped within proceedings. Bold indicates column maxima at the reported precision.

C.2.5 Novelty

Table C.5: Performance of the shortlisted novelty models on the original validation split

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy
A1, without numerical features	0.696	0.720	0.686	0.359	0.678	0.710	0.667
A1, with numerical features	0.719	**0.810**	**0.750**	**0.529**	**0.762**	**0.801**	**0.779**
A2, without numerical features	0.712	0.730	0.655	0.328	0.661	0.716	0.676
A2, with numerical features	**0.721**	0.800	0.724	0.495	0.740	0.790	0.733
A3, without numerical features	0.645	0.320	0.495	-0.018	0.278	0.358	0.363
A3, with numerical features	0.650	0.780	0.738	0.476	0.738	0.767	0.667

C.3 Novelty: model specifications and repeated validation

The novelty training–validation sample contains 637 attacks on 450 distinct claims from 165 cases at depth ≤ 2, and 863 attacks on 625 claims from 167 cases at full depth. Owner wins account for 70.8% and 71.0%, respectively. Cases are excluded when the full text of cited prior art, such as product catalogues, could not be located.

The three best-performing models all use frozen text encoders. In descending order of validation MCC, their inputs are: prior art, the literal claim, and the patent description (A1); prior art, the literal claim, and the complete claim set (A2); and prior art, the expanded claim, and the patent description (A3). Although all three architectures include the prior art, A1 performs best using the literal claim without support from the complete claim set.

A1 and A2 use an architecture analogous to INS-A1 in the insufficiency analysis (Subsection 5.2.1), except that they combine three embeddings rather than two. A3 instead uses simple embedding concatenation, as in INS-A2 of the insufficiency analysis, but without an additional MLP layer. Performance on the original validation set is reported in Table C.5 in Appendix C.

Table C.6 reports median performance across 50 case-grouped splits for each depth regime.

Table C.6: Novelty: median performance across validation splits

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy
Panel A: Depth ≤ 2 (50 splits)							
A1, without numerical features	0.569	0.630	0.517	0.042	0.493	0.613	0.653
A1, with numerical features	0.588	0.646	0.522	0.064	0.508	0.621	0.651
A2, without numerical features	0.591	**0.650**	0.523	0.067	0.513	**0.638**	0.648
A2, with numerical features	0.589	0.642	0.516	0.049	0.503	0.627	**0.667**
A3, without numerical features	0.577	0.366	0.504	0.014	0.353	0.383	0.409
A3, with numerical features	**0.623**	0.591	**0.575**	**0.148**	**0.552**	0.582	0.573
Panel B: Full depth (50 splits)							
A1, without numerical features	0.558	0.675	0.507	0.019	0.478	0.665	0.665
A1, with numerical features	0.571	0.687	0.517	0.058	0.496	0.673	0.667
A2, without numerical features	0.569	0.682	0.522	0.059	0.506	0.674	0.660
A2, with numerical features	**0.573**	0.684	0.533	0.085	0.504	0.680	0.664
A3, without numerical features	**0.573**	**0.713**	0.500	0.000	0.422	**0.693**	**0.697**
A3, with numerical features	0.569	0.620	**0.559**	**0.123**	**0.551**	0.618	0.611

Bold indicates column maxima within each panel.

C.4 UKIPO and combined test results

This subsection reports external-test performance on the UKIPO sample and on the combined court and UKIPO sample. Court-only results are reported in the main text. Where two depth regimes are available, Panel A reports depth ≤ 2 and Panel B reports full depth (≤ 100).

C.4.1 Ex-ante patent-level prediction

Table C.7: External test performance: UKIPO

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Precision (owner loss)	Recall (owner loss)
TF–IDF, without numerical features	0.551	**0.541**	**0.546**	**0.093**	**0.540**	**0.510**	0.625
TF–IDF, with numerical features	0.496	0.459	0.471	-0.064	0.441	0.450	0.675
Word2Vec + linear SVM (L_1), without numerical features	0.521	0.518	0.518	0.036	0.517	0.488	0.525
Word2Vec + linear SVM (L_1), with numerical features	0.519	0.529	0.535	0.070	0.528	0.500	0.625
ChatGPT (GPT-5.6 Sol), zero-shot	**0.607**	0.506	0.526	0.073	0.450	0.486	**0.875**

$N = 85$: 45 owner wins/survivals and 40 owner losses after binary coding.

Precision and recall treat owner loss as the positive outcome. Bold indicates the highest value in each column.

The ChatGPT prompt is reproduced in Section E.1.

Table C.8: External test performance: overall

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Precision (owner loss)	Recall (owner loss)
TF–IDF, without numerical features	0.565	0.576	0.577	0.156	0.575	0.561	0.648
TF–IDF, with numerical features	0.516	0.521	0.523	0.047	0.512	0.511	0.662
Word2Vec + linear SVM (L_1), without numerical features	0.563	0.556	0.556	0.112	0.555	0.547	0.577
Word2Vec + linear SVM (L_1), with numerical features	0.582	**0.590**	**0.591**	**0.183**	**0.589**	**0.575**	0.648
ChatGPT (GPT-5.6 Sol), zero-shot*	**0.633**	0.521	0.526	0.075	0.452	0.508	**0.887**

$N = 144$: 73 owner wins/survivals and 71 owner losses after binary coding.

Precision and recall treat owner loss as the positive outcome. Bold indicates the highest value in each column.

*Includes one manually obtained ChatGPT prediction; all other predictions are from the API.

ChatGPT ROC–AUC uses 143 observations; the manual prediction has no recorded probability.

The ChatGPT prompt is reproduced in Section E.1.

C.4.2 Insufficiency

Table C.9: Insufficiency: external-test performance on UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 37$)								
INS-A1, text only	0.528	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
INS-A1, text + numerical	0.522	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
INS-A2, text only	0.531	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
INS-A2, text + numerical	**0.578**	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
ChatGPT (GPT-5.6 Sol), zero-shot	0.483	0.595	0.478	-0.130	0.373	0.750	**0.000**	**0.000**
Panel B: Full depth (≤ 100; $n = 60$)								
INS-A1, text only	0.586	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
INS-A1, text + numerical	0.553	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
INS-A2, text only	0.596	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
INS-A2, text + numerical	**0.619**	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
ChatGPT (GPT-5.6 Sol), zero-shot	0.428	0.550	0.485	-0.114	0.355	0.763	**0.000**	**0.000**

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Supervised models are refitted on all training and validation data at the matching depth, using a 20-epoch schedule and stopping at epoch 7 (INS-A1) or 9 (INS-A2). ChatGPT is zero-shot; its predictions are subsetted by depth. Precision and recall refer to owner loss on the insufficiency ground; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The ChatGPT prompt is reproduced in Section E.2.

Table C.10: Insufficiency: external-test performance on combined court and UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 83$)								
INS-A1, text only	0.652	0.578	0.539	0.089	0.524	0.690	0.500	0.286
INS-A1, text + numerical	0.648	0.614	0.578	0.176	0.569	0.706	0.571	0.343
INS-A2, text only	0.650	0.590	0.553	0.120	0.542	0.698	0.524	0.314
INS-A2, text + numerical	**0.662**	**0.627**	**0.592**	**0.206**	**0.586**	**0.726**	**0.591**	**0.371**
ChatGPT (GPT-5.6 Sol), zero-shot	0.619	0.590	0.530	0.095	0.474	0.688	0.556	0.143
Panel B: Full depth (≤ 100; $n = 124$)								
INS-A1, text only	0.651	0.613	0.573	0.210	0.530	0.739	0.706	0.218
INS-A1, text + numerical	0.666	**0.661**	**0.622**	**0.352**	0.589	**0.784**	**0.882**	0.273
INS-A2, text only	0.661	0.621	0.606	0.220	**0.605**	0.705	0.591	**0.473**
INS-A2, text + numerical	**0.669**	0.613	0.589	0.197	0.581	0.705	0.600	0.382
ChatGPT (GPT-5.6 Sol), zero-shot	0.542	0.532	0.489	-0.033	0.423	0.688	0.400	0.109

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Supervised models are refitted on all training and validation data at the matching depth, using a 20-epoch schedule and stopping at epoch 7 (INS-A1) or 9 (INS-A2). ChatGPT is zero-shot; its predictions are subsetted by depth. Precision and recall refer to owner loss on the insufficiency ground; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The ChatGPT prompt is reproduced in Section E.2.

C.4.3 Obviousness

Table C.11: Obviousness: external-test performance on UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 232$)									
OBV-A1, text only	0.493	**0.810**	0.497	-0.031	**0.448**	**0.788**	**0.694**	0.000	0.000
OBV-A1, text + numerical	0.427	0.448	0.401	-0.154	0.382	0.432	0.527	0.124	0.326
OBV-A2, text only	0.527	0.806	0.495	-0.044	0.446	0.782	0.689	0.000	0.000
OBV-A2, text + numerical	0.425	0.418	0.364	-0.211	0.354	0.378	0.475	0.103	0.279
ChatGPT (GPT-5.6 Sol), zero-shot	**0.594**	0.405	**0.608**	**0.195**	0.403	0.437	0.435	**0.229**	**0.930**
Panel B: Full depth (≤ 100; $n = 379$)									
OBV-A1, text only	0.378	**0.847**	0.495	-0.037	**0.459**	**0.822**	**0.699**	0.000	0.000
OBV-A1, text + numerical	0.358	0.269	0.474	-0.046	0.267	0.274	0.421	0.137	0.764
OBV-A2, text only	0.431	0.842	0.492	-0.048	0.457	0.814	0.693	0.000	0.000
OBV-A2, text + numerical	0.363	0.266	0.480	-0.036	0.265	0.272	0.437	0.139	0.782
ChatGPT (GPT-5.6 Sol), zero-shot	**0.613**	0.327	**0.569**	**0.119**	0.324	0.356	0.434	**0.167**	**0.909**

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. Supervised models are trained at the same depth as the evaluation sample. ChatGPT is zero-shot; its predictions are subsetted by depth. Metrics are computed at the attack level. Patent-weighted accuracy is the mean of within-patent attack accuracies, with patents grouped within proceedings. Precision and recall refer to owner loss on the obviousness attack; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The ChatGPT prompt is reproduced in Section E.3.

Table C.12: Obviousness: external-test performance on combined court and UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 460$)									
OBV-A1, text only	0.472	**0.743**	0.499	-0.027	0.426	**0.738**	**0.703**	0.000	0.000
OBV-A1, text + numerical	0.519	0.530	0.460	-0.072	**0.456**	0.513	0.614	0.214	0.316
OBV-A2, text only	0.488	0.741	0.497	-0.039	0.426	0.734	0.701	0.000	0.000
OBV-A2, text + numerical	0.519	0.526	0.457	-0.077	0.453	0.499	0.595	0.211	0.316
ChatGPT (GPT-5.6 Sol), zero-shot	**0.638**	0.452	**0.599**	**0.199**	0.452	0.465	0.441	**0.304**	**0.897**
Panel B: Full depth (≤ 100; $n = 685$)									
OBV-A1, text only	0.438	**0.728**	0.495	-0.051	0.421	**0.722**	0.683	0.000	0.000
OBV-A1, text + numerical	0.516	0.425	0.554	0.110	0.425	0.411	0.521	0.292	0.829
OBV-A2, text only	0.459	0.727	0.494	-0.056	0.421	0.719	**0.685**	0.000	0.000
OBV-A2, text + numerical	0.521	0.410	0.539	0.080	0.410	0.393	0.514	0.284	0.812
ChatGPT (GPT-5.6 Sol), zero-shot	**0.633**	0.439	**0.589**	**0.188**	**0.439**	0.447	0.453	**0.309**	**0.906**

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. Supervised models are trained at the same depth as the evaluation sample. ChatGPT is zero-shot; its predictions are subsetted by depth. Metrics are computed at the attack level. Patent-weighted accuracy is the mean of within-patent attack accuracies, with patents grouped within proceedings. Precision and recall refer to owner loss on the obviousness attack; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The ChatGPT prompt is reproduced in Section E.3.

C.4.4 Claim-level validity

Tables C.13 and C.14 extend Table 12 to the UKIPO and combined samples. Each panel uses the same claims and outcome labels for all eight methods.

Table C.13: Claim-level validity: external-test performance on UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (claim invalid)	Recall (claim invalid)
Panel A: Depth ≤ 2 ($n = 172$ claims)								
IND (mean)	0.352	0.407	0.373	-0.227	0.369	0.542	0.169	0.298
ILSE	0.620	0.651	0.594	0.178	0.587	0.591	0.386	0.468
ILSE+T	0.520	0.477	0.494	-0.011	0.458	0.574	0.269	0.532
CUS	**0.647**	0.692	0.516	0.046	0.497	0.620	0.333	0.128
CUS+T	0.602	**0.744**	0.565	0.220	0.556	**0.654**	**0.615**	0.170
CUS+SAB	0.568	0.733	0.557	0.180	0.547	0.633	0.533	0.170
CUS+T+SAB	0.634	0.733	**0.637**	**0.291**	**0.643**	0.621	0.513	0.426
ChatGPT (GPT-5.6 Sol), zero-shot	0.489	0.395	0.484	-0.030	0.395	0.472	0.264	**0.681**
Panel B: Full depth (≤ 100; $n = 293$ claims)								
IND (mean)	0.280	0.317	0.396	-0.195	0.316	**0.538**	0.188	0.549
ILSE	0.560	0.488	0.495	-0.009	0.456	0.508	0.238	0.507
ILSE+T	0.463	0.300	0.375	-0.235	0.299	0.503	0.178	0.521
CUS	0.574	0.539	0.538	0.065	0.500	0.519	0.271	0.535
CUS+T	0.472	0.427	0.478	-0.039	0.414	0.489	0.229	0.577
CUS+SAB	**0.642**	0.563	**0.645**	**0.251**	**0.549**	0.473	**0.333**	**0.803**
CUS+T+SAB	0.612	**0.584**	0.548	0.084	0.525	0.531	0.286	0.479
ChatGPT (GPT-5.6 Sol), zero-shot	0.440	0.334	0.427	-0.139	0.333	0.475	0.205	0.606

The supervised specifications use numerical features and follow the refitting protocol in Table 12. ChatGPT (GPT-5.6 Sol) is evaluated zero-shot on the same claims; its predictions are subsetted by dependency depth. The classification threshold is 0.5. Precision and recall treat claim invalidity on at least one retained insufficiency or obviousness attack as the positive outcome. Patent-weighted accuracy averages within-patent claim accuracies, with patents grouped within proceedings. Bold indicates column maxima within each panel at the reported precision.

Table C.14: Claim-level validity: external-test performance on combined court and UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (claim invalid)	Recall (claim invalid)
Panel A: Depth ≤ 2 ($n = 305$ claims)								
IND (mean)	0.444	0.479	0.452	-0.093	0.451	0.577	0.302	0.361
ILSE	0.706	0.646	0.651	0.289	0.635	**0.616**	0.500	0.667
ILSE+T	0.688	0.541	0.580	0.157	0.540	0.587	0.414	0.713
CUS	0.712	0.692	0.634	0.291	0.639	0.602	0.588	0.435
CUS+T	**0.727**	0.725	0.661	0.363	0.669	0.612	**0.667**	0.444
CUS+SAB	0.702	0.652	0.545	0.133	0.518	0.597	0.528	0.176
CUS+T+SAB	0.710	**0.728**	**0.708**	**0.411**	**0.706**	0.599	0.611	0.639
ChatGPT (GPT-5.6 Sol), zero-shot	0.601	0.456	0.531	0.067	0.450	0.501	0.373	**0.787**
Panel B: Full depth (≤ 100; $n = 476$ claims)								
IND (mean)	0.416	0.418	0.453	-0.093	0.418	0.568	0.326	0.579
ILSE	0.656	0.534	0.571	0.139	0.533	0.533	0.412	0.702
ILSE+T	0.672	0.435	0.506	0.014	0.428	0.551	0.363	0.760
CUS	0.681	0.557	0.604	0.208	0.557	0.516	0.434	0.772
CUS+T	0.642	0.489	0.555	0.118	0.486	0.510	0.395	0.789
CUS+SAB	0.722	0.584	**0.650**	**0.311**	0.583	0.519	0.459	**0.883**
CUS+T+SAB	**0.745**	**0.613**	0.641	0.271	**0.611**	**0.576**	**0.475**	0.737
ChatGPT (GPT-5.6 Sol), zero-shot	0.583	0.439	0.507	0.015	0.434	0.513	0.364	0.749

The supervised specifications use numerical features and follow the refitting protocol in Table 12. ChatGPT (GPT-5.6 Sol) is evaluated zero-shot on the same claims; its predictions are subsetted by dependency depth. The classification threshold is 0.5. Precision and recall treat claim invalidity on at least one retained insufficiency or obviousness attack as the positive outcome. Patent-weighted accuracy averages within-patent claim accuracies, with patents grouped within proceedings. Bold indicates column maxima within each panel at the reported precision.

C.4.5 Insufficiency: separate versus combined encoding

The following tables extend the comparison in Table 8 to the UKIPO and combined samples.

Table C.15: Insufficiency: separate versus combined encoding on UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 37$)								
INS-A1, separate	0.522	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
INS-A2, separate	**0.578**	**0.622**	**0.500**	**0.000**	**0.383**	**0.778**	**0.000**	**0.000**
INS-X1, combined	0.516	0.568	0.457	-0.187	0.362	0.764	**0.000**	**0.000**
INS-X2, combined	0.525	0.568	0.457	-0.187	0.362	0.764	**0.000**	**0.000**
INS-X3, combined	0.568	0.432	0.348	-0.377	0.302	0.729	**0.000**	**0.000**
Panel B: Full depth (≤ 100) ($n = 60$)								
INS-A1, separate	0.553	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
INS-A2, separate	**0.619**	**0.567**	**0.500**	**0.000**	**0.362**	**0.789**	**0.000**	**0.000**
INS-X1, combined	0.528	0.433	0.382	-0.343	0.302	0.737	**0.000**	**0.000**
INS-X2, combined	0.537	0.433	0.382	-0.343	0.302	0.737	**0.000**	**0.000**
INS-X3, combined	0.584	0.433	0.382	-0.343	0.302	0.737	**0.000**	**0.000**

INS-A1 encodes the expanded claim and description separately. INS-A2 additionally encodes the complete claim set separately. INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All specifications include numerical features, use dropout 0.1 and equal patent weights ($\gamma = 1$), and use a 20-epoch learning-rate schedule with training stopped at epoch 7 for INS-A1 and epoch 9 for the other specifications. Training and evaluation depths match within each panel. All descriptions use the first 16,384 tokens, as in the separate-encoding baseline. The classification threshold is 0.5. The separate-encoding results are reproduced from Table C.9. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

Table C.16: Insufficiency: separate versus combined encoding on combined court and UKIPO observations

Model	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 83$)								
INS-A1, separate	0.648	0.614	0.578	0.176	0.569	0.706	0.571	0.343
INS-A2, separate	**0.662**	**0.627**	**0.592**	**0.206**	**0.586**	**0.726**	**0.591**	**0.371**
INS-X1, combined	0.617	0.566	0.532	0.071	0.523	0.688	0.478	0.314
INS-X2, combined	0.611	0.554	0.522	0.047	0.514	0.677	0.458	0.314
INS-X3, combined	0.614	0.482	0.451	-0.103	0.443	0.626	0.346	0.257
Panel B: Full depth (≤ 100) ($n = 124$)								
INS-A1, separate	0.666	**0.661**	**0.622**	**0.352**	**0.589**	**0.784**	**0.882**	0.273
INS-A2, separate	**0.669**	0.613	0.589	0.197	0.581	0.705	0.600	**0.382**
INS-X1, combined	0.590	0.556	0.526	0.061	0.502	0.689	0.500	0.255
INS-X2, combined	0.586	0.556	0.528	0.064	0.508	0.693	0.500	0.273
INS-X3, combined	0.598	0.548	0.517	0.040	0.490	0.686	0.481	0.236

INS-A1 encodes the expanded claim and description separately. INS-A2 additionally encodes the complete claim set separately. INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All specifications include numerical features, use dropout 0.1 and equal patent weights ($\gamma = 1$), and use a 20-epoch learning-rate schedule with training stopped at epoch 7 for INS-A1 and epoch 9 for the other specifications. Training and evaluation depths match within each panel. All descriptions use the first 16,384 tokens, as in the separate-encoding baseline. The classification threshold is 0.5. The separate-encoding results are reproduced from Table C.10. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

C.4.6 Obviousness: separate versus combined encoding

Table C.17: Obviousness: separate versus combined encoding on UKIPO observations

Model	Training depth	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 232$)										
OBV-A1, separate	2	0.427	0.448	0.401	-0.154	0.382	0.432	0.527	0.124	**0.326**
OBV-A2, separate	2	0.425	0.418	0.364	-0.211	0.354	0.378	0.475	0.103	0.279
OBV-X1, combined	2	0.629	0.685	0.448	-0.113	0.443	0.699	0.612	0.083	0.070
OBV-X2, combined	2	**0.631**	**0.694**	**0.471**	**-0.061**	**0.469**	**0.709**	**0.642**	**0.132**	0.116
Panel B: Full depth (≤ 100) ($n = 379$)										
OBV-A1, separate	100	0.358	0.269	0.474	-0.046	0.267	0.274	0.421	0.137	0.764
OBV-A2, separate	100	0.363	0.266	0.480	-0.036	0.265	0.272	0.437	0.139	**0.782**
OBV-X1, combined	100	0.591	**0.712**	0.500	-0.001	0.497	0.732	0.636	0.145	0.200
OBV-X2, combined	100	**0.596**	**0.712**	**0.507**	**0.013**	**0.503**	**0.734**	**0.637**	**0.154**	0.218

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All four specifications include numerical features. The separate-encoding models are trained at the evaluation depth and use patent-size weighting with $\gamma = 1/16$. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100). Both combined-encoding models use uniform loss weights and stop at epoch 2. Training and evaluation depths therefore match within each panel, although the separate- and combined-encoding specifications differ in loss weighting. The OBV-A1 and OBV-A2 results are reproduced from Table C.11. Metrics are computed at the attack level. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

Table C.18: Obviousness: separate versus combined encoding on combined court and UKIPO observations

Model	Training depth	ROC–AUC	Accuracy	Balanced accuracy	MCC	Macro-F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 460$)										
OBV-A1, separate	2	**0.519**	0.530	**0.460**	**-0.072**	**0.456**	0.513	**0.614**	**0.214**	**0.316**
OBV-A2, separate	2	**0.519**	0.526	0.457	-0.077	0.453	0.499	0.595	0.211	**0.316**
OBV-X1, combined	2	0.385	0.550	0.411	-0.176	0.412	0.561	0.526	0.125	0.128
OBV-X2, combined	2	0.385	**0.554**	0.420	-0.159	0.421	**0.567**	0.541	0.139	0.145
Panel B: Full depth (≤ 100) ($n = 685$)										
OBV-A1, separate	100	0.516	0.425	**0.554**	**0.110**	0.425	0.411	**0.521**	**0.292**	**0.829**
OBV-A2, separate	100	**0.521**	0.410	0.539	0.080	0.410	0.393	0.514	0.284	0.812
OBV-X1, combined	100	0.358	**0.547**	0.427	-0.144	0.428	**0.558**	0.499	0.162	0.171
OBV-X2, combined	100	0.358	0.546	0.428	-0.142	**0.429**	**0.558**	0.498	0.165	0.177

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All four specifications include numerical features. The separate-encoding models are trained at the evaluation depth and use patent-size weighting with $\gamma = 1/16$. All models in Panel A are trained and evaluated at depth ≤ 2; all models in Panel B are trained and evaluated at full depth (≤ 100). Both combined-encoding models use uniform loss weights and stop at epoch 2. Training and evaluation depths therefore match within each panel, although the separate- and combined-encoding specifications differ in loss weighting. The OBV-A1 and OBV-A2 results are reproduced from Table C.12. Metrics are computed at the attack level. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

D LLM Prompts

Sections D.1–D.5 reproduce the four extraction prompts and the extra prompt. Sections E.1–E.3 reproduce the zero-shot prediction prompts. Source wording, examples, and output instructions are retained; document formatting is adapted for the appendix.

D.1 First prompt: patent-level extraction

IP JUDGMENT EXTRACTION PROMPT — PATENT ONLY — RIGHT-LEVEL VALIDITY DATASET

Right-level UK patent-validity dataset — revised with a mandatory Google Patents visual citation example

You are extracting structured data from UK patent judgments at the patent-right level.

STRICT RULE: NEVER USE "" IN TEXT FIELDS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

This prompt produces only the validity dataset. Do not produce an infringement dataset or infringement rows.

This is a right-level dataset, not a claim-level dataset.

* Each row corresponds to one disputed patent or patent application, aggregated at patent level.

* Do not create separate rows merely for individual claims.

* Claim-level reasoning must be summarized inside fields 18, 25, 27, and 29.

ABSOLUTE OUTPUT RESTRICTIONS

* Output exactly one fenced code block labelled csv.

* Put all CSV rows inside that csv code block.

* Do not include a header.

* Do not include notes, explanations, citations, footnotes, source references, or any text before or after the csv code block.

* Every row must contain exactly 32 fields.

* Wrap every field in double quotes.

* Escape all internal double quotes using double double-quotes.

* Lists must look like "[""A"", ""B""]" when placed inside a CSV field.

* Dictionaries must use double quotes for all internal keys and strings, escaped correctly for CSV.

* No field may contain a URL unless that field explicitly asks for a URL.

* If the case must be skipped, output exactly one CSV row. Put "skip" in every field except field 30, which must contain the judgment URL.

* Do not hallucinate.

* If a value is not available from the judgment or required public research, leave it as "".

* Date format must be YYYY-MM-DD.

* If only month and year are available, use YYYY-MM.

* If only year is available, use YYYY.

* If a date-list field has no available information, output "[]", not "".

* Field 30 must contain the raw judgment URL only.

* Every CSV record must occupy exactly one physical line in the output.

* Never insert an actual newline or carriage-return character inside any CSV field.

* Whenever this prompt specifies \n, output the two literal characters backslash and n, not an actual line break.

* Field 26 must separate passages using the literal text \n—\n.

* In field 29, place the required patent-level outcome first, followed by the literal text \n and then the remainder of the summary.

SCOPE FILTER — ONLY THREE CASE

* be careful do not make double quote """

* link of patent must not be empty. same as all field of citations. if GB versions not available, find WO version

Extract a row only if the case belongs to one of these three types:

1. An infringement action in which the court actually considers or decides patent validity. Extract only the validity result. Do not extract infringement.

2. A revocation action or counterclaim in which patent validity is considered or decided.

3. A patent-application case in which the court considers patentability or another validity-type issue, including novelty, inventive step/obviousness, insufficiency, added matter, excluded subject matter/ non-patentability, industrial applicability, priority entitlement, or another ground bearing on whether the application may proceed to grant.

An express concession that the whole patent is invalid counts as a validity outcome under Type 1 or Type 2 even if the court does not conduct a full merits analysis of individual validity grounds. Include the patent at right level. A mere withdrawal, abandonment, non-maintenance, or agreed revocation without an express concession of invalidity does not count by itself.

For Type 3 patent-application cases, the Comptroller/UK IPO is mandatory but not necessarily the only defendant/respondent or adverse party:

* Field 6 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every separately represented third-party respondent/opponent identified in the case.

* Field 8 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every third-party opponent that directly opposes or challenges the patent application in this row.

* Field 10 must contain the Comptroller object and separate verified objects for all qualifying third-party opponents in field 8, subject to the historical parent/group collapse rules.

* Do not omit a qualifying third-party opponent merely because the Comptroller is already present.

* Do not add a third party merely because it appears in the caption; it must directly oppose or challenge the application.

* Field 28 = 0 when the applicant is the claimant/appellant.

Skip:

* a pure infringement or non-infringement case with no live validity issue considered, decided, or expressly conceded invalid;

* registered-design cases;

* damages, inquiry as to damages, costs-only, consequential-order, enforcement-only, or other post-merits judgments;

* a merely interlocutory judgment that does not finally or summarily decide a live validity, revocation, refusal, or patentability issue;

* permission-to-appeal decisions following a non-merits interlocutory ruling;

* any strike-out, dismissal, stay, jurisdiction, service, forum, standing, party, pleading-amendment, patent-amendment-procedure, costs, or case-management ruling that does not directly and summarily decide patent validity, revocation, refusal, or patentability;

* a row where the revocation claim, counterclaim, or only live invalidity argument is withdrawn, abandoned, not maintained, or no longer pursued and the court consequently does not decide it;

* any case outside Types 1–3 above.

SUMMARY JUDGMENT / STRIKE-OUT

Apply this section before extracting any row. First identify (a) who made the strike-out or summary-disposition application, (b) who won that application, (c) whether the application directly concerned patent validity, revocation, refusal, or patentability, and (d) whether a later hearing or judgment in the same case decided validity.

DIRECT-VALIDITY REQUIREMENT

* The strike-out or summary-disposition application must directly seek a final or summary determination of a live patent-validity, revocation, refusal, or patentability issue.

* A threshold merits assessment may count even though it is shorter than a full trial, but the court must assess the validity challenge itself rather than merely decide a collateral procedural question.

* Always skip a strike-out or other application concerning only jurisdiction, service, forum conveniens, a stay, standing, joinder or identity of parties, duplication or utility of claims, permission to amend pleadings, permission or procedure for patent amendment without deciding validity, costs, timetabling, disclosure, evidence, or other case management.

MANDATORY STRIKE-OUT DECISION MATRIX

* Challenger applies and challenger wins: INCLUDE. Treat the result as an adverse-side right-level win; field 13 = 0 and field 25 = 1. This applies when the ruling summarily disposes of the patent or the live validity issue against the owner.

* Challenger applies and patent owner wins: INCLUDE only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, SKIP this preliminary row and use the later validity decision instead.

* Patent owner applies and patent owner wins: INCLUDE only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, SKIP this preliminary row and use the later validity decision instead.

* Patent owner applies and challenger wins: SKIP. A refusal to strike out the challenger's validity case merely allows that case to continue and is not itself a patent-validity outcome.

* If the identity of the applicant, the winner, the direct validity issue, or the existence of a later same-case validity decision cannot be established reliably, do not guess. SKIP.

WITHDRAWN OR ABANDONED REVOCATION ARGUMENTS

* If a party withdraws, abandons, does not maintain, or no longer pursues a revocation claim, revocation counterclaim, or validity argument and the court consequently does not assess or decide it, do not treat the withdrawal as a validity outcome.

* If no other live validity, revocation, refusal, or patentability issue is finally or summarily decided for that patent, SKIP the entire right-level row.

* If only one ground is withdrawn but one or more other grounds are actually decided, keep the row, leave the withdrawn ground blank in field 27, and code only the grounds the court decides.

* A formal statement that a patent is valid, maintained, or not revoked does not override this rule when it

merely records the consequence of an unopposed withdrawal or amendment and the court does not assess any live validity ground.

OUTPUT FOR INCLUDED PRE-TRIAL RULINGS

* Output one right-level row per disputed patent or patent application.

* Fill all right-level fields that can be established and make field 29 state that the result is a summary or strike-out outcome rather than a full merits judgment.

* Do not invent claim-level rows or validity grounds that the court did not assess.

ROW UNIT RULE

Each row must correspond to:

* one disputed patent or patent application;

* one owner/applicant/enforcement side; and

* one adverse validity challenger side.

Do not split rows merely because:

* different claims of the same patent have different outcomes;

* the same side contains multiple co-owners, co-applicants, exclusive licensees, enforcement-side parties, or joint validity challengers; or

* the same patent is attacked on multiple validity grounds.

Split rows only when the legal/economic unit changes: a different patent/application, a different owner/applicant/enforcement side, or a different adverse validity challenger side.

If different claims of the same patent have different outcomes, aggregate at patent level. For field 25, count only claims or claim sets that were actually attacked and substantively decided. Unattacked claims, unargued claims, withdrawn or abandoned attacks, and unresolved or undecided claims do not count as surviving claims for field 25. Apply the mandatory attacked-claim and amendment rules under field 25 below.

WHOLE-PATENT INVALIDITY CONCESSION — RIGHT-LEVEL RULE

* If the judgment, final order, or reliable litigation record expressly states that the patent owner concedes that the whole patent is invalid, include one right-level row for that patent.

* A whole-patent invalidity concession is a valid adverse validity outcome even if no full merits analysis or ground-by-ground decision follows.

* Field 25 must be 1 because the patent owner loses the whole patent.

* Identify the exact historical patent/version covered by the concession.

* Field 18 must state "patent as a whole; whole patent conceded invalid" and list the independent claims of the conceded version if they can be verified.

* Field 27 must record the identified grounds. If no ground is identified, set "Others" = 1 and leave the named grounds blank.

* Field 29 must state that invalidity was conceded, identify when the concession occurred, and distinguish it from a judicial merits finding.

* An express whole-patent invalidity concession recorded before the substantive validity trial still counts.

* Mere withdrawal of a revocation application, abandonment of a patent, non-maintenance of claims, or agreed revocation without an express admission/concession that the patent is invalid does not count and must not be coded as a right-level validity loss.

PATENT-VERSION RULE — MANDATORY

The extracted patent/application must be the version legally operative and actually addressed by the judgment on the judgment date.

* Identify whether the judgment concerns the application claims, granted claims, a previously amended specification, or proposed amended claims.

* Use the correct patent/application number and publication/grant suffix applicable at that date, for example A, A1, B, or B1 where relevant.

* If the judgment does not identify the version clearly, research the official UKIPO/EPO record and contemporaneous patent documents.

* Do not silently use the current Google Patents version when it differs from the version before the court.

* Field 15 must name the correct version.

* Field 18 must identify the actual claim version attacked or decided.

* Field 29 must explain any old/granted/amended/proposed-amended distinction and identify which version was operative.

* If the court substantively assesses both the granted and amended versions, record both in fields 18 and 29 but retain one right-level row.

* If the correct historical version cannot be verified, do not guess; state the uncertainty briefly in field 29.

REFERENCE DATES

* FIRST_HEARING_DATE = the earliest hearing date in field 2 for the judgment being extracted.

* FIRST_JUDGMENT_DATE = the earliest judgment date in field 1 for the judgment being extracted.

* ENTITY_REFERENCE_DATE for fields 9 and 10 = FIRST_JUDGMENT_DATE.

* The four forward-citation counts use only FIRST_HEARING_DATE and FIRST_JUDGMENT_DATE as specified in fields 19, 20, 31, and 32.

FIELDS

1. "date_of_judgment"

List of judgment date(s), in YYYY-MM-DD format.

Example: ["2024-07-15"].

2. "date_of_hearing"

List of hearing date(s), in YYYY-MM-DD format.

If unavailable, output [].

Do not use ranges. Expand ranges into separate dates where possible.

3. "date_filing_the_complaint"

List of claim-form, revocation-application, counterclaim, or appeal filing date(s).

If unavailable, output [].

If only month and year are available, use YYYY-MM.

If only year is available, use YYYY.

4. "judges"

Names of all judges. Return a list of strings.

Do not include prefixes such as Sir, Mr, Mrs, The Hon, or Lord Justice.

If the judgment uses only a judicial title, identify the actual name if reliable public research permits.

5. "claimants"

Names of all claimants/appellants in the judgment caption or case title. Return a list of strings.

6. "defendants"

Names of all defendants/respondents in the judgment caption or case title. Return a list of strings.

For every Type 3 patent-application case, the list must include "Comptroller-General of Patents, Designs and Trade Marks". Also include every other defendant/respondent that separately opposes the application. The Comptroller is mandatory, not exclusive.

FIELDS 7–10: LEGAL PARTIES AND ECONOMIC ENTITY REPRESENTATIVES

Fields 5 and 6 are caption-level fields.

Fields 7 and 8 are row-level legal-party fields.

Fields 9 and 10 are row-level economic-entity fields measured at ENTITY_REFERENCE_DATE.

Fields 7–10 must not mechanically follow claimant/defendant status.

* Field 7 identifies the owner/applicant/enforcement legal side for the patent/application in field 15.

* Field 8 identifies the adverse validity challenger legal side.

* Fields 9 and 10 may collapse legal subsidiaries or related parties in fields 7 and 8 to their common parent/group representative under the mandatory parent rules below.

* Therefore, the standardized parent name in field 9 or 10 need not be identical to the legal-party name in field 7 or 8.

7. "right_owner_or_applicant_side"

List only the legal parties that own, apply for, exclusively license, or directly enforce the patent/application in field 15.

Rules:

* This is a patent-right owner/applicant-side field, not a claimant/defendant field.

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* Do not include parties active only on another patent, another issue, damages, costs, or a formal/nominal role.

* If aligned co-owners, co-applicants, exclusive licensees, or enforcement-side parties are treated together, list them together.

* If the owner/applicant side is a defendant, still list it here.

* If ownership or applicant status cannot be determined, output [].

8. "other_side_or_validity_challenger_side"

List only the legal parties that attack, oppose, seek revocation of, or directly challenge the validity/patentability of the patent/application in field 15.

Rules:

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* If several parties jointly challenge the same right and the court treats them together, list them together.

* For every Type 3 patent-application case, include "Comptroller-General of Patents, Designs and Trade Marks" and every third-party opponent that directly opposes or challenges the application in this row.

* The Comptroller is mandatory but does not replace or absorb a separate private or public opponent.

* If the adverse side cannot be determined, output [].

9. "right_owner_or_applicant_party_type_ticker_and_headquarter"

Return a dictionary indexed consecutively from "1". Each object must contain exactly these four fields and

no others:

{"1":{"name":"","type":1,"ticker_at_time":[],"headquarter":0}}

10. "other_side_or_validity_challenger_party_type_ticker_and_headquarter"

Use the same JSON structure and the same mandatory verification rules as field 9.

For every Type 3 patent-application case, field 10 must contain at least:

{"1":{"name":"Comptroller-General of Patents, Designs and Trade Marks","type":3,"ticker_at_time":[], "headquarter":0}}

If field 8 contains one or more third-party opponents, add consecutively numbered objects for them. Verify and, where required, collapse each opponent under the historical entity/parent/group rules below. The Comptroller remains a separate mandatory object and must never be collapsed with another opponent.

MANDATORY FIELD 9 / FIELD 10 VERIFICATION RULES

Verify every entry as of ENTITY_REFERENCE_DATE. Do not use current ownership, listing status, ticker, or headquarters when it differs from the historical position on that date.

JSON format:

* Keys are "1", "2", and so on.

* Every object has exactly: "name", "type", "ticker_at_time", and "headquarter".

* "type" and "headquarter" must be numbers, never strings.

* "ticker_at_time" must always be a list, for example ["NYSE:ABC"]. Use [] when there was no verified ticker.

* If field 7 or field 8 is [], return {} for the corresponding field 9 or field 10.

Type coding:

* 0 = individual/person.

* 1 = private or non-listed company/entity.

* 2 = listed company. This includes a legal party that is a subsidiary of a listed parent at ENTITY_ REFERENCE_DATE; collapse it to the listed parent.

* 3 = government, UKIPO, Comptroller, or public authority.

Headquarter coding:

* 0 = United Kingdom.

* 1 = European Union excluding the United Kingdom, with Switzerland included in code 1.

* 2 = outside the United Kingdom and the code-1 region.

* For an individual/person, always use headquarter = 0 unless the judgment, the input, or reliable public research explicitly establishes a different residence or domicile.

* For the Comptroller or UKIPO, use headquarter = 0.

Parent/group representative rule:

* If a legal party was a subsidiary of a listed parent at ENTITY_REFERENCE_DATE, collapse it to the listed parent and use the parent's legal name, historical ticker, and headquarters.

* If multiple legal entities on the same side belonged to the same corporate group or ultimate parent at ENTITY_REFERENCE_DATE, collapse them into one entry.

* A UK subsidiary of Pfizer is represented by Pfizer Inc., not by the UK subsidiary.

* A German subsidiary of Clariant is represented by Clariant AG, not by the German subsidiary.

* If the ultimate parent/group was private, collapse to the private parent/group when the relationship is clear.

* If both parent and subsidiary appear on the same side, normally retain only the parent representative after collapse.

* Do not collapse entities merely because their names are similar. Require reliable evidence of common control or parentage at ENTITY_REFERENCE_DATE.

* If the relevant parent cannot be verified, retain the legal entity separately and use the safest verifiable type/ headquarter coding. Mention "insufficient evidence to collapse" briefly in field 29.

Historical ticker rule:

* The ticker must be the ticker that existed at ENTITY_REFERENCE_DATE.

* Use an exchange prefix, for example NYSE:, NASDAQ:, LSE:, EPA:, AMS:, FWB:, SIX:, SWX:, TSE:, TYO:, KRX:, BSE:, NSE:, BIT:, or EBR:.

* If a company had not yet completed its IPO at ENTITY_REFERENCE_DATE, use type = 1 and ticker_ at_time = [].

* If a company had already been acquired or delisted before ENTITY_REFERENCE_DATE, use the acquiring/group parent at that date when the parent rule applies.

* Do not use a present-day ticker for an earlier date.

Name rule:

* Standardize the legal name without unnecessary rewriting.

* When collapsing, use the parent's legal name.

* Preserve the correct suffix, such as Inc., Ltd, plc, AG, GmbH, S.p.A., N.V., Corporation, or Limited.

11. "claimant_lawyers"

Names of all lawyers representing the claimants/appellants. Return a list of strings.

If unavailable, output [].

12. "defendant_lawyers"

Names of all lawyers representing the defendants/respondents. Return a list of strings.

If unavailable, output [].

13. "stage"

Return:

* 0 = a summary judgment, strike-out judgment, or other dispositive pre-trial ruling that is INCLUDED under the mandatory SUMMARY JUDGMENT / STRIKE-OUT decision matrix above.

* 1 = merits judgment after trial or a substantive final hearing.

Skip merely interlocutory or post-merits judgments under the scope rules above.

14. "right_type"

Always return 0.

0 = patent or patent application.

Registered designs and all other rights are out of scope.

15. "right_name"

Name, title, and full publication/grant/application number of the patent or patent application.

* Use the exact historically correct version actually addressed by the judgment.

* Include prefixes and suffixes such as GB, EP, UK, WO, A, A1, B, or B1 where applicable.

* If the judgment is unclear, verify the contemporaneous version using official UKIPO/EPO records and patent documents.

* Do not substitute a later granted or later amended version for the version before the court.

16. "right_short_description"

A concise factual description of the invention or patent application.

Use the judgment's wording where possible.

Do not speculate.

17. "google_patent_url"

Raw Google Patents URL for the patent or patent application.

You must research this online.

If no Google Patents page exists, use the best official UKIPO/EPO patent-record URL.

If no reliable URL is found, leave "".

18. "patent_unit"

Begin with "patent as a whole" or "patent application as a whole".

Then identify:

* the claims actually attacked, defended, amended, proposed for amendment, or decided;

* whether each relevant claim set is the application, old/granted, previously amended, or proposed-amended version; and

* whether revocation/refusal applies to the whole right even though only representative claims were analysed.

Examples:

* "patent as a whole; validity attacked on granted claims 1, 3, and 5"

* "patent as a whole; granted claim 1 invalid; proposed amended claim 1 allowed"

* "patent application as a whole; application claims 1–20 refused for obviousness"

* "patent as a whole; whole patent conceded invalid; independent granted claims 1, 10, and 15"

* "patent as a whole; attacked claims not individually identified"

Do not infer that unargued claims were individually invalid merely because the whole patent was revoked. An express whole-patent invalidity concession is recorded as a right-level loss under the special rule above; listing its verified independent claims in field 18 does not create separate rows in this right-level dataset.

FORWARD-CITATION RULES FOR FIELDS 19, 20, 31, AND 32

You must research Google Patents and use only the family-level "Families Citing this family" table.

General rules for all four counts:

* Count citing families, not individual publication rows.

* Each citing family counts once per field.

* Do not use the separate publication-level "Cited by" table for these four fields.

* Do not subtract self-family or self-citation entries shown by Google Patents.

* Apply the relevant cutoff date inclusively: a date equal to the cutoff qualifies.

* If there are no qualifying citing families, return 0.

* If FIRST_HEARING_DATE is unavailable, leave fields 19 and 31 empty rather than inventing a date.

19. "forward_citations_first_hearing_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_HEARING_DATE.

20. "forward_citations_first_judgment_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_JUDGMENT_DATE.

21. "backward_citations_1_family_to_family"

Count only the family-to-family portion of the backward citations shown by Google Patents: the rows under "Family Cites Families" or below the "Family To Family Citations" divider.

* Count the rows in this family-to-family portion only; each displayed cited family counts once.

* Exclude every ordinary patent-citation row in the first table, meaning the rows above the "Family To Family Citations" divider or before the separate "Family Cites Families" heading.

* Do not use the forward-looking "Families Citing this family" section. The direction of the words matters: "Family Cites Families" is backward; "Families Citing this family" is forward.

* Field 21 is therefore field 22 with the first ordinary patent-citation table removed.

* If the family-to-family portion is empty, return 0 even when the first ordinary patent-citation table contains rows.

* Do not impose a first-hearing or first-judgment cutoff.

* If there are no such citations, return 0.

22. "backward_citations_2_all_rows_in_patent_citations"

FIELD 22 IS ALWAYS THE NUMBER IN PARENTHESES IMMEDIATELY BESIDE "Patent Citations": "Patent Citations (N)" means field 22 = N. Extract the parenthesized N, not any number beside another heading.

* Mandatory verification formula: field 22 = number of ordinary rows in the first patent-citation table + field 21.

* The parenthesized N beside "Patent Citations" is the value to extract for field 22. Verify it by counting all backward rows: the ordinary first-table rows plus the family-to-family rows.

* Never use the forward-looking "Cited By (N)" total for field 22. "Cited By" is not a source for either backward field.

* Count the rows in each backward table exactly once within that table. Do not subtract a row merely because a related citation appears elsewhere.

* Do not impose a first-hearing or first-judgment cutoff.

* Field 22 must be greater than or equal to field 21. The difference field 22 - field 21 must equal the number of ordinary rows in the first patent-citation table.

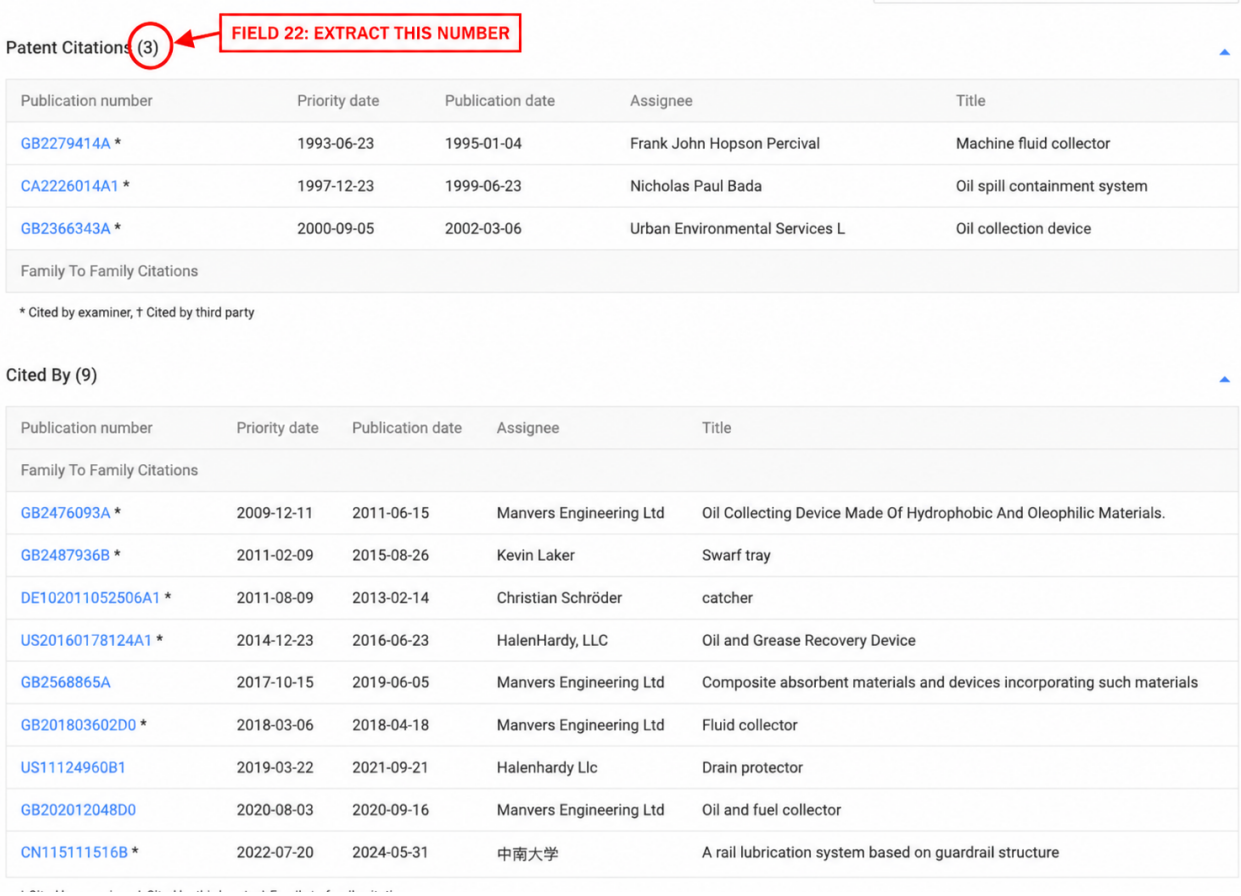
* If both the first ordinary table and the family-to-family portion are empty, return 0.

Mandatory examples:

* If the first table has 3 ordinary rows, the family-to-family portion has 0 rows, and "Cited By (9)" is shown, then field 21 = 0 and field 22 = 3. The number 9 is irrelevant.

MANDATORY VISUAL EXAMPLE — FIELDS 21–22

READ THE DIRECTIONS LITERALLY: FIELD 22 is always the parenthesized number immediately beside "Patent Citations" — "Patent Citations (N)" means field 22 = N. Backward 1 excludes the first ordinary patent-citation table; backward 2 includes the whole backward-citation block. Always verify field 22 >= field 21 and never use "Cited By (N)".



GB2428032B — the first patent-citation table has 3 ordinary backward rows; the family-to-family portion is empty; "Cited By (9)" is forward and irrelevant. Therefore field 21 = 0 and field 22 = 3.

* If the page shows "Patent Citations (61)", the first ordinary table has 1 row, and the family-to-family portion has 60 rows, then field 21 = 60 and field 22 = 61. The parenthesized 61 is field 22; any "Cited By" total is irrelevant.

23. "grant_date"

Date the patent was granted, in YYYY-MM-DD format.

For a pending or refused application that never reached grant, leave "".

24. "priority_date"

Priority date of the patent/application, in YYYY-MM-DD format.

You must research this online.

If unavailable, leave "".

25. "right_level_winner_owner_vs_other"

Code the outcome from the perspective of the patent owner or applicant:

* 0 = owner/applicant wins at right level: every attacked claim in the operative claim set survives every live validity ground asserted and decided against it; or, after amendment, every attacked claim in the final replacement amended claim set that is substantively assessed survives every live ground asserted and decided against it. A fully successful replacement amended claim set is coded 0 even if old/granted claims were deleted, abandoned, conceded invalid, or held invalid.

* 0.5 = mixed or partial outcome among attacked claims: at least one attacked claim is affirmatively upheld

and survives every live validity ground directed to that claim, but the result does not qualify for 0 because at least one other attacked claim or attacked claim set fails or remains unresolved. A claim that wins one ground but loses another, or has a material live attack left undecided, does not count as surviving.

* 1 = adverse side wins at right level: all attacked claims that are substantively decided fail, and no attacked claim is affirmatively upheld through all live grounds directed to it; this remains 1 even if other claims remain in the patent solely because they were not attacked, were not pursued, or were left undecided. Also code 1 for unconditional revocation, refusal/rejection of the application, or an express concession that the whole patent is invalid, unless a final replacement amended claim set is substantively assessed and fully survives.

MANDATORY ATTACKED-CLAIM OUTCOME RULES:

* Determine the outcome claim-by-claim first, then aggregate only across attacked claims.

* Count only claims or claim sets actually attacked on validity/patentability. A claim that was never attacked cannot rescue the owner from 1 and cannot create 0.5.

* An attacked claim counts as a survivor only if the tribunal affirmatively resolves every live validity ground that it actually decides against that claim in the owner/applicant's favour. If any decided ground succeeds against the claim, that claim is lost. If a material attack on that claim is left unresolved or undecided, do not count that claim as a survivor for field 25.

* Do not infer survival from silence. Unattacked, unargued, withdrawn, abandoned, not-maintained, or undecided claims/grounds do not create an owner win for field 25.

* A bare "revoked unless amended" order does not automatically mean 0.5. If all attacked claims decided in the judgment lose and no replacement amended claim set is substantively assessed and maintained in that judgment, code 1.

* AMENDMENT OVERRIDE: if the old/granted claims fail or are deleted, but a replacement amended claim set is then substantively assessed and every attacked amended claim survives every live ground decided against it, code 0, not 0.5. If the attacked amended set is mixed, code 0.5; if all attacked amended claims fail, code 1.

26. "product_or_process_judicial_quotes"

This field must contain raw judicial text, not a human-written summary.

Extract every materially relevant passage in which the judge describes:

* the patented invention, product, process, manufacturing method, material, component, apparatus, device, or technical field;

* a commercial embodiment; or

* a prior-art product, process, or embodiment relevant to the validity dispute.

Rules:

* Quote literally from the judgment.

* Do not paraphrase, summarize, simplify, or add words.

* Prefer fuller technical passages over generic labels.

* Put all relevant passages in the same field.

* Separate passages using the literal text \n—\n. Do not insert actual line breaks.

* Do not add ellipses unless the ellipsis appears in the judgment.

* Escape internal double quotes correctly for CSV.

* If different claim versions are in issue, include the judicial passages that identify or compare those versions. Put the human-written explanation of the version history in field 29, not here.

* Leave empty only if the judgment genuinely contains no relevant product/process/invention/prior-art description.

27. "issue_dictionary"

Return this dictionary exactly:

{

"Lack of novelty":"",

"Lack of inventive step / obviousness":"",

"Insufficiency":"",

"Added matter":"",

"Excluded subject matter / non-patentability":"",

"Priority not entitled":"",

"Lack of industrial applicability":"",

"Others":""

}

For each ground:

* 0 = patent owner/applicant wins on that ground at right level.

* 0.5 = mixed or partial outcome on that ground at right level.

* 1 = patent owner/applicant loses on that ground at right level.

* "" = the ground was not live or not decided in the present judgment.

If a ground is withdrawn, abandoned, not maintained, or not pursued and the court does not decide it, leave that ground as "". If no other ground or validity issue is decided, SKIP the entire row under the withdrawal rule above.

Code every live ground separately. Do not invent grounds.

The dictionary must be substantively consistent with field 25 but need not mechanically equal it.

For a whole-patent invalidity concession, code every expressly identified validity ground. If no specific ground is identified, set "Others" = 1 and leave all named grounds as "".

28. "claimant_is_owner_side"

Return:

* 0 if the claimant/appellant side is the patent owner or applicant side.

* 1 if the claimant/appellant side is not the patent owner or applicant side.

For revocation claims brought by a challenger, this will usually be 1.

For patent-application appeals brought by the applicant, this must be 0.

If unclear, leave "".

29. "summary"

Keep the summary under 500 words.

The first segment, before the literal text \n, must state the patent-level validity outcome clearly, such as:

* "whole patent maintained";

* "whole patent revoked";

* "whole patent conceded invalid";

* "patent partially revoked";

* "patent maintained only in amended or narrowed form";

* "revoked unless amended";

* "patent application allowed"; or

* "patent application refused".

The summary must also:

* identify the case as Type 1, 2, or 3 under the scope filter;

* identify whether the judgment follows trial/substantive final hearing or is a dispositive pre-trial ruling;

* name the patent/application and state the correct historical version considered by the judgment;

* identify the owner/applicant side and adverse validity challenger side;

* explain why the case is within scope;

* if invalidity was conceded, state expressly that the concession counts as the right-level validity outcome, identify its timing, and distinguish it from a judicial merits finding;

* identify the claims and claim versions attacked or decided without treating unargued claims as individually invalid;

* distinguish application, old/granted, previously amended, and proposed-amended claims where relevant;

* explain the result on every live validity ground;

* explain precisely why field 25 is 0.5 if the outcome is partial or mixed;

* explain why fields 7 and 8 contain the correct row-level legal parties when their alignment differs from claimant/defendant status;

* briefly state any Fields 9–10 parent/group collapse that is not obvious from fields 7–8;

* state "insufficient evidence to collapse" if a possible parent relationship could not be verified at ENTITY_REFERENCE_DATE;

* identify relevant prior art and the claims against which it was compared;

* distinguish same-case judgments from background/comparator judgments where useful;

* explain multiple complaint dates if field 3 contains more than one date; and

* contain no URLs.

30. "link"

Raw URL of the judgment only.

31. "forward_citations_first_hearing_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_HEARING_DATE.

Apply all general forward-citation rules stated above field 19.

32. "forward_citations_first_judgment_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_JUDGMENT_DATE.

Apply all general forward-citation rules stated above field 19.

FINAL VALIDATION BEFORE OUTPUT

Before returning the csv code block, verify all of the following:

1. The case belongs to exactly one of Types 1–3.

2. The output contains validity only and no infringement row.

3. Type 3 includes the Comptroller/UKIPO in fields 6, 8, and 10 and also retains every qualifying third-party opponent in those fields.

4. An express whole-patent invalidity concession is included with field 25 = 1, while withdrawal, abandonment, non-maintenance, or agreed revocation without an express invalidity concession is not misclassified.

4A. Every summary judgment or strike-out row passes the mandatory applicant/winner matrix, directly concerns validity, and has been checked for a later same-case validity hearing or judgment.

4B. If revocation or the only live validity argument was withdrawn or abandoned and the court did not decide it, the row is skipped; a withdrawn individual ground is left blank when other grounds were decided.

4C. Field 25 follows the mandatory attacked-claim rule: no attacked claim affirmatively survives all of its live grounds = 1; at least one attacked claim affirmatively survives all of its live grounds, but the attacked set is not a complete win = 0.5; all attacked claims in the operative set win = 0; and a fully successful substantively assessed replacement amended claim set = 0 even if old/granted claims were deleted or invalid.

5. The patent/application version is correct as of the judgment date and is the version actually addressed by the judgment.

6. Fields 9 and 10 use historical entity, parent, ticker, and headquarter information as of FIRST_ JUDGMENT_DATE.

7. Fields 9 and 10 use numeric type/headquarter values and list-valued ticker_at_time.

8. Forward fields 19, 20, 31, and 32 are family-to-family counts and use the correct date field and cutoff date.

9. Field 21 counts only the backward family-to-family portion and excludes every ordinary row in the first patent-citation table; it never uses the forward "Families Citing this family" section.

10. Field 22 is always the parenthesized N immediately beside "Patent Citations". Verify that N equals the number of ordinary first-table rows plus field 21; never use "Cited By (N)".

11. Field 22 is greater than or equal to field 21, and field 22 minus field 21 equals the number of rows in the first ordinary patent-citation table. If either check fails, recount before output.

12. Every row has exactly 32 double-quoted fields.

13. Every CSV record occupies exactly one physical line, with no actual newline or carriage-return character inside any field.

14. The answer contains exactly one csv code block and no text outside it.

D.2 Second prompt: claim-level validity attacks

Validity-Attack Extraction Prompt

Claim-level patent validity rows | eight attack categories | max-parent dependency_depth | prior-art unit IDs

Updated version: dependency_depth is computed by the max-parent rule. Independent claims are 0. A dependent claim has dependency_depth = 1 + the maximum dependency_depth of its direct parent claims. claim_text_expanded must be a standalone technical definition, must preserve alternative dependency paths as alternatives rather than cumulative limitations, and must contain no patent-claim cross-reference. An express concession must be classified under a specific attack category whenever the judgment identifies one; attack category 8 is only the residual category for expressly conceded invalidity with no identified ground in categories 1-7. For novelty and obviousness, each prior-art document/component is output on its own CSV row, while the final field prior_art_unit_id groups consecutive component rows that the judgment treats together as one prior-art unit. Mandatory discrepancy warnings are now a controlling exception to the eight-block-only output rule: every input claim with zero output rows must be named visibly, and every mismatch between issue_dictionary and the judgment must also be reported. Claim-set completeness is checked only within the patent or patents actually submitted in the current request: warn for missing or extra input claims within each submitted patent, but never warn about claims from other patents in the same judgment that were not submitted.

1. Inputs

You will receive three inputs:

One or more claims text files, covering either one patent from the judgment or multiple patents from the judgment.

A BAILII, Casemine, archive, or other judgment text file or judgment URL.

One or more CSV input rows.

Each CSV input row contains, in order:

claims_text_file_link

judgment_url

google_patents_url

target_claim_label

index

issue_dictionary

Example input row:

file:///Users/nghiapham/Documents/claim/text/EP0958447B1_claims.txt

https://www.bailii.org/ew/cases/EWPCC/2006/49.html

https://patents.google.com/patent/EP0958447B1/en

claim 1

10000

'{"Lack of novelty":0,"Lack of inventive step / obviousness":0,"Insufficiency":"","Added matter":"","Excluded subject matter / non-patentability":"","Priority not entitled":"","Lack of industrial applicability":"","Others":"","Conceded invalidity":""}'

The issue_dictionary is to double check. If there is a number rather than null (“”””) it means it had this ground when you did this the first time. It is not ground truth, you might have made a mistake the first time but make sure to check it. Its possible you hallucinated things that didn't exist, it is also possible you forgot a ground of attack.

2. Task

Your task is to extract validity-attack data for the target claim identified in target_claim_label. If multiple CSV input rows are provided, process each target claim independently and preserve the first five fields of each row exactly.

3. Eight possible validity attacks

Section name | Attack type

novelty/ số 1 | Novelty

obviousness/ số 2 | Obviousness / inventive step

added matter/ số 3 | Added matter

lack of industrial applicability/ số 4 | Lack of industrial applicability

excluded subject matter/ số 5 | Excluded subject matter

extension by amendment/ số 6 | Extension by amendment

insufficiency/ số 7 | Insufficiency

concede invalid/ số 8 | Conceded invalidity - unspecified ground only

4. Output format

Output exactly eight sections, in the exact order shown above.

Under each section name, output exactly one CSV code block.

If a validity attack is actually addressed for the target claim, put the relevant CSV row or rows inside that attack's CSV code block.

If a validity attack is not actually addressed for the target claim, still output that attack section and still output a CSV code block, but leave the CSV code block completely empty.

Do not output JSON.

Do not include a CSV header.

Do not include explanation, comments, bullet points, or extra text outside the eight attack section names and the eight CSV code blocks, except for the mandatory discrepancy WARNING lines required by Section 4A.

4A. Mandatory discrepancy warnings - explicit exception to the eight-block rule

Before final output, determine the submitted-patent scope, build an expected-claim inventory for each submitted patent, and build an internal claim-coverage ledger for every CSV input row. For each input target claim, identify every validity ground actually addressed by the judgment and count all output rows across the eight CSV blocks. Check issue_dictionary against the judgment, but treat the judgment as ground truth.

Submitted-patent scope: the current request may contain all target claims for only one patent from a multi-patent judgment, or it may contain all target claims for multiple or all patents in that judgment. Treat only the patent or patents represented by the claims text file(s), Google Patents URL(s), and CSV input rows deliberately supplied in the current request as submitted. Do not assume that every patent mentioned in the judgment has been submitted merely because the judgment discusses it.

Perform the completeness check separately for each submitted patent. If the current request submits only Patent A, check missing and extra claims only for Patent A. Do not issue any missing-claim, missing-row, or missing-patent warning for Patent B, Patent C, or any other patent from the same judgment that was not submitted in the current request. If multiple patents are submitted, repeat the check independently for each one.

For each submitted patent, build expected_claim_inventory from every claim or separately treated claim version of that patent whose validity the judgment actually addresses on at least one of the eight grounds. Distinct granted, amended, proposed-amended, corrected, or other claim versions count as distinct targets only when the judgment treats them separately. Do not include claims of an unsubmitted patent in this inventory.

Compare the CSV input rows for each submitted patent against that patent's expected_claim_inventory. Match by patent identity and the complete target_claim_label, including any version qualifier such as granted, amended, proposed amended, corrected, claim-set label, or dependency qualifier. Do not match merely by bare claim number when the judgment distinguishes versions.

If an expected claim or separately treated claim version of a submitted patent has no CSV input row, output exactly one visible warning in this format: WARNING - MISSING INPUT CLAIM: patent="[patent identifier]"; expected_target_claim_label="[expected target claim label]"; the judgment addresses this claim of a patent submitted in the current request on at least one of the eight validity grounds, but no CSV input row was supplied; no CSV row can be produced for this missing input claim.

If a CSV input row does not correspond to any claim or separately treated claim version in expected_claim_ inventory for its submitted patent, output exactly one visible warning in this format: WARNING - EXTRA INPUT CLAIM: patent="[patent identifier]"; target_claim_label="[target_claim_label]"; index="[index]"; this input claim is not an expected validity target for this submitted patent in the judgment; review or remove this CSV input row.

A duplicate CSV input row for the same submitted patent and the same complete target_claim_label is also extra unless the judgment genuinely treats the rows as distinct claim versions and the labels distinguish those versions. Multiple output rows generated from one legitimate input claim because of different prior-art units or component documents do not make the input claim extra.

The missing/extra completeness check concerns the CSV input claim set, not the complete numbered claim set printed in the patent. A claim that merely exists in the claims text file is not expected unless the judgment actually addresses its validity on at least one of the eight grounds. Conversely, a claim of a submitted patent that the judgment does address on such a ground must be reported as missing if its CSV input row is absent.

If an input target claim produces zero CSV rows because the judgment does not attack that claim on any of the eight validity grounds, output exactly one visible warning for that input row in this format: WARNING - ZERO OUTPUT ROW: target_claim_label="[target_claim_label]"; index="[index]"; the judgment does not attack this target claim on any of the eight validity grounds; no CSV row is output for this input claim.

The ZERO OUTPUT ROW warning is mandatory even when issue_dictionary is entirely empty. An input target claim that produces no CSV row must never silently disappear from the final answer.

If the judgment attacks an input target claim on a ground whose issue_dictionary field is empty, output one visible warning in this format: WARNING - MISSED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary leaves "[attack category]" empty, but the judgment attacks this target claim on that ground; the corresponding CSV row(s) are included.

If issue_dictionary lists a ground for an input target claim but the judgment does not attack that target claim on that ground, output one visible warning in this format: WARNING - HALLUCINATED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary lists "[attack category]", but the judgment does not attack this target claim on that ground; no CSV row is output for that ground.

Apply the missing-input-claim, extra-input-claim, zero-output-row, missed-ground, and hallucinated-ground warning tests as specified above. Output one WARNING line for each applicable discrepancy. Place all WARNING lines in the final answer immediately before novelty/ số 1. These WARNING lines are mandatory and are the only permitted text outside the eight section names and eight CSV code blocks. This exception overrides every instruction in this prompt that says "nothing else", "no extra text", or "only the eight sections". Never hide a required warning in analysis, reasoning, commentary, a CSV block, or any content that may be collapsed.

If none of these discrepancies exists, output no WARNING line and follow the normal eight-section-only format.

5. Row granularity

For each validity attack that is actually addressed for the target claim, output the CSV row or rows required by the rules below.

For novelty and obviousness, first identify each distinct prior-art unit considered by the judgment. A prior-art unit may consist of one document, multiple documents treated together as one combined reference/disclosure/attack, or Common general knowledge. Output one CSV row per component document, not one row per prior-art unit. A one-document unit therefore produces one row; a multi-document unit produces one row for each component, with all of those component rows grouped by the same prior_art_unit_id. Exception: if the patent owner expressly concedes that the target claim lacks novelty or is obvious but the judgment identifies no prior-art unit for that concession, output exactly one row for the target claim under that specific attack and leave prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id empty.

For obviousness/inventiveness, Common general knowledge may be output as its own single-row unit if the judgment actually addresses it as part of the obviousness analysis for the target claim. Give that row its own prior_art_unit_id.

Do not combine Common general knowledge and a named prior-art document in the same CSV row. If the judgment treats them together as one combined obviousness unit, output separate component rows and assign the same prior_art_unit_id; otherwise assign different IDs.

Do not collapse genuinely distinct prior-art units together. If the judgment treats two or more documents together as one prior-art reference, one combined disclosure, or one combined attack, output each component document on its own CSV row and give every component row of that unit the same prior_art_unit_id.

Keep the component rows for one unit consecutive.

Treat multiple documents as the same prior-art unit when they are copies, translations, versions, parts, appendices, or evidentiary records of the same underlying prior-art disclosure and the judgment analyzes them as one unit. In that case, output separate component rows with the same prior_art_unit_id. Do not assign the same ID merely because documents concern the same technology, belong to the same patent family, have the same author, or are cited near each other if the judgment analyzes them as distinct prior art.

For a multi-document prior-art unit, prior_art_name_less_than_20_words must identify only the specific component document represented by that row. Do not concatenate multiple component names with " + ", semicolons, pipes, or any other delimiter. Keep each component name under 20 words.

For novelty and obviousness, assign prior_art_unit_id as a zero-based integer string in judgment order, starting at "0" separately for each target claim and each attack section. All rows belonging to the same prior-art unit must have the same ID and must be consecutive. Different prior-art units must have different IDs. Example: if documents A and B are treated together as one prior-art unit and document C is a separate unit, output A with prior_art_unit_id = "0", B immediately after it with prior_art_unit_id = "0", and C with prior_art_unit_id = "1".

Do not group different validity attacks together. prior_art_unit_id is local to the target claim and attack section and does not link rows across novelty and obviousness.

For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output one CSV row for the attack if that attack is actually addressed for the target claim, unless the judgment clearly treats multiple distinct attacks of the same type separately for the same target claim.

Apply ground-specific classification before concede invalid. If the judgment states that the patent owner concedes, accepts, or no longer contests invalidity for lack of novelty, obviousness, added matter, lack of industrial applicability, excluded subject matter, extension by amendment, or insufficiency, output the row only in the corresponding specific section (1-7), with patent_owner_loses = "1". The fact that the result arose by concession does not create an additional concede invalid row.

Use concede invalid (section 8) only as a residual category: the judgment must expressly concede, accept, or no longer contest the invalidity of the target claim, and must not identify any corresponding ground in sections 1-7 for that concession. Do not infer a concession from amendment, withdrawal, silence, non-appearance, or the general result of the case. Do not infer a specific ground from an unspecified concession.

For every residual concede invalid row, patent_owner_loses must be "1" and prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id must all be empty. Classify separately claim by claim: a ground-specific concession for one target claim does not determine the classification of another target claim.

6. Output row schema

Each output row must begin with the exact same five fields from the input CSV row, in the same order. After those five original fields, append the following extra fields in this exact order:

claim_text_literal

claim_text_expanded

dependency_depth

patent_owner_loses

prior_art_name_less_than_20_words

amended_claim

prior_art_link_status

prior_art_full_text_link

prior_art_unit_id

7. Field definitions

Field | Instruction

claim_text_literal | Output the literal text of the target claim. Prefer the wording as reproduced in the judgment. If the judgment does not reproduce the full claim text, use the claims text file. Do not paraphrase. For amended_claim = "1", the controlling text is the amended, proposed-amended, corrected, or otherwise modified claim version reproduced, approved, or analyzed in the judgment, including any schedule, annex, or order. Use the claims text file only as a fallback for wording that the judgment does not reproduce and only after confirming that the relevant wording was not amended, deleted, renumbered, disclaimed, or replaced. Never substitute superseded wording from the old claim set.

claim_text_expanded | For amended_claim = "1", first reconstruct the controlling amended claim set from the judgment, including all renumbering, changed dependency references, inserted limitations, deletions, substitutions, provisos, and disclaimers. Build claim_text_expanded from that amended claim set and use the judgment's amended wording wherever it is available. Do not trace a direct parent or ancestor by the numbering or dependency graph of a superseded claims text file. The claims text file is only a fallback for the exact wording of a portion that the judgment does not reproduce and that is demonstrably unchanged. Never mix wording or dependency nodes from different claim versions in one expansion. If dependency_depth = "0", this must be exactly identical to claim_text_literal. If dependency_depth is greater than "0", construct a standalone technical definition that preserves every substantive technical limitation of the target claim and every inherited limitation required by each applicable dependency path. First identify the target claim's direct parents from its dependency wording. Recursively expand each direct parent along its own path until an independent claim is reached; do not stop at the immediate parent. Remove or minimally rewrite only patent-claim cross-references and dependency scaffolding such as "Claim 1:", "according to claim 4", "claims 2-4", or "the solution of claim 5". Preserve the substantive wording literally as far as possible and add only the minimum bridging language needed to produce grammatical standalone text. Do not remove the word "claim" when it is part of the substantive technical subject matter, such as an insurance-claim processing system. For alternative direct parents p1, p2, ..., first construct one complete expansion for each path. The result must be logically equivalent to [E(p1) + N(c)] OR [E(p2) + N(c)] OR ..., where E(p) is the recursively expanded parent text and N(c) contains the substantive limitations newly added by the target claim. Present these branches in one field using "or", "any one of the following alternatives", or equivalent standalone wording. Identical wording shared by every branch may be factored out only when the result remains exactly logically equivalent. Never combine branch-specific limitations cumulatively. Every limitation newly added by the target claim must apply to every alternative. Neutral bridging language such as "no additional limitation as to dosage form" may be introduced only when needed to describe a branch that adds no limitation on that dimension; it must not add new technical substance.

dependency_depth | Output the dependency depth using the max-parent rule: independent claim = "0"; dependent claim = 1 + the maximum dependency_depth of all direct parents selected by the target claim's dependency wording. Claims inherited only through a direct parent are ancestors and enter recursively; do not add them again as direct parents. Use "1", "2", "3", etc. Do not cap at "2" unless explicitly instructed. See Section 8 for examples. For amended_claim = "1", identify every direct parent and calculate every depth from the controlling amended claim set, never from the old claims file.

patent_owner_loses | Output "0" if the patent owner wins on this validity attack for this target claim. Output "1" if the patent owner loses. For novelty and obviousness, decide the outcome separately for each prior-art unit or Common general knowledge unit. If one prior-art unit has multiple component rows, repeat the same patent_owner_loses value on every row sharing that prior_art_unit_id because those rows represent one combined attack. A ground-specific concession and a residual concede invalid row both require "1" because the patent owner has expressly conceded, accepted, or ceased to contest invalidity of the target claim on the relevant basis. Do not use outcomes from other claims. Do not guess from the general result of the case.

prior_art_name_less_than_20_words | For novelty and obviousness, output the name of the specific prior-art document/component represented by this row, or "Common general knowledge" for a Common

general knowledge row. If multiple documents form one prior-art unit, do not combine their names in one field; each component must have its own row and its own concise name under 20 words. The shared prior_art_unit_id, not name concatenation, indicates that multiple rows belong to one combined prior-art unit. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For all other validity attacks, output an empty CSV field.

amended_claim | Output "1" if the target claim is an amended claim. Output "0" if the target claim is not an amended claim.

prior_art_link_status | For novelty and obviousness only, assess this field separately for the specific component document represented by the row. Output "2" if prior_art_full_text_link provides full text for that component; output "1" if it provides only an abstract, summary, extract, citation record, bibliographic record, citation in another document, or another identifiable non-full-text description; output "0" if no identifiable source link is available for that component. For a multi-document prior-art unit, do not calculate one aggregate link status for the whole unit; each component row has its own status. For Common general knowledge, output "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field rather than "0". For all other validity attacks, output an empty CSV field.

prior_art_full_text_link | For novelty and obviousness, output the single best available link for the specific prior-art document/component represented by this row. Prefer full text. For patent prior art, prefer a Google Patents full-text link if the patent number is clearly identified. For non-patent prior art, use the best available reliable link if full text is unavailable, including an abstract, summary, bibliographic record, citation record, or another supplied document that clearly identifies the same specific reference. Never place links for two or more prior-art components in one CSV field, whether separated by semicolons, " | ", commas, "+", or any other delimiter. If multiple documents form one prior-art unit, output separate component rows and group them with the same prior_art_unit_id. For Common general knowledge and all non-novelty/non-obviousness attacks, output an empty CSV field. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, also output an empty CSV field.

prior_art_unit_id | For novelty and obviousness only, use this final field to identify which component rows belong to the same prior-art unit. For each target claim and each attack section separately, number distinct prior-art units in judgment order using zero-based integer strings: "0", "1", "2", and so on. All component rows of one multi-document unit must share the same ID and be consecutive. A one-document unit has one row with its own ID. Common general knowledge, when output as its own unit, also receives its own ID. Do not reuse an ID for a different unit within the same target claim and attack section. Numbering may restart at "0" for a different target claim or for the other attack section. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output an empty CSV field.

8. Dependency-depth rules

dependency_depth = "0": target claim is independent; claim_text_expanded must be identical to claim_text_literal.

Mandatory source-of-truth rule for amended claims: when amended_claim = "1", treat the amended or proposed-amended claim set stated in the judgment, schedule, annex, or order as the controlling and closed dependency graph. Every claim number in a dependency reference means the claim with that number in that amended set. NEVER trace the target, a direct parent, or an ancestor through the numbering or dependency wording of the old or granted claims file.

For an amended claim, use the judgment's wording for the target claim and for every amended parent or ancestor wherever that wording is provided. The claims text file may supply only exact wording omitted from the judgment and confirmed to be unchanged. If the judgment changes claim 1 but leaves a dependent claim's own wording unchanged, expand that dependent claim through the amended claim 1, not the old claim 1.

Apply every amendment that changes substantive scope, even if it appears outside the numbered claim text: include inserted disclaimers, provisos, substitutions, deletions, and limitations stated in the judgment, schedule, annex, or order. Do not omit such text merely because it is absent from the claims text file.

Renumbering/version check: before expanding an amended target, make an internal mapping of each amended claim number to its direct parent or parents. If the judgment omits, deletes, consolidates, or renumbers old claims, do not carry the old numbering forward and do not assume that amended claim N corresponds to old claim N.

Mandatory trace check for amended claims with dependency_depth of "2" or greater: write an internal path ledger for every alternative path, listing the target, each direct parent, every ancestor, the claim-set version/ source used for that node, and the calculated depth. Verify that every path ends at an independent claim in the same amended claim set and that no node was silently taken from a superseded claim set. If the amended dependency path cannot be reconstructed unambiguously from the available materials, do not guess or import the old path.

For any dependent claim, first identify its direct parents from the dependency wording in the applicable controlling claim set. A direct parent is a claim expressly selected by the dependency phrase. An earlier claim inherited only through a direct parent is an ancestor, not an additional direct parent. Calculate recursively from that same controlling claim set: dependency_depth(independent claim) = 0; dependency_ depth(dependent claim) = 1 + max(dependency_depth(direct parent) for every direct parent selected by the target claim).

If a claim refers to "any preceding claim", "any one of the preceding claims", "any of claims 1 to N", or similar wording, treat every claim in that referenced range as a direct-parent alternative for the target claim. Ancestors of those alternatives enter only through recursive expansion.

dependency_depth = "1": every expressly listed parent claim is independent, e.g. claim 3 according to claim 1 or claim 2 where both claim 1 and claim 2 are independent.

dependency_depth = "2": at least one expressly listed parent has dependency_depth "1" and no expressly listed parent has a greater depth, e.g. claim 3 according to claim 1 or claim 2 where claim 1 is independent and claim 2 depends on claim 1.

Use "3", "4", etc. for deeper dependency chains. Do not cap at "2" unless explicitly instructed.

Do not use the shortest path where alternative direct parents have different depths. Use the maximum direct-parent depth because claim_text_expanded must account for every dependency alternative selected by the target claim.

Do not create extra rows merely because a dependent claim has alternative dependency paths, unless the judgment actually treats those alternatives differently for validity. Preserve the "or" dependency language in one expanded text where appropriate.

When expanding a dependent target with alternative direct parents, construct each complete path separately before combining the paths with "or". Write claim_text_expanded as one standalone technical definition using wording such as "any one of the following alternatives: ...; ...; or ...". Do not label branches as patent claims and do not leave patent-claim cross-references in the expanded text.

If preserving literal dependency wording would leave a patent-claim cross-reference inside claim_text_ expanded, rewrite only the dependency-reference wording into standalone language. Preserve every substantive technical limitation. The first priority is logical equivalence: no missing inherited limitation, no new technical limitation, and no misleading cumulative combination of alternative branches.

Alternative-path check: if the direct parents are p1, p2, ..., verify that the expanded text means [E(p1) + N(c)] OR [E(p2) + N(c)] OR ..., not E(p1) + E(p2) + ... + N(c).

Worked example: claim 1 is independent; claim 2 depends on claim 1; claims 3 and 4 each depend on claim 2; and claim 5 depends on any one of claims 2-4 while adding a concentration limitation. The direct parents of claim 5 are claims 2, 3, and 4. Its longest path is 5 -> 3 or 4 -> 2 -> 1, so dependency_depth = "3".

For the worked example, the three alternatives are: the composition inherited through claim 2 with no additional dosage-form limitation; the same composition inherited through claim 3 and restricted to a cream; or the same composition inherited through claim 4 and restricted to a gel. The concentration limitation added by claim 5 applies to all three alternatives.

A correct claim_text_expanded for the worked example is: "A pharmaceutical composition containing calcipotriol monohydrate: (1α, 3β, 5Z, 7E, 22E, 24S)-24-cyclopropyl-9,10-secochola-5,7,10(19),22-tetraene-1, 3,24-triol, monohydrate, and having an active-component content of 1-100 μg/g of the composition, where either (i) no further limitation is imposed on the dosage form; (ii) the composition is a cream; or (iii) the composition is a gel."

9. Prior-art link rules

Do not leave prior_art_full_text_link empty merely because the available source is only an abstract, summary, extract, citation record, bibliographic record, or citation in another document.

The link must identify the same specific prior-art reference being compared.

Do not output an unrelated modern page, generic search result, or source that does not clearly identify the prior-art reference.

If a prior-art component is identified but no identifiable link is available for that component, output an empty prior_art_full_text_link and set that row's prior_art_link_status to "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit at all, leave prior_art_name_less_ than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id empty.

When multiple documents form one prior-art unit, output one consecutive CSV row per component document. Put only that component's best available link in prior_art_full_text_link. Never put multiple component links side by side in one field. Give all component rows of the same unit the same prior_art_unit_id.

For a multi-document prior-art unit, determine prior_art_link_status separately on each component row: "2" if full text is available for that component, "1" if only an identifiable non-full-text source is available for that component, and "0" if no identifiable source link is available for that component. Do not downgrade or upgrade one component because of link availability for another component.

10. Important rules

Always output all eight attack sections and all eight CSV code blocks.

Do not omit an attack section merely because the attack is not addressed.

If an attack is not addressed for the target claim, its CSV code block must be empty.

Do not infer an attack merely because it is mentioned elsewhere in the judgment.

The validity attack must be applied to the target claim.

A concession must first be assigned to any specific validity ground expressly identified by the judgment. For example, a claim conceded to be obvious belongs only in obviousness/ sṓ 2, with patent_owner_loses = "1"; do not duplicate it in concede invalid/ sṓ 8. Apply the same rule to every specific ground in sections 1-7.

A concede invalid row requires an express concession or acceptance of invalidity as to the target claim and is permitted only when no specific ground in sections 1-7 is identified for that concession. Do not infer it from a concession concerning another claim, from a proposed amendment, or from the patent-level outcome.

For residual concede invalid, patent_owner_loses must be "1". The prior-art name, link-status, link, and prior_art_unit_id fields must be empty.

Preserve the first five input fields exactly as CSV fields.

Quote every CSV field with double quotes.

Escape internal double quotes by doubling them.

Do not put line breaks inside any CSV field. Each CSV row must be one single physical line.

For an empty CSV field, output "".

The prior_art_link_status field must appear immediately before prior_art_full_text_link, and prior_art_ unit_id must appear immediately after prior_art_full_text_link as the final field of every output row.

For novelty and obviousness rows tied to an identified prior-art unit, prior_art_link_status must be "2", "1", or "0" for the specific component row and prior_art_unit_id must be a zero-based integer string identifying

its unit. For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id must all be empty. For all other validity attacks, prior_art_link_status and prior_art_unit_id must be empty CSV fields.

Do not output null, None, N/A, unknown, or any other placeholder.

Be careful not to output malformed CSV such as quadruple quotes.

The final answer must contain exactly eight section names and exactly eight CSV code blocks. The only permitted additional text is the mandatory discrepancy WARNING lines required by Section 4A.

11. Mandatory internal validation before final output

Create the submitted-patent scope, per-patent expected_claim_inventory, and claim-coverage ledger required by Section 4A before drafting the final answer. The ledger must contain one entry for every CSV input row, keyed by patent identity, target_claim_label, and index.

For each submitted patent, compare expected_claim_inventory against the CSV input rows for that patent. Emit one MISSING INPUT CLAIM warning for every expected claim or separately treated claim version with no input row, and one EXTRA INPUT CLAIM warning for every supplied row that does not correspond to the expected inventory or is an unjustified duplicate.

Recheck the patent-scope boundary before emitting completeness warnings. Never emit a missing warning for a claim belonging only to a patent from the same judgment that was not submitted in the current request.

For every input row, count its CSV rows across all eight attack blocks. If the count is zero, verify that the final answer contains the mandatory ZERO OUTPUT ROW warning for that input row, even when issue_dictionary is entirely empty.

Compare the judgment-determined attack grounds with issue_dictionary for every input target claim. Emit a MISSED GROUND warning for every judgment-only ground and a HALLUCINATED GROUND warning for every issue_dictionary-only ground.

Do not finalize while any submitted patent's expected claim is missing without a MISSING INPUT CLAIM warning, any extra input claim lacks an EXTRA INPUT CLAIM warning, any input target claim has silently disappeared, or any other required discrepancy warning is absent. Required WARNING lines must be visible immediately before novelty/ sṓ 1 in the final answer, not merely mentioned in analysis or commentary.

This validation checklist is internal only. Do not output the checklist. Output any mandatory discrepancy WARNING lines required by Section 4A, followed by exactly the required eight sections and eight CSV code blocks.

Reconstruct the dependency chain for every target claim before drafting rows. For an unamended claim, use the applicable claims text. For amended_claim = "1", use the controlling amended claim set from the judgment, schedule, annex, or order; use the claims text file only for wording that is omitted and confirmed unchanged.

For every amended_claim = "1" row, verify that the target, every direct parent, and every ancestor belong to the same controlling amended claim set. Fail the row if any dependency node or claim number was taken from the old or granted claim set merely because it was available in the claims text file.

For every amended claim with dependency_depth of "2" or greater, inspect the complete internal path ledger node by node and branch by branch. Verify the amended numbering, direct-parent choices, maximum depth, independent endpoint, and source/version for every node before accepting claim_text_expanded.

For every amended claim, compare claim_text_literal and claim_text_expanded against the judgment's amendment language and verify that all inserted limitations, disclaimers, provisos, substitutions, deletions, and renumbering effects are reflected. Do not allow old wording to overwrite or dilute amended wording.

If dependency_depth = "0", verify claim_text_expanded is exactly identical to claim_text_literal.

If dependency_depth > "0", verify claim_text_expanded is not identical to claim_text_literal.

If dependency_depth > "0", verify claim_text_expanded preserves every substantive technical limitation of

the target dependent claim and incorporates every inherited limitation required by each applicable dependency path. Patent-claim cross-references and dependency scaffolding may and must be removed or minimally rewritten.

If dependency_depth > "0", verify claim_text_expanded contains no patent-claim cross-reference or dependency scaffolding such as "claim 1", "claims 2-4", or "according to claim 3". Do not reject a row merely because the word "claim" forms part of the substantive technical subject matter.

If the target has alternative dependency paths, verify claim_text_expanded preserves the alternative logic using “or”, “any one of the following”, or equivalent standalone wording, rather than stacking all parent alternatives cumulatively.

For multi-level dependencies, do not stop at the immediate parent claim. Continue until an independent claim is reached.

Verify the numeric dependency_depth follows the recursive max-parent rule: independent claims are 0; dependent claims equal 1 plus the maximum dependency_depth of every direct parent selected by the dependency wording.

If dependency_depth and claim_text_expanded are inconsistent, the row is invalid and must be fixed before output.

Verify amended_claim is correct for the target claim. If amended_claim = "1", also verify that claim_text_literal, claim_text_expanded, dependency_depth, and every dependency path were derived from the same controlling amended claim set rather than the old claims file.

Verify patent_owner_loses is decided separately for each attack and each prior-art unit. If a unit has multiple component rows, verify every row sharing its prior_art_unit_id repeats the same patent_owner_loses value.

For novelty and obviousness, verify that every specific prior-art document/component has its own CSV row. If multiple documents are treated together as one prior-art unit, verify that their rows are consecutive and share the same prior_art_unit_id. Verify that genuinely distinct prior-art units have different IDs.

For every multi-document prior-art unit, verify that no component names or links are concatenated in one field; each row contains only one component name, one component-specific prior_art_link_status, and at most one best link. Verify that all component rows share the same zero-based prior_art_unit_id, are consecutive, and that IDs increase in judgment order starting at "0" for each target claim and attack section.

For every express concession, verify first whether the judgment identifies a specific ground in sections 1-7. If it does, verify that the row appears only in that specific section, patent_owner_loses is "1", and no duplicate row appears in concede invalid/ sṓ 8.

For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, verify that exactly one row is output for the target claim in the specific section and that prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id are all empty; do not invent a prior-art name, unit ID, or use link status "0".

For every concede invalid row, verify that the concession is express, applies to the target claim, identifies no ground in sections 1-7, and has patent_owner_loses = "1".

For every residual concede invalid row, verify that prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id are empty.

Verify no CSV row contains line breaks, every field is double-quoted, internal double quotes are escaped by doubling them, and no malformed quadruple quotes appear.

12. Required final answer skeleton

[zero or more mandatory WARNING lines under Section 4A, only when applicable]

novelty/ sṓ 1

“‘csv

“‘

obviousness/ số 2

“‘csv

“‘

added matter/ số 3

“‘csv

“‘

lack of industrial applicability/ số 4

“‘csv

“‘

excluded subject matter/ số 5

“‘csv

“‘

extension by amendment/ số 6

“‘csv

“‘

insufficiency/ số 7

“‘csv

“‘

concede invalid/ số 8

“‘csv

“‘

Final check: make sure claim_expanded is the same for the same claims

Final check: with the exception of novelty and obviousness, each claim only has 1 row for other fields.

Final check: if the judgment copies the exact prior art then it is also type 2

Final check: same prior arts must have the same link. For example link for prior art A used for claim 1 must have the same link as prior art A used for claim 2

Final check: verify the submitted-patent scope, each submitted patent's expected_claim_inventory, and the claim-coverage ledger. Every expected validity target of every submitted patent must have an input row or a visible MISSING INPUT CLAIM warning; every extra input claim must have an EXTRA INPUT CLAIM warning. Do not check or warn about missing claims from patents in the judgment that were not submitted in the current request. Every input target claim must either produce at least one CSV row or a visible ZERO OUTPUT ROW warning. Every ground mismatch between issue_dictionary and the judgment must produce a visible MISSED GROUND or HALLUCINATED GROUND warning. Place these warnings immediately before novelty/ số 1; this requirement overrides every "nothing else" rule in this prompt.

D.3 Third prompt: patent text extraction

Patent extraction

attention: before extracting make sure this is the correct patent in the case

You are given a Google Patents link or a patent PDF. Extract only the English patent text and create exactly two downloadable UTF-8 plain text files.

IMPORTANT SOURCE RULE:

- If a Google Patents link is provided, use the Google Patents HTML text as the primary source.

- Use the PDF only as a fallback when the HTML text is missing, incomplete, or clearly corrupted.

- Use OCR only if both HTML text and PDF text layer are unavailable or unusable.

- Do not OCR chemical structures, formula images, drawings, or tables into garbage text.

File naming rule:

- Infer the patent publication number from the patent, e.g. EP0383425B1.

- Create exactly:

1. EP0383425B1_description.txt

2. EP0383425B1_claims.txt

Do not create JSON, CSV, DOCX, or PDF. Create only .txt files.

1. DESCRIPTION FILE

Put only the English Description section into the description file.

Start at the heading "Description", "Specification", or the equivalent English heading that begins the substantive patent description.

Stop immediately before the heading "Claims".

Do not include:

- front-page bibliographic metadata,

- abstract-only front matter,

- publication number repeated in headers,

- page numbers,

- column line numbers,

- OCR artifacts,

- drawings-only pages,

- table-only pages,

- German/French claim translations,

- "Patentansprüche",

- "Revendications".

2. CLAIMS FILE

Put only the English Claims section into the claims file.

Start at the heading "Claims".

Stop immediately before any non-English claim heading, such as "Patentansprüche" or "Revendications", or before drawings-only/table-only pages.

Preserve claim numbers exactly.

Preserve claim dependencies exactly.

Keep each claim separated by a blank line.

3. TABLES

Do not reproduce tables.

Do not convert tables into plain text.

Do not include table image content.

Keep table references in the description text, e.g. "Table I shows..." or "as shown in Table II".

Keep table captions/titles if they appear as part of the English description text.

For table-only pages or table blocks, insert only:

[TABLES_NOT_INCLUDED]

Do not describe the tables from visual inspection unless the surrounding description text itself describes them.

4. FORMULAS / CHEMICAL STRUCTURES / EQUATIONS

The goal is to produce clean NLP-friendly patent text for embedding models. Do not preserve chemical-structure garbage, layout-based formulas, or OCR formula fragments.

Keep ordinary machine-readable chemical and technical text, including:

- chemical names, material names, compound names, polymer names, and reagent names;

- simple molecular formulas only when they appear inline as ordinary text, e.g. H2O, CO2, NaCl, HCl;

- substituent definitions, e.g. R is hydrogen, alkyl or aryl;

- chemical class or range expressions, e.g. C1-C6 alkyl, C1-C4 alcohol, halogen, aryl, alkoxy;

- composition ranges, ratios, concentrations, percentages, units, temperatures, pressures, pH ranges, and reaction conditions;

- simple arithmetic equations or inequalities using +, -, ×, /, =, <, >, ≤, ≥, powers, percentages, ratios, ranges, and units.

Remove chemical formulas or structures that are not useful as ordinary text embeddings, including:

- chemical structure drawings;

- Markush structures;

- reaction schemes;

- structural formulas represented by bonds, rings, branches, or diagram layout;

- isolated chemical fragments such as =C—C—C=, =C—C—H, —CH2—CH2—, unless they occur as clean inline text inside a normal sentence;

- multi-line chemical formulas;

- formula images;

- OCR-corrupted chemical formulas;

- ASCII-art or layout-based chemical structures.

Replacement rules for chemical content:

- If a chemical structure, Markush formula, reaction scheme, or structural formula is removed, write:

[CHEMICAL_STRUCTURE_REMOVED]

- If a formula is OCR-corrupted or uncertain, write:

[FORMULA_OCR_REMOVED]

- If several adjacent chemical-structure lines belong to the same formula block, replace the whole block with a single placeholder only.

- Do not convert chemical structure images into LaTeX.

- Do not attempt to reconstruct chemical structures from visual inspection.

- Do not include ASCII-art formula fragments.

- Do not invent missing formula content.

For mathematical formulas:

- Keep simple arithmetic equations, ratios, powers, inequalities, and simple systems of equations if they are clean and machine-readable.

- Keep simple expressions such as x + y = z, a^2 + b^2 = c^2, A:B = 1:3, or $0.1 \leq x \leq 5$.

- Remove complex mathematical notation, including integrals, derivatives, summations/products, vectors, matrices, tensors, and multi-line displayed equations that cannot be represented cleanly as simple text.

- Replace removed complex mathematical formulas with:

[COMPLEX_FORMULA_REMOVED]

5. FIGURES / DRAWINGS

Do not include image files or attempt to reproduce drawings.

Keep figure references in the description text, e.g. "Figure 1 shows..."

Keep figure captions if they appear in the English description.

For drawings-only pages or drawing-only blocks, insert only:

[DRAWINGS_NOT_INCLUDED]

Do not describe the figures from visual inspection unless the description text itself describes them.

6. OCR / SCANNED PDF RULES

The PDF may be image-only or scanned.

Use OCR/vision only if needed.

Preserve the correct reading order, especially for two-column patent pages.

Fix broken line wraps and obvious OCR hyphenation, for example:

"con-\ntact" -> "contact"

"polymerisa-\ntion" -> "polymerisation"

But do not change technical terms, claim scope, numbers, reference numerals, units, or legal wording.

If ordinary text is too blurry or uncertain, mark it with:

[OCR_UNCLEAR: ...]

Do not guess.

Do not use OCR to transcribe tables, chemical structures, complex formulas, or drawings into noisy text.

7. CLEANING RULES

Remove repeated headers, footers, page numbers, publication numbers, and column numbers.

Normalize whitespace.

Keep paragraph order.

Keep reference numerals in parentheses, e.g. (20,22;50,51).

Do not summarize.

Do not paraphrase.

Do not modernize spelling.

Do not translate.

8. OUTPUT RULES

Create the two .txt files as downloadable files.

Do not paste the full extracted contents into the chat unless asked.

After creating the files, report only:

- the two filenames,
- the download links,
- the file path name in the form "text/filename",
- whether Google Patents HTML text was used,
- whether PDF text layer was used,
- whether OCR was needed,
- whether any [OCR_UNCLEAR] markers were inserted,
- whether any tables/formulas/figures were found and how they were handled.

Give the filenames in an easy way for me to copy-paste.

Here is a guidance

After giving me the links for download, give me the file path names in the form:

text/filename

you need 2 fields, both in csv

first field for description

second field for claim file

Final check. the name of the file should match the right version.

D.4 Fourth prompt: prior-art extraction

Prior Art Extraction - Single-Row Input / Eight-Field Output

Purpose: Process one prior-art dataset row, resolve the actual prior-art document from the judgment URL, create the required text files, and return one eight-field CSV row.

If cant find patent then say skip in all 4 fields that you have to fill in. But try your best.

INPUT FORMAT

You are given exactly one CSV row containing exactly four fields, in this order:

1. index
2. judgment_url
3. prior_art_name_less_than_20_words
4. prior_art_full_text_link

index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link

Example:

17,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070,Oshlack,https://patents.google.com/patent/EP0253104A1/en

The judgment_url is the judgment in which the prior art is discussed. The prior_art_full_text_link is only a candidate source and may be correct, incorrect, incomplete, a different patent-family member, a later or earlier patent publication, a non-patent document, or a placeholder source.

Do not expect the full judgment text to be pasted into the prompt. Retrieve and read the judgment directly from judgment_url. The judgment is the primary authority for identifying the prior-art document actually

considered by the court.

IMMUTABLE INPUT FIELDS RULE

Fields 1-4 are immutable. The output row MUST begin with the exact original input row string, copied verbatim, character-for-character. Do not parse and reserialize the first four fields in a way that changes their textual representation. Construct the final row by taking the original four-field input row exactly as supplied and appending only: ,field5,field6,field7,field8.

Do not correct, normalize, canonicalize, trim, rewrite, replace, translate, re-quote, unquote, change capitalization, alter URL paths, add or remove trailing slashes, change language suffixes, or otherwise modify any character in fields 1-4, even when a field is inaccurate.

Any corrected or resolved prior-art URL belongs ONLY in field 8 (resolved_prior_art_link). If field 4 points to the wrong patent publication or uses a non-/en Google Patents URL, field 4 must still remain exactly unchanged; only field 8 may contain the corrected canonical URL.

1. TASK OVERVIEW

For the single supplied CSV row:

1. Read the judgment at judgment_url.
2. Identify the exact prior-art reference represented by prior_art_name_less_than_20_words.
3. Determine whether the prior art is a patent document or a non-patent document.
4. Resolve the correct full-text source.
5. Extract the required text.
6. Create the required downloadable UTF-8 .txt file or files.
7. Output exactly one CSV code block containing exactly one line and exactly eight fields.

The first four output fields are IMMUTABLE. They must be copied from the input row verbatim, character-for-character, with no normalization or reserialization. Append exactly four new fields:

1. description_file_path
2. claims_file_path
3. patent_full_file_path
4. resolved_prior_art_link

index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link,description_file_path,claims_file_path,patent_full_file_path,resolved_prior_art_link

Do not output a CSV header.

2. IDENTIFY THE PRIOR ART FROM THE JUDGMENT

Read the judgment before relying on the candidate prior-art link. Find the passages discussing the prior art named in prior_art_name_less_than_20_words.

Determine as precisely as possible:

the prior-art name or shorthand used by the court;

whether it is a patent or non-patent reference;

the document actually relied upon by the court;

publication number, application number, or patent number where applicable;

author, title, inventor, applicant, proprietor, or other identifying information;

jurisdiction and patent-family relationship where applicable;

publication or version information;

the date of judgment.

The judgment is the primary authority for prior-art identity. The candidate prior_art_full_text_link is only a starting point and must not be assumed to be correct.

3. PATENT PRIOR ART

If the prior art is a patent document, resolve the exact patent publication relied upon by the judgment.

The candidate Google Patents link may point to:

the wrong patent-family member;

the wrong jurisdiction;

a granted patent instead of the published application relied upon in the judgment;

an application instead of a granted patent discussed by the judgment;

a later publication, correction, republication, grant, or amended version;

a document that did not yet exist on the judgment date;

a document with a similar title, inventor, applicant, or priority number but a different publication identity.

3.1 Resolution hierarchy

Express publication number

If the judgment expressly gives a publication number and kind code, use that exact publication. Example: if the judgment identifies EP1234567A1, use EP1234567A1. Do not replace it with a later B1 merely because that version is now easier to access.

Publication number without kind code

If the judgment identifies a patent or application number but omits the publication kind code, use bibliographic history and the judgment context to determine the relevant published version.

Expressly discussed version

If the judgment expressly discusses the granted patent, use the granted version if it existed by the judgment date. If it discusses the published application, use that publication instead of a later grant.

Judgment-date restriction

Never resolve to a publication, grant, correction, amendment, or republication that first became public after the judgment date.

Latest relevant pre-judgment version

If the judgment does not expressly specify the publication version, select the latest relevant public version that existed by the judgment date, belongs to the correct patent family, corresponds to the document actually discussed by the judgment, and does not materially change the identity of the relied-upon prior art.

Jurisdiction

Prefer the jurisdiction expressly used in the judgment. Do not replace an EP publication with WO, GB with US, WO with a national-stage patent, or one national publication with another merely because another family member has cleaner text.

Family-equivalent fallback

A family-equivalent English patent may be used as a text source only when the resolved publication has no usable English text. The final resolved link and filenames must still identify the publication supported by the judgment.

Ambiguity

Do not invent a publication identity. If exact resolution is genuinely uncertain, choose the best-supported version only when the evidence strongly favors it.

4. PATENT SOURCE PRIORITY

For patent prior art, use sources in this order:

1. Google Patents HTML for the resolved publication.

2. PDF text layer for the resolved publication.

3. Official patent-office HTML or text.

4. English family-equivalent publication, only as a justified fallback.

5. OCR or visual transcription only as a last resort.

For patent prior art, resolved_prior_art_link (field 8) MUST be the canonical Google Patents English URL for the resolved publication, and MUST end in /en:

https://patents.google.com/patent/PUBLICATION_NUMBER/en

Patent field-8 rule: always report exactly https://patents.google.com/patent/PUBLICATION_NUMBER/en for the resolved patent publication. Do not use /zh, /de, /fr, another language suffix, a bare Google Patents URL without /en, a PDF URL, or an alternative patent-office URL in field 8 when a canonical Google Patents page exists.

This rule applies to field 8 only. Field 4 remains immutable and must never be changed to match field 8.

5. PATENT OUTPUT FILES

If the prior art is a patent, create exactly three downloadable UTF-8 plain-text files:

INDEX-PUBLICATION_NUMBER_description.txt

INDEX-PUBLICATION_NUMBER_claims.txt

INDEX-PUBLICATION_NUMBER_patent_full.txt

Example for input index 17:

17-EP0253104A1_description.txt

17-EP0253104A1_claims.txt

17-EP0253104A1_patent_full.txt

The dataset paths placed in the output CSV must be:

text_prior_art/17-EP0253104A1_description.txt

text_prior_art/17-EP0253104A1_claims.txt

text_prior_art/17-EP0253104A1_patent_full.txt

The index prefix is mandatory. Use the resolved publication number, not automatically the publication number appearing in the candidate link.

6. PATENT DESCRIPTION FILE

The Description file must contain only the complete cleaned English Description. Start at Description, Specification, or the equivalent heading beginning the substantive patent description. Stop immediately before Claims.

Do not include:

front-page bibliographic metadata or abstract-only front matter;

search reports or examination reports;

headers, footers, page numbers, repeated patent titles, repeated publication numbers, or column line numbers;

non-English duplicate sections;

judgment text or prior-art URLs.

Do not summarize or paraphrase.

7. PATENT CLAIMS FILE

The Claims file must contain only the complete cleaned English Claims section. Preserve claim numbers, dependencies, ordering, and substantive wording.

Do not silently substitute:

later amended claims;

later granted claims;

later corrected claims;

claims from another family member;

for the claims belonging to the resolved publication. Do not summarize or paraphrase.

8. PATENT FULL FILE

The patent-full file must contain the complete Description followed by the complete Claims, using this structure:

Description

[complete cleaned English Description]

Claims

[complete cleaned English Claims]

It must contain the same substantive patent text as the two individual files. Do not add metadata, summaries, judgment text, source reports, URLs, or commentary.

9. NON-PATENT PRIOR ART

If the judgment shows that the prior art is not a patent, do not create Description or Claims files.

Instead:

1. Resolve the actual prior-art document.

2. Locate the best available full-text source.

3. Create exactly one downloadable UTF-8 plain-text full-text file.

4. Leave description_file_path and claims_file_path empty.

5. Populate patent_full_file_path with the non-patent full-text file path.

6. Populate resolved_prior_art_link with the resolved source URL.

The non-patent full-text file should be named:

INDEX-SHORT_IDENTIFIER_full.txt

Examples:

150-Roy_full.txt

151-Kabachnik_1978_full.txt

152-Frontline_spray_full.txt

Use a short filename-safe identifier derived from the prior-art name. Replace spaces with underscores, remove filename-invalid characters, and do not make the identifier unnecessarily long.

The dataset path must be:

text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt

For non-patent prior art, the last four output fields therefore have this pattern:

,,text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt,RESOLVED_SOURCE_URL

Example complete row:

150,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213,Roy,https://pubmed.ncbi.nlm.nih.gov/8742940/,,,text_prior_art/150-Roy_full.txt,https://pubmed.ncbi.nlm.nih.gov/8742940/

10. NON-PATENT FULL-TEXT EXTRACTION

For non-patent prior art, extract the substantive text of the document itself. Preserve the original substantive wording. Do not summarize or paraphrase.

Remove:

website navigation, menus, advertisements, and unrelated page material;

repeated headers and footers;

citation-manager UI, download controls, and cookie notices;

obvious HTML extraction artifacts.

Where reasonably available, preserve the title, authors, section headings, and substantive body text. Do not include the judgment merely because the candidate link points to the judgment. If the candidate link is only a placeholder and the actual document can be identified from the judgment, resolve and use the actual document.

11. TABLES

Do not reproduce tables as noisy plain text. For a continuous table block, insert:

[TABLES_NOT_INCLUDED]

Use one placeholder per continuous table block. Keep ordinary textual references to tables.

12. CHEMICAL STRUCTURES AND FORMULAS

Keep ordinary machine-readable technical text, including chemical names, compound names, material names, simple molecular formulas, concentrations, percentages, units, ranges, and simple equations.

For chemical structures, Markush structures, reaction schemes, or diagrammatic formulas that cannot be represented reliably as ordinary text, use:

[CHEMICAL_STRUCTURE_REMOVED]

For OCR-corrupted formulas, use:

[FORMULA_OCR_REMOVED]

For complex mathematical formulas that cannot be cleanly represented, use:

[COMPLEX_FORMULA_REMOVED]

Do not invent missing formula content.

13. FIGURES AND DRAWINGS

Do not reproduce drawings or figure images. Keep textual references to figures. For drawing-only material, use:

[DRAWINGS_NOT_INCLUDED]

Do not infer technical details from drawings.

14. OCR

Use OCR only as a last resort. Fix obvious OCR line-wrap and hyphenation artifacts without changing substantive wording. If ordinary text cannot be read reliably, use:

[OCR_UNCLEAR: ...]

Do not guess.

15. CLEANING RULES

Normalize whitespace and obvious extraction artifacts. Preserve paragraph order, technical terminology, legal wording, numbers, units, claim wording, spelling conventions, and useful document structure.

Do not:

summarize;

paraphrase;

translate;

modernize;

rewrite;

add analysis;

change terminology to match the judgment.

16. VALIDATION

Immutable fields validation

Before output, compare the literal prefix through field 4 against the supplied input row and verify that every character is identical. The output prefix through field 4 must be exactly the original input row. Only fields 5-8 may be newly generated.

For patent prior art

The resolved patent is the prior art discussed in the judgment.

The publication belongs to the correct patent family.

The relevant version existed on or before the judgment date.

No improper later grant or correction was substituted.

The jurisdiction matches the judgment unless a justified text fallback was necessary.

resolved_prior_art_link identifies the selected publication and, for patent prior art, is exactly the canonical Google Patents English URL ending in /en.

All three filenames contain the correct index and resolved publication number.

Description and Claims belong to the same resolved version.

The patent-full file contains the same substantive Description and Claims as the two separate files.

For non-patent prior art

The resolved document corresponds to the prior art discussed in the judgment.

The full-text source represents the actual document rather than merely a secondary citation when the primary document is available.

The filename begins with the supplied index.

Exactly one full-text file is created.

description_file_path and claims_file_path are empty. For non-patent prior art, resolved_prior_art_link is the resolved non-patent source URL; the /en requirement does not apply.

17. FINAL RESPONSE

Provide the downloadable file links first.

For patent prior art, provide exactly three downloadable files. For non-patent prior art, provide exactly one downloadable full-text file.

After the download link or links, output exactly one CSV code block. The CSV code block must:

NON-NEGOTIABLE OUTPUT CONSTRUCTION: use the exact original input row as the literal prefix of the output row, then append fields 5-8. Never regenerate fields 1-4 from parsed values.

use the csv language;

contain exactly one line;

contain exactly eight fields;

contain no header;

contain no commentary;

contain no numbering;

begin with the exact original four-field input row verbatim, character-for-character; do not reconstruct, normalize, or reserialize fields 1-4;

append description_file_path, claims_file_path, patent_full_file_path, and resolved_prior_art_link as fields 5-8; for patent prior art, field 8 must be the canonical Google Patents URL ending in /en.

Patent example

17,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070,Oshlack,https://patents.google.com/patent/EP0253104A1/en,text_prior_art/17-EP0253104A1_description.txt,text_prior_art/17-EP0253104A1_claims.txt,text_prior_art/17-EP0253104A1_patent_full.txt,https://patents.google.com/patent/EP0253104A1/en

Non-patent example

150,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213,Roy,https://pubmed.ncbi.nlm.nih.gov/8742940/,,,text_prior_art/150-Roy_full.txt,https://pubmed.ncbi.nlm.nih.gov/8742940/

The final CSV code block must be the last thing in the response. Do not output any additional code block after it.

D.5 Extra prompt

IP JUDGMENT EXTRACTION PROMPT — STEP 2 — CLAIM-LEVEL VALIDITY ROWS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

You are converting patent-level validity rows produced by Step 1 into claim-level validity rows for UK patent judgments.

This prompt produces only the validity dataset. Do not produce infringement rows.

CORE PURPOSE

Step 1 has already extracted and verified the reusable case-level and patent-level information, including dates, judges, parties, lawyers, historical entity classifications, parents, tickers, headquarters, patent metadata, citation counts, and source URLs.

Do not research, reconstruct, recode, standardize, or otherwise change those reusable fields in Step 2.

Your tasks in Step 2 are only to:

1. identify every patent claim or separately assessed claim version that was actually attacked, decided, expressly conceded invalid, or otherwise given a clean claim-level validity outcome;

2. create one output row for each qualifying claim unit;

3. copy every reusable field exactly from the corresponding Step-1 patent-level row;

4. complete the claim-specific fields 18, 25, 27, and 29; and

5. keep every overall claim-level outcome in field 25 strictly binary: "0" or "1".

This prompt does not create a claim_index. Any stable claim_index needed for later matching must be assigned outside this prompt by a persistent registry or deterministic data-processing script. Never invent a

claim_index.

INPUTS

You will receive:

1. A judgment text file or judgment URL.
2. The claims text file for each patent or patent application covered by the supplied Step-1 row or rows.
3. One or more Step-1 patent-level CSV rows.

Each Step-1 patent-level CSV row has exactly 32 fields in this order:

1. date_of_judgment
2. date_of_hearing
3. date_filing_the_complaint
4. judges
5. claimants
6. defendants
7. right_owner_or_applicant_side
8. other_side_or_validity_challenger_side
9. right_owner_or_applicant_party_type_ticker_and_headquarter
10. other_side_or_validity_challenger_party_type_ticker_and_headquarter
11. claimant_lawyers
12. defendant_lawyers
13. stage
14. right_type
15. right_name
16. right_short_description
17. google_patent_url
18. patent_unit
19. forward_citations_first_hearing_priority_date
20. forward_citations_first_judgment_priority_date
21. backward_citations_1_family_to_family
22. backward_citations_2_all_rows_in_patent_citations
23. grant_date
24. priority_date
25. right_level_winner_owner_vs_other
26. product_or_process_judicial_quotes
27. issue_dictionary
28. claimant_is_owner_side
29. summary
30. link
31. forward_citations_first_hearing_publication_date
32. forward_citations_first_judgment_publication_date

COPY-EXACTLY RULE

For every claim-level row derived from a Step-1 patent-level row, copy these fields exactly and character-for-character from that Step-1 row:

Fields 1–17, fields 19–24, field 26, field 28, and fields 30–32.

Do not change spelling, capitalization, ordering, lists, dictionaries, dates, names, entity coding, quotation content, citation counts, URLs, or CSV escaping in those copied fields.

Only fields 18, 25, 27, and 29 may be newly completed or rewritten in Step 2.

In the Step-2 output, these four positions change to claim-level meanings:

* field 18 = claim_unit;

* field 25 = claim_level_winner_owner_vs_other;

* field 27 = the nine-key claim-level issue_dictionary defined below; and

* field 29 = the claim-level summary defined below.

If several claim-level rows come from the same Step-1 patent-level row, repeat the copied fields exactly in every such row.

ABSOLUTE OUTPUT RESTRICTIONS

* Output exactly one fenced code block labelled csv.

* Put every output row for every patent, claim, and claim version inside the same csv code block.

* Do not include a header.

* Do not include notes, explanations, citations, footnotes, source references, or any text before or after the csv code block.

* Every output row must contain exactly 32 fields.

* Wrap every field in double quotes.

* Escape every internal double quote by doubling it.

* Do not produce malformed CSV such as quadruple quotes.

* Lists and dictionaries copied from Step 1 must retain their existing valid CSV escaping.

* No CSV field may contain an actual carriage return or newline.

* Each CSV row must occupy exactly one physical line.

* If field 29 needs several statements, put them on the same physical line separated by spaces.

* Preserve literal backslash-n sequences already contained in copied fields; do not convert them into actual line breaks.

* Do not output JSON outside a CSV field.

* Do not hallucinate.

* If no qualifying claim-level row exists for a supplied patent-level row, do not create a claim row for it.

* If no qualifying claim-level row exists for any supplied patent-level row, output one empty csv code block.

CLAIM-LEVEL ROW UNIT — MANDATORY

Each output row must correspond to:

* one patent or patent application;

* exactly one identifiable claim number;

* exactly one identifiable claim version or other outcome-homogeneous claim unit;

* the owner/applicant/enforcement side already recorded in the Step-1 row; and

* the adverse validity-challenger side already recorded in the Step-1 row.

Create a row only for a claim or claim version that meets one of the following conditions:

* at least one validity or patentability ground was raised against that claim or claim version, AND the judgment adjudicates its outcome on that ground;

* its invalidity was expressly conceded or expressly no longer contested, subject to the concession rules below; or

* it is a corresponding independent claim in another patent to which the judgment applies its validity determination for a representative patent. Assign the same outcome to that corresponding claim.

Do not create a row for a claim merely because it:

* appears in the patent;

* is mentioned as dependency or technical background;

* is relevant only to infringement;

* is referred to only as an unargued claim;

* is withdrawn, abandoned, not maintained, or not separately defended without a clean validity outcome; or

* belongs to a patent that was revoked when the judgment does not identify or necessarily decide that claim at claim level.

If claims 1–3 are separately identifiable but decided together, output three rows: one for claim 1, one for claim 2, and one for claim 3.

Never write a grouped unit such as "claims 1–3" when the individual claim numbers can be identified.

If the judgment genuinely refers only to "the asserted claims", "the relevant claims", or equivalent language and individual claim numbers cannot be identified after checking the judgment and claims text, output one fallback row with field 18 = "attacked claims not individually identified" and explain the limitation in field 29.

Do not create duplicate rows for the same claim unit merely because several validity grounds were argued. All grounds for the same outcome-homogeneous claim unit belong in field 27 of the same row.

OLD / GRANTED / AMENDED CLAIM VERSIONS

Treat each separately assessed claim version as a distinct claim unit.

* An old/granted claim and an amended claim are different claim units.

* Two separately assessed proposed amendments to the same numbered claim are different claim units.

* Conditional and unconditional amended versions are different claim units when both are substantively assessed.

* If the judgment considers only the amended claim and gives no separate outcome to the old/granted claim, output only the amended claim row.

* If the old/granted claim is invalid and a separately assessed amended version survives, output two rows: field 25 = "1" for the old/granted row and field 25 = "0" for the amended row.

* Do not use one mixed row to combine old/granted and amended outcomes.

Use precise field-18 labels, for example:

* "application claim 1"

* "old/granted claim 1"

* "amended claim 1"

* "conditional amended claim 1"

* "conditional amended claim 12 — revised amendment 1"

* "conditional amended claim 12 — revised amendment 2"

* "old/granted claim 1; whole patent conceded invalid"

WHOLE-PATENT INVALIDITY CONCESSION — SPECIAL RULE

If the judgment, final order, or reliable litigation record expressly states that the patent owner concedes that the whole patent is invalid:

* identify the exact patent version covered by the concession;

* identify every independent claim in that exact version from the claims text or contemporaneous official patent document;

* output one row for each independent claim;

* set field 25 = "1" for every such row;

* determine whether the concession expressly identifies one or more ordinary validity grounds;

* if the concession identifies an ordinary ground, code that ground = 1 and leave "Conceded invalidity" = "";

* if the concession identifies several ordinary grounds, code each identified ground = 1 and leave "Conceded invalidity" = "";

* if the concession identifies no ordinary validity ground, set "Conceded invalidity" = 1; and

* explain in field 29 that the row was created under the express whole-patent invalidity-concession rule.

Do not add dependent claims solely because the whole patent was conceded invalid. Add a dependent claim only if it also has a separately identifiable attack, decision, or individual invalidity concession.

A withdrawal, abandonment, non-maintenance, agreed revocation, or decision not to defend a patent does not trigger this rule unless the record expressly concedes or accepts invalidity.

If the whole-patent concession is clear but the independent claim numbers cannot be verified after best efforts, output one fallback row with field 18 = "independent claims not individually identified; whole patent conceded invalid" and field 25 = "1". Apply the same ground-specific concession rule: code an identified ordinary ground = 1 and leave "Conceded invalidity" = ""; use "Conceded invalidity" = 1 only when no ordinary ground of concession is identified. Explain the failed verification in field 29. Never guess claim numbers.

BINARY OUTCOME RULE FOR FIELD 25 — MANDATORY

Field 25, claim_level_winner_owner_vs_other, must always be either "0" or "1".

Never output "0.5" in field 25.

Use:

* "0" = the patent owner/applicant wins for the precise claim unit in field 18. That claim unit remains valid or patentable as presented.

* "1" = the adverse side wins for the precise claim unit in field 18. That claim unit is invalid, revoked, refused, rejected, expressly conceded invalid, or otherwise does not remain in valid form.

If the owner loses on any ground that legally invalidates the claim unit, field 25 must be "1", even if the owner wins on other grounds.

If the owner wins on every live ground and the claim unit survives, field 25 must be "0".

Do not use "0.5" to describe partial survival, narrowing, severance, different alternatives, different versions, or different claim constructions.

OUTCOME-SPLITTING RULE — MANDATORY

Every row must be outcome-homogeneous.

If the court divides what would otherwise appear to be one numbered claim or claim version into two separately identifiable claim units and gives them different overall outcomes, create separate rows.

For example, if the judgment treats one separable part, alternative, category, construction, or version of

claim 1 as valid and another as invalid:

* create one row for the valid claim unit with field 25 = "0"; and

* create one row for the invalid claim unit with field 25 = "1".

Field 18 must distinguish the two units precisely, using the judgment's terminology where possible. Field 29 must explain the distinction and why the rows were split.

Do not keep one row and assign field 25 = "0.5".

Split only when the judgment itself supplies a legally meaningful and separately identifiable division. Do not invent artificial claim fragments merely to force a binary outcome.

If a genuinely mixed result remains within a single validity ground after every legally meaningful claim unit has been separated, field 27 may use 0.5 for that ground. Field 25 must nevertheless remain "0" or "1" according to the final legal effect on the claim unit.

FIELD 18 — CLAIM_UNIT

Identify exactly one claim number and one claim version or other outcome-homogeneous unit.

Field 18 must:

* name the precise claim number when identifiable;

* identify whether it is application, old/granted, previously amended, conditional amended, unconditional amended, or another separately assessed version;

* distinguish court-defined claim parts, alternatives, constructions, or categories when the outcome-splitting rule applies; and

* remain concise but sufficiently specific to prevent two different rows from appearing to represent the same claim unit.

FIELD 25 — CLAIM_LEVEL_WINNER_OWNER_VS_OTHER

Output only "0" or "1" under the binary outcome rule above.

FIELD 27 — ISSUE_DICTIONARY

Return this dictionary with exactly these nine keys and no others:

{

"Lack of novelty":"",

"Lack of inventive step / obviousness":"",

"Insufficiency":"",

"Added matter":"",

"Excluded subject matter / non-patentability":"",

"Priority not entitled":"",

"Lack of industrial applicability":"",

"Others":"",

"Conceded invalidity":""

}

For the precise claim unit in field 18, code each ordinary validity ground as follows:

* 0 = the patent owner/applicant wins on that ground.

* 0.5 = the result on that ground is genuinely mixed or partial within the same outcome-homogeneous claim unit and cannot be represented more accurately by applying the mandatory outcome-splitting rule.

* 1 = the patent owner/applicant loses on that ground.

* "" = the ground was not live or was not decided for that claim unit.

The value 0.5 is permitted only inside field 27. It is never permitted in field 25.

CONCEDED INVALIDITY — FALLBACK CODING ONLY

"Conceded invalidity" is a fallback dataset category for an express invalidity concession that does not identify any ordinary validity ground.

* First determine whether the patent owner expressly concedes, accepts, admits, or expressly no longer contests invalidity on a specific ordinary ground.

* If the concession is expressly based on lack of novelty, set "Lack of novelty" = 1 and leave "Conceded invalidity" = "".

* If the concession is expressly based on lack of inventive step or obviousness, set "Lack of inventive step / obviousness" = 1 and leave "Conceded invalidity" = "".

* Apply the same rule to insufficiency, added matter, excluded subject matter/non-patentability, priority not entitled, lack of industrial applicability, or an identifiable ground falling within "Others".

* If several ordinary grounds are expressly conceded, code every conceded ordinary ground = 1 and leave "Conceded invalidity" = "".

* Do not double-code the same concession under both an ordinary ground and "Conceded invalidity".

* Set "Conceded invalidity" = 1 only when the patent owner expressly concedes, accepts, admits, or expressly no longer contests the invalidity of the precise claim unit without identifying any ordinary validity ground for that concession.

* Whenever "Conceded invalidity" = 1, field 25 must be "1".

* "Conceded invalidity" may contain only 1 or "".

* Never set "Conceded invalidity" to 0 or 0.5.

* If no express concession exists, leave it as "".

* If an opponent alleges that a concession occurred but the court finds that no concession or admission existed, leave "Conceded invalidity" as ""; do not code 0.

* Do not infer conceded invalidity from amendment, withdrawal, abandonment, silence, non-appearance, non-maintenance, deletion, failure to defend, or the patent-level result alone.

* If an express concession identifies no ordinary validity ground, set "Conceded invalidity" = 1 and leave the ordinary grounds according to whether they were separately live or decided. Do not automatically set "Others" = 1 merely because the concession lacks a named ground.

* A ground separately litigated or decided is still coded on its own merits. The no-double-coding rule applies to the basis of the concession, not to an independent judicial decision on another ground.

Field 27 must be substantively consistent with field 25:

* If any ordinary ground is coded 1, field 25 must be 1.

* If "Conceded invalidity" is coded 1, field 25 must be 1.

* Field 25 may be 1 even when no ordinary ground is coded 1, for example when conceded invalidity is the only adverse basis.

* If an ordinary ground is coded 1 because it was expressly conceded, "Conceded invalidity" must be "".

* Different ordinary grounds may have different results in the same row. For example, novelty may be 0 and obviousness may be 1; field 25 is then 1.

FIELD 29 — CLAIM-LEVEL SUMMARY

Rewrite field 29 for each claim-level row. Keep it under 500 words and on one physical line.

The first sentence must state:

* the overall binary outcome for the precise claim unit; and

* the patent/application-level consequence, where available.

Examples:

* "old/granted claim 1 valid; whole patent maintained."

* "old/granted claim 1 invalid; whole patent revoked."

* "old/granted claim 1 invalid; amended claim 1 considered separately."

* "amended claim 1 valid; patent maintained in amended form."

* "independent old/granted claim 1 conceded invalid; whole patent conceded invalid."

* "application claim 1 refused; patent application refused."

The summary must also:

* identify the precise claim number, claim version, and any court-defined split unit in field 18;

* explain every live validity ground and its result for that unit;

* explain any express concession of invalidity, identify its ground when one was specified, and state when the ground-neutral "Conceded invalidity" fallback was used;

* explain why two or more rows were created when the outcome-splitting rule applies;

* distinguish old/granted and amended versions;

* explain separately whether the whole patent/application was maintained, revoked, amended, narrowed, refused, restored, or upheld;

* avoid treating unargued claims as individually invalid;

* contain no URLs; and

* contain no actual line breaks.

Do not repeat entity-research explanations, ticker research, citation methodology, or other metadata work already completed in Step 1 unless it is strictly necessary to explain the claim-level result.

FINAL VALIDATION BEFORE OUTPUT

Before returning the single csv code block, verify all of the following:

1. Every output row derives from a supplied Step-1 patent-level row.

2. Fields 1–17, 19–24, 26, 28, and 30–32 are copied exactly from the corresponding Step-1 row.

3. Only fields 18, 25, 27, and 29 have been newly completed or rewritten.

4. Every separately identifiable attacked, decided, or expressly conceded claim version has its own row.

5. Claims mentioned only as background, dependency, infringement, abandonment, withdrawal, non-maintenance, or unargued claims have not been added without a clean claim-level validity outcome.

6. Old/granted and amended versions are separate whenever they receive separate consideration or outcomes.

7. Every row is outcome-homogeneous.

8. If one numbered claim or version received different outcomes for separately identifiable units, those units were split into separate rows.

9. Field 25 contains only "0" or "1" in every row.

10. Field 25 never contains "0.5" or any other value.

11. Field 27 contains exactly the nine required keys.

12. The value 0.5 appears only inside field 27 and only when a ground is genuinely mixed after all meaningful row splitting.

13. "Conceded invalidity" contains only 1 or "".

14. Whenever "Conceded invalidity" = 1, field 25 = "1" and the concession identifies no ordinary validity ground.

15. If the concession identifies an ordinary validity ground, that ground is coded 1 and "Conceded invalidity" is "".

16. The same concession is never double-coded under both an ordinary ground and "Conceded invalidity".

17. A rejected allegation of concession is not coded as conceded invalidity or as a conceded ordinary ground.

18. No claim_index was invented.

19. Every row contains exactly 32 double-quoted CSV fields.

20. Every row occupies exactly one physical line and contains no actual carriage return or newline.

21. All rows appear inside one csv code block with no header.

22. The final answer contains exactly one csv code block and no text outside it.

E ChatGPT zero-shot prompts

E.1 Patent

You are performing a zero-shot prediction task about patent-level outcomes in UK patent validity or patentability proceedings.

Your task is to predict whether all claims at issue would fail on one or more applicable grounds of invalidity or non-patentability.

You are supplied with the full patent claim set and the patent description.

A claim fails if at least one applicable ground succeeds against it. Consider the following seven grounds:

1. Lack of novelty (anticipation).
2. Lack of inventive step / obviousness.
3. Insufficiency.
4. Lack of industrial applicability.
5. Excluded subject matter / non-patentability.
6. Added matter.
7. Impermissible extension of protection by post-grant amendment.

The “skilled person” is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. “Common general knowledge” means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied patent texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any

case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent's litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

If a claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Prediction target:

- 0 = SURVIVES: at least one claim at issue survives or is allowed, including partial success.
- 1 = INVALIDATED: all claims at issue fail, including refusal of all claims at issue in an application.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with `FINAL_JSON:` followed by a JSON object containing exactly these two fields:

- "prediction": an integer, either 0 or 1.
- "probability_invalidated": your estimated probability that all claims at issue fail, expressed as a number from 0 to 1.

Use prediction = 1 if and only if `probability_invalidated` ≥ 0.5.

FULL CLAIM SET

`{patent_claims}`

PATENT DESCRIPTION

`{patent_description}`

E.2 Insufficiency

You are performing a zero-shot prediction task about UK patent insufficiency.

Your task is to predict whether the TARGET CLAIM would be found invalid for insufficiency.

You are supplied with the target claim, the full patent claim set, and the patent description.

Insufficiency means that the patent disclosure, together with the skilled person's common general knowledge, does not enable the claimed invention to be performed across substantially the full scope of the target claim without undue burden.

The "skilled person" is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. "Common general knowledge" means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied patent texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent's litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

The supplied TARGET CLAIM is the claim to be assessed. Use the full patent claim set only as context to help interpret the target claim and its dependencies. The target claim may not appear verbatim in that claim set. Do not treat its absence as evidence of insufficiency, and do not replace it with a different claim from the claim set. Where their wording differs, assess the target claim as supplied.

If the target claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Prediction target:

- 0 = NOT INSUFFICIENT: the target claim survives an insufficiency challenge.
- 1 = INSUFFICIENT: the target claim fails for insufficiency.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with `FINAL_JSON:` followed by a JSON object containing exactly these two fields:

- "prediction": an integer, either 0 or 1.
- "probability_insufficient": your estimated probability that the target claim fails for insufficiency, expressed as a number from 0 to 1.

Use prediction = 1 if and only if `probability_insufficient` ≥ 0.5.

TARGET CLAIM

`{target_claim}`

FULL CLAIM SET

`{patent_claims}`

PATENT DESCRIPTION

`{patent_description}`

E.3 Obviousness

You are performing a zero-shot prediction task about UK patent obviousness.

Your task is to predict whether the TARGET CLAIM would be found invalid for obviousness on the basis of the supplied prior-art document or documents.

You are supplied with the target claim, the full patent claim set, and one to three prior-art documents for a single obviousness attack.

Obviousness means that the skilled person, using their common general knowledge and without hindsight, would have arrived at the claimed invention from the supplied prior art without exercising inventive ingenuity.

The "skilled person" is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. "Common general knowledge" means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied claim and prior-art texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent's litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

The supplied TARGET CLAIM is the claim to be assessed. Use the full patent claim set only as context to help interpret the target claim and its dependencies. The target claim may not appear verbatim in that claim set. Do not treat its absence as evidence of obviousness, and do not replace it with a different claim from the claim set. Where their wording differs, assess the target claim as supplied.

If the target claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Return one prediction for the supplied attack as a whole. Where the attack relies on combining documents, assess whether the skilled person would have had reason, without hindsight, to consider their teachings together. Finding individual claim features across different documents is not, by itself, sufficient to establish obviousness.

Prediction target:

- 0 = NOT OBVIOUS: the target claim survives this obviousness attack.
- 1 = OBVIOUS: the target claim fails for obviousness on the basis of this attack.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with `FINAL_JSON:` followed by a JSON object containing exactly these two fields:

- "prediction": an integer, either 0 or 1.
- "probability_obvious": your estimated probability that the target claim fails for obviousness on the basis of this attack, expressed as a number from 0 to 1.

Use prediction = 1 if and only if `probability_obvious` ≥ 0.5.

TARGET CLAIM

`{target_claim}`

FULL CLAIM SET

`{patent_claims}`

PRIOR-ART DOCUMENTS FOR THIS ATTACK

`{numbered_prior_art_documents}`

References

Arlot, S. and Celisse, A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79.

Ashtor, J. H. (2022). Modeling patent clarity. *Research Policy*, 51(2):104415.

Branstetter, L. G., Chatterjee, C., and Higgins, M. J. (2016). Regulation and welfare: Evidence from paragraph IV generic entry in the pharmaceutical industry. *RAND Journal of Economics*, 47(4):857–890.

Chen, W., Gao, Y., Zhang, J., King, I., and Lyu, M. R. (2019). Title-guided encoding for keyphrase generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 6268–6275.

Conneau, A., Kiela, D., Schwenk, H., Barrault, L., and Bordes, A. (2017). Supervised learning of universal sentence representations from natural language inference data. In *Proceedings of EMNLP*, pages 670–680.

Courts and Tribunals Judiciary (n.d.a). Court of appeal judges. `https://www.judiciary.uk/about-the-judiciary/who-are-the-judiciary/judges/coa-judges/`. Accessed August 31, 2026.

Courts and Tribunals Judiciary (n.d.b). Work of the intellectual property enterprise court. `https://www.judiciary.uk/courts-and-tribunals/business-and-property-courts/business-list-general-chancery/intellectual-property-list/intellectual-property-enterprise-court-ipec/work/`. Accessed August 31, 2026.

Cowart, T. W., Lirely, R., and Avery, S. (2014). Two methodologies for predicting patent litigation outcomes: Logistic regression versus classification trees. *American Business Law Journal*, 51(4):843–877.

Dietterich, T. G., Lathrop, R. H., and Lozano-Pérez, T. (1997). Solving the multiple instance problem with axis-parallel rectangles. *Artificial Intelligence*, 89(1-2):31–71.

Farre-Mensa, J., Hegde, D., and Ljungqvist, A. (2020). What is a patent worth? evidence from the U.S. patent "lottery". *Journal of Finance*, 75(2):639–682.

Galasso, A. and Schankerman, M. (2015). Patents and cumulative innovation: Causal evidence from the courts. *Quarterly Journal of Economics*, 130(1):317–369.

Gao, X. K., Yao, F., Zhao, K., He, B., Kumar, A., Krishnan, V., and Shang, J. (2024). Beyond scaling: Predicting patent approval with domain-specific fine-grained claim dependency graph. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5218–5234, Bangkok, Thailand. Association for Computational Linguistics.

Hall, B. H., Jaffe, A. B., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hicks, J. D. (2021). *New Approaches to Old Questions in Intellectual Property*. Ph.d. dissertation, University of California, Berkeley. Chapter 2: How Predictable Are Patent Validity Decisions?

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Ilse, M., Tomczak, J., and Welling, M. (2018). Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2127–2136. PMLR.

Janicke, P. M. and Ren, L. (2006). Who wins patent infringement cases? *AIPLA Quarterly Journal*, 34(1):1–44.

Knappich, V., Friedrich, A., Hätty, A., and Razniewski, S. (2025). PEDANTIC: A dataset for the automatic examination of definiteness in patent claims. In *Proceedings of the 6th Workshop on Patent Text Mining and Semantic Technologies*.

Knappich, V., Hätty, A., Razniewski, S., and Friedrich, A. (2026). Is it novel and why? fine-grained patent novelty prediction based on passage retrieval. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery.

Kogan, L., Papanikolaou, D., Seru, A., and Stoffman, N. (2017). Technological innovation, resource allocation, and growth. *Quarterly Journal of Economics*, 132(2):665–712.

Kumar, V., Gupta, Y., Chemmengath, S., Sen, J., Chakrabarti, S., Bharadwaj, S., and Pan, F. (2023). Multi-row, multi-span distant supervision for Table+Text question answering. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8080–8094. Association for Computational Linguistics.

Lee, J., Lee, Y., Kim, J., Kosiorek, A., Choi, S., and Teh, Y. W. (2019). Set Transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3744–3753. PMLR.

Lim, H., Nam, S., Na, S., Cho, J. Y., Yang, J. Y., Shin, H., Lee, Y., Kim, J., Lee, M., and Hong, H. (2025). PANORAMA: A dataset and benchmarks capturing decision trails and rationales in patent examination. In *Advances in Neural Information Processing Systems*, volume 38, pages 57923–57974.

Lo, H.-C. and Chu, J.-M. (2021). Pre-trained transformer-based classification for automated patentability examination. In *2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, pages 1–5. IEEE.

Marco, A. C. (2005). The value of certainty in intellectual property rights: Stock market reactions to patent litigation. Working Paper 82, Vassar College Department of Economics.

Maron, O. and Lozano-Pérez, T. (1998). A framework for multiple-instance learning. In *Advances in Neural Information Processing Systems*, volume 10. MIT Press.

Moss, H., Leslie, D., and Rayson, P. (2018). Using J-K-fold cross validation to reduce variance when tuning NLP models. In *Proceedings of the 27th International Conference on Computational Linguistics*, pages 2978–2989, Santa Fe, New Mexico, USA. Association for Computational Linguistics.

Ngiam, J., Khosla, A., Kim, M., Nam, J., Lee, H., and Ng, A. Y. (2011). Multimodal deep learning. In *Proceedings of the International Conference on Machine Learning*, pages 689–696.

Pakes, A. (1986). Patents as options: Some estimates of the value of holding european patent stocks. *Econometrica*, 54(4):755–784.

Panattoni, L. E. (2011). The effect of paragraph IV decisions and generic entry before patent expiration on brand pharmaceutical firms. *Journal of Health Economics*, 30(1):126–145.

Raghupathi, V., Zhou, Y., and Raghupathi, W. (2018). Legal decision support: Exploring big data analytics approach to modeling pharma patent validity cases. *IEEE Access*, 6:41518–41528.

Raghupathi, W. (2026). Exploring artificial intelligence and machine learning approaches to legal reasoning. *AppliedMath*, 6(2):32.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992.

Sathe, A. and Park, J. (2021). Automatic fact-checking with document-level annotations using BERT and multiple instance learning. In *Proceedings of the Fourth Workshop on Fact Extraction and VERification (FEVER)*, pages 101–107. Association for Computational Linguistics.

Shao, Z., Bian, H., Chen, Y., Wang, Y., Zhang, J., Ji, X., and Zhang, Y. (2021). TransMIL: Transformer based correlated multiple instance learning for whole slide image classification. In *Advances in Neural Information Processing Systems*, volume 34.

Shi, F., Chen, X., Misra, K., Scales, N., Dohan, D., Chi, E. H., Schärli, N., and Zhou, D. (2023). Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227. PMLR.

Suh, D.-C., Manning, Jr., W. G., Schondelmeyer, S. W., and Hadsall, R. S. (2000). Effect of multiple-source entry on price competition after patent expiration in the pharmaceutical industry. *Health Services Research*, 35(2):529–547.

UK Intellectual Property Office (n.d.). Manual of patent practice. `https://www.gov.uk/guidance/manual-of-patent-practice-mopp`. Accessed August 31, 2026.

UK Supreme Court (n.d.). The supreme court of the united kingdom and the supreme court of the united states: A comparative learning tool. `https://www.supremecourt.uk/uploads/the_supreme_court_of_the_united_states_and_the_supreme_court_of_the_united_kingdom_a_comparative_learning_tool_7c149667ed.pdf`. Accessed August 31, 2026.

Xu, X., Frank, E., and Holmes, G. (2025). Multiple instance verification. *Journal of Machine Learning Research*, 26(188):1–46.

Zhang, Y., Li, M., Long, D., et al. (2025). Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.